DWG. No. 310SC90-794

DWG.No. SC90-794 R2// REGD. 2015-05-29
CHKD. R.KOJIMA/ APPD. Y.KOUNO// RE-REGD. 2016-07-21

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REV.	DESCRIPTION	REVD.	CHKD.	APPD.	DATE
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C	TO REFLECT THE DESIGN PROGRESS.	R.KOJIMA	M.SAKAKIBARA	N.MATSUMOTO	2014-10-02
D	TO REFLECT THE DESIGN PROGRESS.	M.SAKAKIBARA	N.MATSUMOTO	N.MATSUMOTO	2014-11-14
0	SUBMISSION AS "FINAL" ENGINEERING DOCUMENT.	R.KOJIMA	M.SAKAKIBARA	M.SAKAKIBARA	2015-05-29
1	TO REFLECT CUSTOMER REQUIREMENTS.	R.KOJIMA	Y.KONO	Y.KONO	2016-01-19
2	TO REFLECT SITE COMMISSIONING RESULTS.	A.DEYNCHAUSTI	R.KOJIMA	Y.KONO	2016-07-18

TURBINE GENERATOR
INTERLOCK BLOCK DIAGRAM

REFERENCE DRAWINGS

1. DIGITAL INTERLOCK CONTROL LOGIC DIAGRAM (K1AU4-IO-005)
2. INSTRUMENT LIST FOR TURBINE AND GENERATOR (310SC90-799)
3. ATS UNIT SEQUENCE CONTROL PROGRAM (310SC90-795)
4. ATS CRITERIA LIST AND CRITERIA TABLE (310SC90-796)
5. INPUT/OUTPUT LIST BETWEEN CUSTOMER EQUIPMENT AND MHPS EQUIPMENT (310SC90-793)
6. ELEMENTARY WIRING DIAGRAM OF MOTOR OPERATED VALVE (310SC90-792)
7. MAIN CONTROL SYSTEM CONFIGURATION FOR TURBINE/GENERATOR AND AUXILIARIES (PCE-13-0106)
8. OPERATIONAL CONCEPT AND CONTROL SYSTEM DESCRIPTION FOR STEAM TURBINE (PCE-13-0142)

三菱日立パワーシステムズ株式会社
2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD.

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FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

Drawn : Name/Date	R.KOJIMA	/2013-05-31
Checked : Name/Date	M.SAKAKIBARA	/2013-05-31
Approved: Name/Date	S.Nemoto	/2013-05-31

Title,supplementary title

Turbine Generator
Interlock Block Diagram

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Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	1/99	2016-07-18

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CAD SOFTWARE	AutoCAD	SIZE DESIGNATION	A3

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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
1	——	COVER	——	——	——	——	——	——	——	——	——	
2	——	CONTENTS (1)	——	——	——	——	——	——	——	——	——	
3	——	CONTENTS (2)	——	——	——	——	——	——	——	——	——	
4	——	CONTENTS (3)	——	——	——	——	——	——	——	——	——	
5	——	CONTENTS (4)	——	——	——	——	——	——	——	——	——	
6	——	CONTENTS (5)	——	——	——	——	——	——	——	——	——	
7	——	CONTENTS (6)	——	——	——	——	——	——	——	——	——	
8	——	CONTENTS (7)	——	——	——	——	——	——	——	——	——	
9	——	CONTENTS (8)	——	——	——	——	——	——	——	——	——	
10	——	CONTENTS (9)	——	——	——	——	——	——	——	——	——	
11	——	CONTENTS (10)	——	——	——	——	——	——	——	——	——	
12	——	CONTENTS (11)	——	——	——	——	——	——	——	——	——	

NOTE:
"n" INDICATES UNIT NO. AND IT CHANGES ACCORDING TO EACH UNIT.

TYPE	–			PAGE	2
SERVICE	CONTENTS (1)				
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	2	2016-07-18

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Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

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—— : NOT EXIST　　◉ : INDICATION ONLY

PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
13	——	SYMBOLS AND ABBREVIATIONS	——	——	——	——	——	——	——	——	——	
14	——	TYPICAL LOGIC FOR MOTOR	——	——	——	——	——	——	——	——	——	
15	——	SIMPLIFIED LOGIC FOR MOTOR	——	——	——	——	——	——	——	——	——	
16	——	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, SEAL-IN TYPE)	——	——	——	——	——	——	——	——	——	
17	——	SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, SEAL-IN TYPE)	——	——	——	——	——	——	——	——	——	
18	——	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (POSITION SEATING, SEAL-IN TYPE)	——	——	——	——	——	——	——	——	——	
19	——	SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (POSITION SEATING, SEAL-IN TYPE)	——	——	——	——	——	——	——	——	——	
20	——	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, INCHING TYPE)	——	——	——	——	——	——	——	——	——	
21	——	SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, INCHING TYPE)	——	——	——	——	——	——	——	——	——	
22	——	TYPICAL LOGIC FOR AIR OPERATED VALVE	——	——	——	——	——	——	——	——	——	
23	——	SIMPLIFIED LOGIC FOR AIR OPERATED VALVE	——	——	——	——	——	——	——	——	——	

TYPE	–					PAGE 3
SERVICE	CONTENTS (2)					
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 3	Date of issue 2016-07-18
	MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division					

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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TA-1	——	TURBINE MASTER TRIP / MASTER RESET*	——	——	——	——	——	○	——	○	○**	*TURBINE MASTER RESET HAS NO REMOTE HARD/LOCAL OPERATION **TRIP LEVER IS PROVIDED WITH TRIP DEVICE (MASTER TRIP SOLENOID VALVE) INSTALLED ON MAIN OIL TANK AND TRIP PB INSTALLED ON TURBINE FRONT STANDARD.
TA-2	——	MASTER TRIP SOLENOID (A1),(A2) TEST	——	——	——	——	○	○	——	——	——	
TA-3	——	MASTER TRIP SOLENOID (B1),(B2) TEST	——	——	——	——	○	○	——	——	——	
TA-4	——	–	——	——	——	——	——	——	——	——	——	
TA-5	——	–	——	——	——	——	——	——	——	——	——	
TA-6	——	MAIN STOP VALVE FULL CLOSE (FOR BOILER PROTECTION & ELECTRICAL TRIP)	——	——	——	——	——	——	——	——	——	
TA-7	——	MECHANICAL LOGIC	——	——	——	——	——	——	——	——	——	
TA-8	——	–	——	——	——	——	——	——	——	——	——	
TA-9	——	TRIP LOGIC (1) (TOTAL)	——	——	——	——	——	——	——	——	——	
TA-10	——	TRIP LOGIC (2) (CONDENSER VACUUM LOW TRIP)	——	——	——	——	——	——	——	——	——	
TA-11	——	TRIP LOGIC (3) (MAIN STEAM TEMP. LOW TRIP)	——	——	——	——	——	——	——	——	——	
TA-12	——	TRIP LOGIC (4) (THRUST BEARING FAILURE TRIP)	——	——	——	——	——	——	——	——	——	
TA-13	——	TRIP LOGIC (5) (CONTROL OIL PRESS. VERY LOW TRIP/ BEARING OIL PRESS. VERY LOW TRIP)	——	——	——	——	——	——	——	——	——	
TA-14	——	TRIP LOGIC (6) (LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP/ TURBINE TRIP OIL PRESS. LOW TRIP)	——	——	——	——	——	——	——	——	——	
TA-15	——	TRIP LOGIC (7) (OVERSPEED TRIP/BACK-UP OVERSPEED TRIP)	——	——	——	——	——	——	——	——	——	
TA-16	——	–	——	——	——	——	——	——	——	——	——	
TA-17	——	TRIP LOGIC (8) (BEARING VIBRATION VERY HIGH TRIP (1))	——	——	——	——	——	——	——	——	——	
TA-18	——	TRIP LOGIC (9) (BEARING VIBRATION VERY HIGH TRIP (2))	——	——	——	——	——	——	——	——	——	
TA-19	——	–	——	——	——	——	——	——	——	——	——	
TA-20	——	TRIP LOGIC (10) (HP TURBINE EXHAUST STEAM TEMP. VERY HIGH TRIP)	——	——	——	——	——	——	——	——	——	

TYPE –	PAGE 4
SERVICE CONTENTS (3)	

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	4	2016-07-18

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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TA-21	——	TRIP LOGIC (11) (HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. VERY HIGH TRIP)	——	——	——	——	——	——	——	——	——	
TA-22	——	TRIP LOGIC (12) (HP TURBINE EXHAUST PRESS. VERY HIGH TRIP)	——	——	——	——	——	——	——	——	——	
TA-23	——	–	——	——	——	——	——	——	——	——	——	
TA-24	——	–	——	——	——	——	——	——	——	——	——	
TA-25	——	–	——	——	——	——	——	——	——	——	——	
TA-26	——	–	——	——	——	——	——	——	——	——	——	
TA-27	——	–	——	——	——	——	——	——	——	——	——	
TA-28	——	–	——	——	——	——	——	——	——	——	——	
TA-29	——	MSV(A)&(B) / RSV(A)&(B) OPENING PROCEDURE	——	——	——	——	——	——	——	——	——	
TA-30	——	MSV (A)/(B), CV (A)/(B) PARTIAL CLOSE TEST CHART	——	——	——	——	——	——	——	——	——	
TA-31	——	MAIN STOP VALVE (A) PARTIAL CLOSE TEST	——	——	——	——	○	○	——	——	——	
TA-32	——	MAIN STOP VALVE (B) PARTIAL CLOSE TEST	——	——	——	——	○	○	——	——	——	
TA-33	——	CONTROL VALVE (A) PARTIAL CLOSE TEST	——	——	——	——	○	○	——	——	——	
TA-34	——	CONTROL VALVE (B) PARTIAL CLOSE TEST	——	——	——	——	○	○	——	——	——	
TA-35	——	–	——	——	——	——	——	——	——	——	——	
TA-36	——	CRV (A)/(B) CLOSE TEST CHART	——	——	——	——	——	——	——	——	——	
TA-37	——	COMBINED REHEAT VALVE (A) TEST	——	——	——	——	○	○	——	——	——	
TA-38	——	COMBINED REHEAT VALVE (B) TEST	——	——	——	——	○	○	——	——	——	
TA-39	——	–	——	——	——	——	——	——	——	——	——	
TA-40	——	MSV/CV (A)/(B) SEQUENTIAL FULL CLOSE TEST CHART	——	——	——	——	——	——	——	——	——	

TYPE	–	PAGE	5
SERVICE	CONTENTS (4)		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	5	2016-07-18

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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TA-41	——	MSV/CV(A) SEQUENTIAL FULL CLOSE TEST	—	—	—	—	—	—	—	—	—	
TA-42	——	MSV/CV(B) SEQUENTIAL FULL CLOSE TEST	—	—	—	—	—	—	—	—	—	
TA-43	——	MSV/CV (A)/(B) SEQUENTIAL FULL CLOSE TEST PERMISSIVE	—	—	—	—	—	—	—	—	—	
TB-1	P-n922A	MAIN OIL PUMP (A)	○	—	—	—	—	○	—	—	—	
TB-2	P-n922B	MAIN OIL PUMP (B)	○	—	—	—	—	○	—	—	—	
TB-3	——	–	—	—	—	—	—	—	—	—	—	
TB-4	——	–	—	—	—	—	—	—	—	—	—	
TB-5	P-n923	DC EMERGENCY OIL PUMP	—	—	○	—	—	○	—	—	—	
TB-6	——	–	—	—	—	—	—	—	—	—	—	
TB-7	——	–	—	—	—	—	—	—	—	—	—	
TB-8	——	TURNING GEAR ENGAGE TIME CHART (AUTO MODE)	—	—	—	—	—	—	—	—	—	
TB-9		TURNING GEAR MOTOR	○	—	—	—	—	○	—	—	○*	*TURNING GEAR LOCAL BOX
TB-10		TURNING GEAR ENGAGE SOLENOID	—	—	—	—	○	—	—	—	○*	*TURNING GEAR LOCAL BOX
TB-11	——	–	—	—	—	—	—	—	—	—	—	
TB-12	F-n937	MAIN OIL TANK VAPOUR EXTRACTOR	○	—	—	—	—	○	—	—	—	
TB-13	——	–	—	—	—	—	—	—	—	—	—	
TB-14	——	–	—	—	—	—	—	—	—	—	—	
TB-15	P-n941	OIL PURIFIER PUMP	○	—	—	—	—	○	—	—	—	
TB-16	——	–	—	—	—	—	—	—	—	—	—	
TB-17	——	–	—	—	—	—	—	—	—	—	—	

TYPE –					PAGE 6	
SERVICE CONTENTS (5)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 6	Date of issue 2016-07-18
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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TB-21	P-n931A	CONTROL OIL PUMP (A)	○	——	——	——	——	○	——	——	——	
TB-22	P-n931B	CONTROL OIL PUMP (B)	○	——	——	——	——	○	——	——	——	
TB-23	——	–	——	——	——	——	——	——	——	——	——	
TB-24	——	–	——	——	——	——	——	——	——	——	——	
TB-25	——	–	——	——	——	——	——	——	——	——	——	
TB-26	——	–	——	——	——	——	——	——	——	——	——	
TB-27	——	–	——	——	——	——	——	——	——	——	——	
TB-28	——	–	——	——	——	——	——	——	——	——	——	
TB-29	——	–	——	——	——	——	——	——	——	——	——	
TB-30	——	–	——	——	——	——	——	——	——	——	——	
TB-31	——	–	——	——	——	——	——	——	——	——	——	
TB-32	——	–	——	——	——	——	——	——	——	——	——	
TB-33	——	–	——	——	——	——	——	——	——	——	——	
TB-34	——	–	——	——	——	——	——	——	——	——	——	
TB-35	——	–	——	——	——	——	——	——	——	——	——	
TB-36	——	–	——	——	——	——	——	——	——	——	——	
TB-37	——	–	——	——	——	——	——	——	——	——	——	
TB-38	——	–	——	——	——	——	——	——	——	——	——	
TB-39	——	–	——	——	——	——	——	——	——	——	——	
TB-40	——	–	——	——	——	——	——	——	——	——	——	

TYPE	–				PAGE	7
SERVICE	CONTENTS (6)					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	7	2016-07-18

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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TC-1	MV-STn002A	MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-2	MV-STn002B	MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-3	——	–	——	——	——	——	——	——	——	——	——	
TC-4	——	–	——	——	——	——	——	——	——	——	——	
TC-5	MV-STn008	CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-6	——	–	——	——	——	——	——	——	——	——	——	
TC-7	MV-STn004A	CV(A) ABOVE SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-8	MV-STn004B	CV(B) ABOVE SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-9	——	–	——	——	——	——	——	——	——	——	——	
TC-10	——	–	——	——	——	——	——	——	——	——	——	
TC-11	——	–	——	——	——	——	——	——	——	——	——	
TC-12	MV-STn010A	CRV(A) BELOW SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-13	MV-STn010B	CRV(B) BELOW SEAT SECONDARY DRAIN VALVE	○	——	——	○	——	○	——	——	——	
TC-14	——	–	——	——	——	——	——	——	——	——	——	
TC-15	——	–	——	——	——	——	——	——	——	——	——	
TC-16	——	–	——	——	——	——	——	——	——	——	——	
TC-17	FV-CRn902	VENTILATOR VALVE	——	——	——	——	○	——	——	——	——	
TC-18	FV-GSn039	BLOWDOWN VALVE	——	——	——	——	○	——	——	——	——	
TC-19	XV-CRn901	COLD REHEAT NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-20	MV-ASn011	HP TURBINE WARMING VALVE	○	——	——	○	——	○	——	——	——	

TYPE –					PAGE 8	
SERVICE CONTENTS (7)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 8	Date of issue 2016-07-18
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PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TC-21	——	–	——	——	——	——	——	——	——	——	——	
TC-22	——	–	——	——	——	——	——	——	——	——	——	
TC-23	——	–	——	——	——	——	——	——	——	——	——	
TC-24	——	–	——	——	——	——	——	——	——	——	——	
TC-25	——	–	——	——	——	——	——	——	——	——	——	
TC-26	——	–	——	——	——	——	——	——	——	——	——	
TC-27	——	–	——	——	——	——	——	——	——	——	——	
TC-28		GLAND STEAM EXHAUSTER	○	——	——	——	——	○	——	——	——	
TC-29	——	–	——	——	——	——	——	——	——	——	——	
TC-30	——	–	——	——	——	——	——	——	——	——	——	
TC-31	MV-GSn030	STEAM SEAL FEED INLET VALVE	○	——	——	○	——	○	——	——	——	
TC-32	——	–	——	——	——	——	——	——	——	——	——	
TC-33	——	–	——	——	——	——	——	——	——	——	——	
TC-34	——	–	——	——	——	——	——	——	——	——	——	
TC-35	FV-CDn904	LP EXHAUST HOOD WATER SPRAY VALVE	——	——	——	——	○	○	——	——	——	
TC-36	FV-GSn036	STEAM SEAL DIVERTING VALVE	——	——	——	——	○	○	——	——	——	
TC-37	——	–	——	——	——	——	——	——	——	——	——	
TC-38	——	–	——	——	——	——	——	——	——	——	——	
TC-39	——	–	——	——	——	——	——	——	——	——	——	
TC-40	——	–	——	——	——	——	——	——	——	——	——	

TYPE –	PAGE 9
SERVICE CONTENTS (8)	

Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	9	2016-07-18

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—— : NOT EXIST　　◉ : INDICATION ONLY

PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TC-41	XV-TSn700	NO.2 EXTRACTION STEAM NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-42	XV-TSn710	NO.3 EXTRACTION STEAM PRIMARY NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-43	XV-TSn740	NO.3 EXTRACTION STEAM SECONDARY NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-44	XV-TSn720	NO.4 EXTRACTION STEAM NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-45	XV-TSn730	NO.5 EXTRACTION STEAM NON RETURN VALVE	——	——	——	——	○	○	——	——	——	
TC-46	——	–	——	——	——	——	——	——	——	——	——	
TC-47	——	–	——	——	——	——	——	——	——	——	——	
TC-48	——	–	——	——	——	——	——	——	——	——	——	
TC-49	——	–	——	——	——	——	——	——	——	——	——	
TC-50	——	–	——	——	——	——	——	——	——	——	——	
TC-51	——	–	——	——	——	——	——	——	——	——	——	
TC-52	——	–	——	——	——	——	——	——	——	——	——	
TC-53	——	–	——	——	——	——	——	——	——	——	——	
TC-54	——	–	——	——	——	——	——	——	——	——	——	
TC-55	——	–	——	——	——	——	——	——	——	——	——	
TC-56	——	–	——	——	——	——	——	——	——	——	——	
TC-57	——	–	——	——	——	——	——	——	——	——	——	
TC-58	——	–	——	——	——	——	——	——	——	——	——	
TC-59	——	–	——	——	——	——	——	——	——	——	——	
TC-60	——	–	——	——	——	——	——	——	——	——	——	

TYPE	–				PAGE	10
SERVICE	CONTENTS (9)					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	10	2016-07-18

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Engineering Headquarters / Plant Engineering Division

○ : EXIST　　◎ : OPERATION AND INDICATION
—— : NOT EXIST　　◉ : INDICATION ONLY

PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TD-1	——	TURBINE ALARM (1)	——	——	——	——	——	——	——	——	——	
TD-2	——	TURBINE ALARM (2)	——	——	——	——	——	——	——	——	——	
TD-3	——	TURBINE ALARM (3)	——	——	——	——	——	——	——	——	——	
TD-4	——	TURBINE ALARM (4)	——	——	——	——	——	——	——	——	——	
TD-5	——	–	——	——	——	——	——	——	——	——	——	
TD-6	——	–	——	——	——	——	——	——	——	——	——	
TD-7	——	–	——	——	——	——	——	——	——	——	——	
TD-8	——	–	——	——	——	——	——	——	——	——	——	
TD-9	——	–	——	——	——	——	——	——	——	——	——	
TD-10	——	HOUSE LOAD OPERATION ALARM	——	——	——	——	——	——	——	——	——	
TD-11	——	–	——	——	——	——	——	——	——	——	——	
TD-12	——	–	——	——	——	——	——	——	——	——	——	
TD-13	——	–	——	——	——	——	——	——	——	——	——	
TD-14	——	–	——	——	——	——	——	——	——	——	——	
TD-15	——	–	——	——	——	——	——	——	——	——	——	
TD-16	——	–	——	——	——	——	——	——	——	——	——	
TD-17	——	–	——	——	——	——	——	——	——	——	——	
TD-18	——	–	——	——	——	——	——	——	——	——	——	
TD-19	——	–	——	——	——	——	——	——	——	——	——	
TD-20	——	–	——	——	——	——	——	——	——	——	——	

TYPE	–
PAGE	11
SERVICE	CONTENTS (10)

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	11	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

○ : EXIST　　◎ : OPERATION AND INDICATION
—— : NOT EXIST　　◉ : INDICATION ONLY

PAGE NO.	EQUIPMENT NO.	SERVICE	POWER SUPPLY			VALVE TYPE		OPERATION				REMARKS
			MCC	AC DB	DC DB	MOV	AOV SV	TCS HMI	DCS HMI	REMOTE HARD	LOCAL	
TE-0	——	HP TURBINE WARMING CHART	——	——	——	——	——	——	——	——	——	
TE-1	——	HP TURBINE WARMING START GROUP CONTROL	——	——	——	——	——	○	——	——	——	
TE-2	——	HP TURBINE WARMING STOP GROUP CONTROL	——	——	——	——	——	○	——	——	——	
TE-3	——	HP TURBINE WARMING GC RELATED AUX. IN AUTO	——	——	——	——	——	——	——	——	——	
TE-4	——	–	——	——	——	——	——	——	——	——	——	
TE-5	——	TURBINE GLAND SEAL TIME CHART	——	——	——	——	——	——	——	——	——	
TE-6	——	TURBINE GLAND SEAL START GROUP CONTROL	——	——	——	——	——	——	——	——	——	
TE-7	——	TURBINE GLAND SEAL STOP GROUP CONTROL	——	——	——	——	——	——	——	——	——	
TE-8	——	TURBINE GLAND SEAL GC RELATED AUX. IN AUTO	——	——	——	——	——	——	——	——	——	
TE-9	——	–	——	——	——	——	——	——	——	——	——	
GD-1	——	SPACE HEATER FOR GEN.	○	——	——	——	——	——	——	——	——	
GD-2	——	SPACE HEATER FOR AC EXCITER	——	○	——	——	——	——	——	——	——	
GD-3	——	–	——	——	——	——	——	——	——	——	——	
GD-4	——	–	——	——	——	——	——	——	——	——	——	
GD-5	——	GENERATOR ALARM (1)	——	——	——	——	——	——	——	——	——	
GD-6	——	–	——	——	——	——	——	——	——	——	——	
GD-7	——	–	——	——	——	——	——	——	——	——	——	
GD-8	——	–	——	——	——	——	——	——	——	——	——	
GD-9	——	–	——	——	——	——	——	——	——	——	——	
GD-10	——	–	——	——	——	——	——	——	——	——	——	

TYPE	–	PAGE	12
SERVICE	CONTENTS (11)		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	12	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

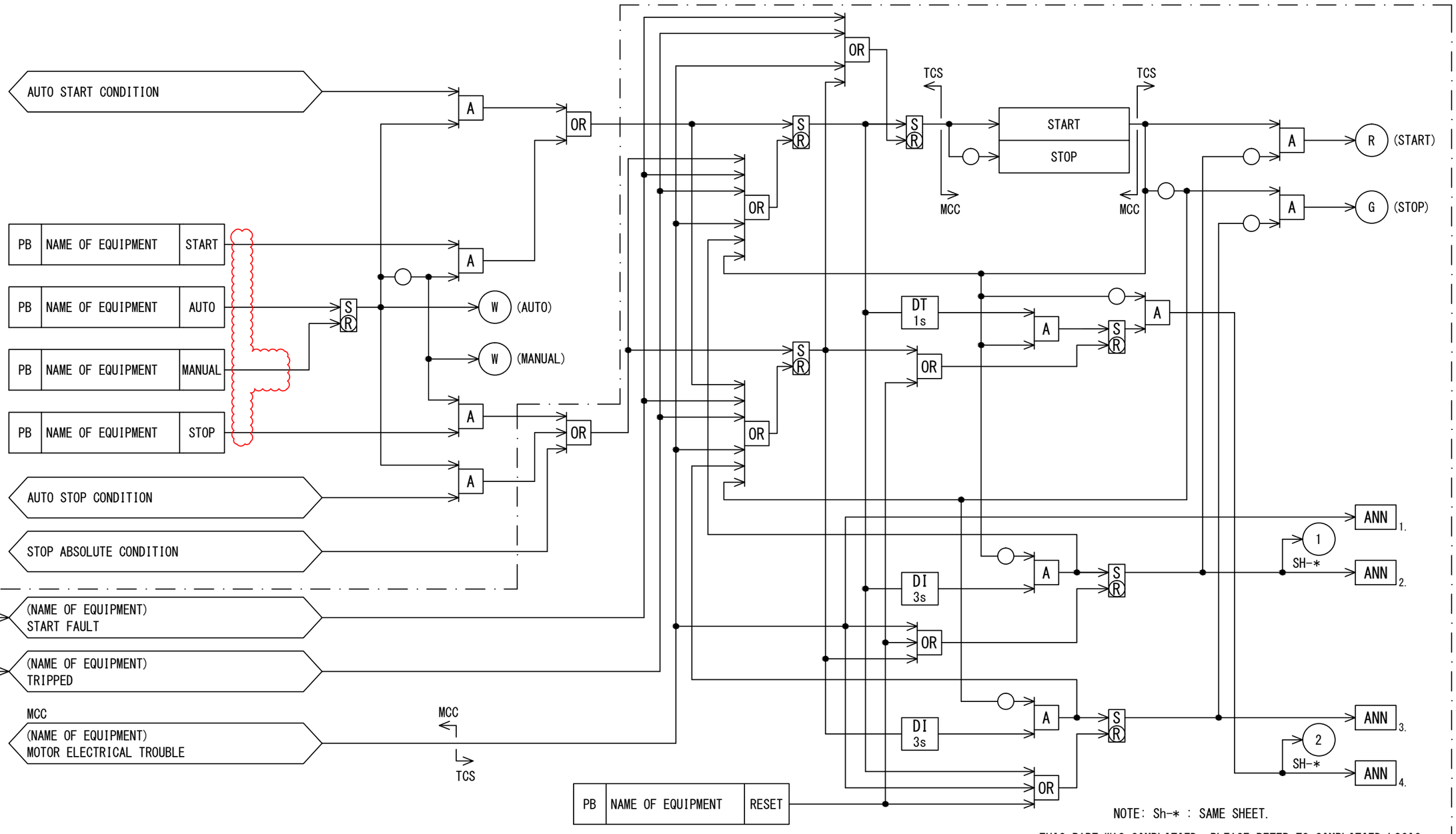
SYMBOL	DESCRIPTION
→	ELECTRICAL SIGNAL (DIGITAL)
- - - - - →	ELECTRICAL SIGNAL (ANALOG)
⊙ ⊙ ⊙ ⊙ →	MECHANICAL SIGNAL
L L L L →	HYDRAULIC SIGNAL
// // // // →	PNEUMATIC SIGNAL
○ ○ ○ ○ →	COMMUNICATION SIGNAL
A	"AND" GATE
OR	"OR" GATE
○	"NOT" GATE
2/3	"2 OUT OF 3" GATE
A → S → B, C → R	MEMORY CIRCUIT IF THERE IS NOT "C" SIGNAL, "A" SIGNAL IS SELF-HOLDED AND IT GOES TO "B". IF THERE IS "C" SIGNAL, THE SELF-HOLD IS RESET AND "A" SIGNAL DOES NOT GO TO "B". ("C" SIGNAL HAS PRIORITY)
DI t	TIMER (DELAY PICK UP) t: SETTING TIME
DT t	TIMER (DELAY DROP OUT) t: SETTING TIME
PO t	TIMER (ONE SHOT) t: SETTING TIME
OPI	OPENING POSITION INDICATOR
CMP	COMPARATOR
FLK T1 : s T2 : s	SIGNAL INPUT 1 0 SIGNAL OUTPUT 1 0 T1 T2 T1 T2 T1 T2 T1
A — (WO) — B C	IF SIGNAL "A" EXISTS (1) AND SIGNAL "C" NOT EXISTS (0), SIGNAL "B" WILL BE OUTPUTTED (1). IF SIGNAL "A" NOT EXISTS (0) AND SIGNAL "C" EXISTS (1), SIGNAL "B" WILL NOT BE OUTPUTTED (0). IF SIGNAL "A" EXISTS (1) AND SIGNAL "C" EXISTS (1), SIGNAL "B" WILL NOT BE OUTPUTTED (0). IF SIGNAL "A" NOT EXISTS (0) AND SIGNAL "C" NOT EXISTS (0), SIGNAL "B" WILL NOT BE OUTPUTTED (0).

SYMBOL	DESCRIPTION
Y	CONDITION Y : INSTRUMENT TAG No.
A Z Z	OPERATION OF EQUIPMENT OR STATUS Z : STATUS A : NAME OF EQUIPMENT
ANN	OUTPUT SIGNAL TO ALARM
ATS	OUTPUT SIGNAL TO AUTOMATIC TURBINE START-UP/SHUTDOWN FUNCTION
MID	"MEDIAN" GATE
R (N)	INDICATING LIGHT R : RED LIGHT G : GREEN LIGHT W : WHITE LIGHT A : AMBER LIGHT O : ORANGE LIGHT (N) : LOCATION
A SH-N	LOGIC CONTINUATION (DESTINATION) A : CONNECTION POINT N : PAGE No.
A SH-N	LOGIC CONTINUATION (DEPARTURE) A : CONNECTION POINT N : PAGE No.
SYSTEM or EQUIPMENT SYSTEM or EQUIPMENT	SCOPE OF SUPPLY (FOR HARDWIRED SIGNAL)
SYSTEM or EQUIPMENT SYSTEM or EQUIPMENT	SCOPE OF SUPPLY (FOR DATA LINK (NETWORK) SIGNAL)
REMOTE	OPERATION AT TCS
FG	FUNCTION GENERATOR
+ △ −	DIFFERENCE
LS	LOW SELECTOR

TYPE	–				PAGE	13
SERVICE	SYMBOLS AND ABBREVIATIONS					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	13	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
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DWG. No. 310SC90-794



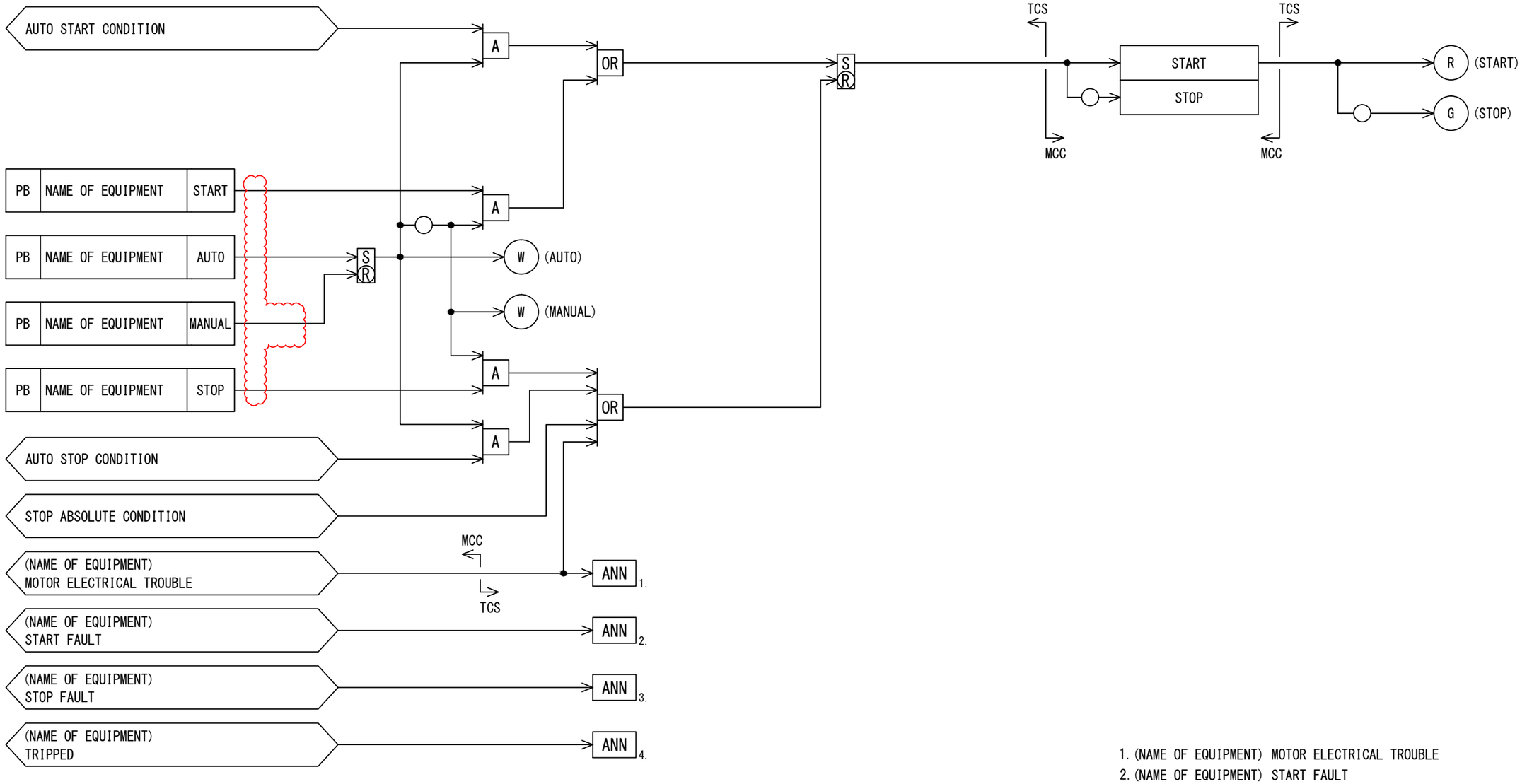
1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) START FAULT
3. (NAME OF EQUIPMENT) STOP FAULT
4. (NAME OF EQUIPMENT) TRIPPED

TYPE	–	PAGE	14
SERVICE	TYPICAL LOGIC FOR MOTOR		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	14	2016-07-18

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Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

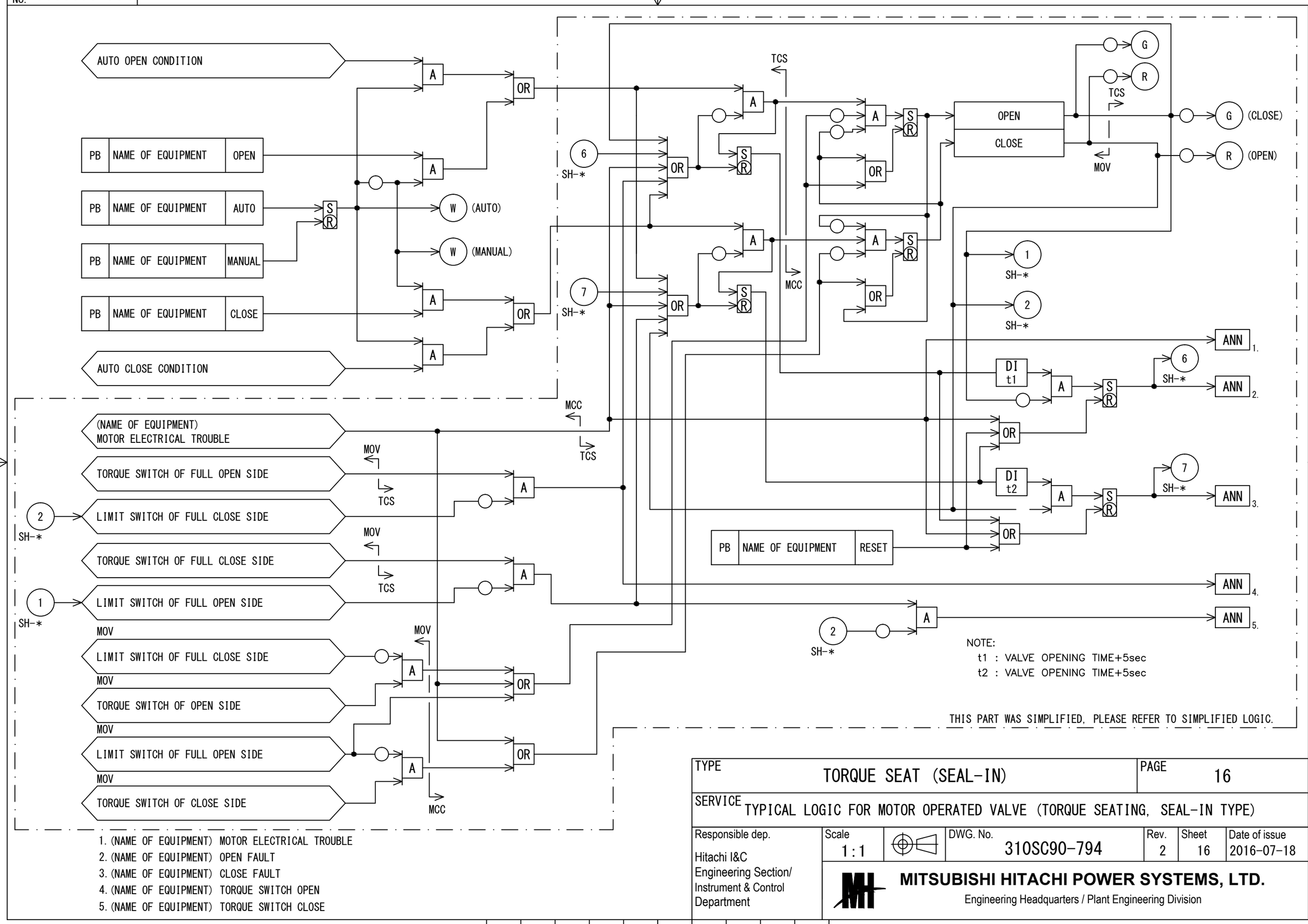


1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) START FAULT
3. (NAME OF EQUIPMENT) STOP FAULT
4. (NAME OF EQUIPMENT) TRIPPED

TYPE –					PAGE 15	
SERVICE SIMPLIFIED LOGIC FOR MOTOR						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 15	Date of issue 2016-07-18

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Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



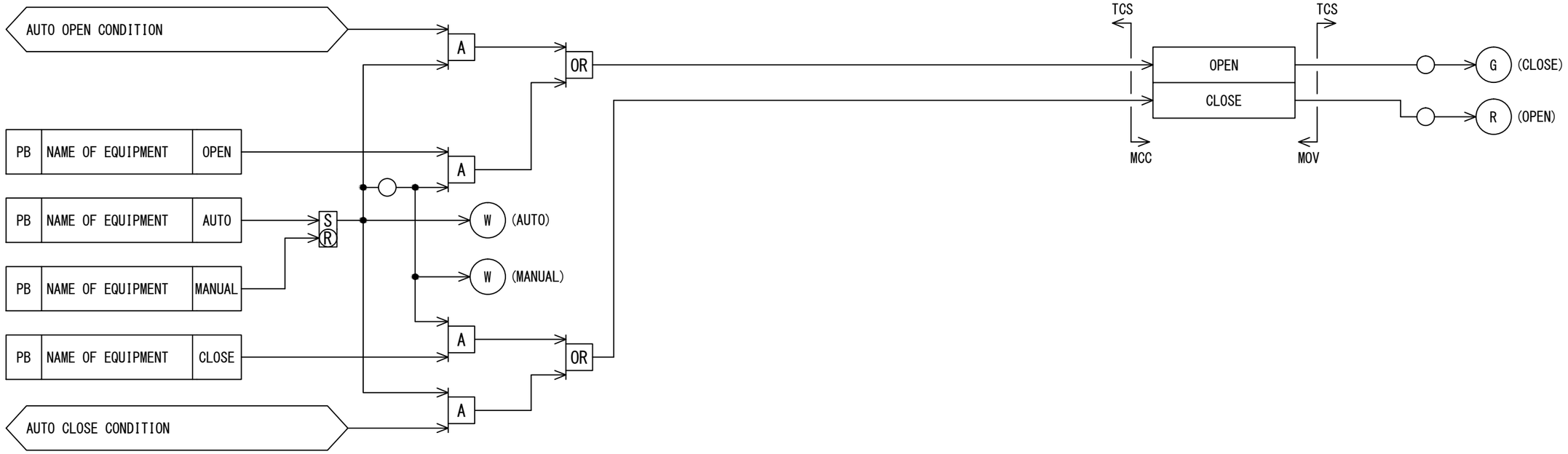
1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) OPEN FAULT
3. (NAME OF EQUIPMENT) CLOSE FAULT
4. (NAME OF EQUIPMENT) TORQUE SWITCH OPEN
5. (NAME OF EQUIPMENT) TORQUE SWITCH CLOSE

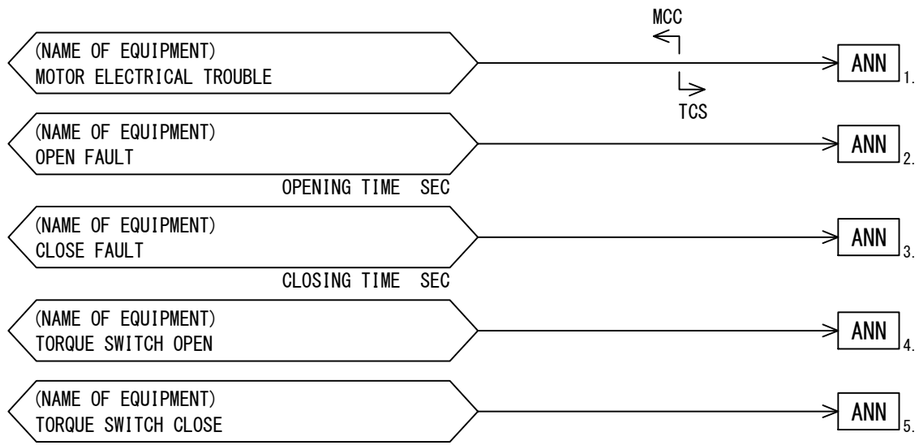
TYPE	TORQUE SEAT (SEAL-IN)	PAGE 16
SERVICE	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, SEAL-IN TYPE)	

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	16	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) OPEN FAULT
3. (NAME OF EQUIPMENT) CLOSE FAULT
4. (NAME OF EQUIPMENT) TORQUE SWITCH OPEN
5. (NAME OF EQUIPMENT) TORQUE SWITCH CLOSE

TYPE: TORQUE SEAT (SEAL-IN)					PAGE: 17	
SERVICE: SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, SEAL-IN TYPE)						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 17	Date of issue: 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

AUTO OPEN CONDITION

PB | NAME OF EQUIPMENT | OPEN

PB | NAME OF EQUIPMENT | AUTO

PB | NAME OF EQUIPMENT | MANUAL

PB | NAME OF EQUIPMENT | CLOSE

AUTO CLOSE CONDITION

A, OR, S (SET) / R (RESET), W (AUTO), W (MANUAL)

6 SH-* , 7 SH-*

TCS, MCC, MOV

G (CLOSE), R (OPEN)

OPEN / CLOSE

1 SH-*, 2 SH-*

DI t1, DI t2

ANN 1., ANN 2., ANN 3., ANN 4., ANN 5.

PB | NAME OF EQUIPMENT | RESET

(NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE

TORQUE SWITCH OF FULL OPEN SIDE

2 SH-* LIMIT SWITCH OF FULL CLOSE SIDE

TORQUE SWITCH OF FULL CLOSE SIDE

1 SH-* LIMIT SWITCH OF FULL OPEN SIDE

MOV LIMIT SWITCH OF FULL CLOSE SIDE

MOV TORQUE SWITCH OF OPEN SIDE

MOV LIMIT SWITCH OF FULL OPEN SIDE

MOV TORQUE SWITCH OF CLOSE SIDE

NOTE:
t1 : VALVE OPENING TIME+5sec
t2 : VALVE OPENING TIME+5sec

THIS PART WAS SIMPLIFIED, PLEASE REFER TO SIMPLIFIED LOGIC.

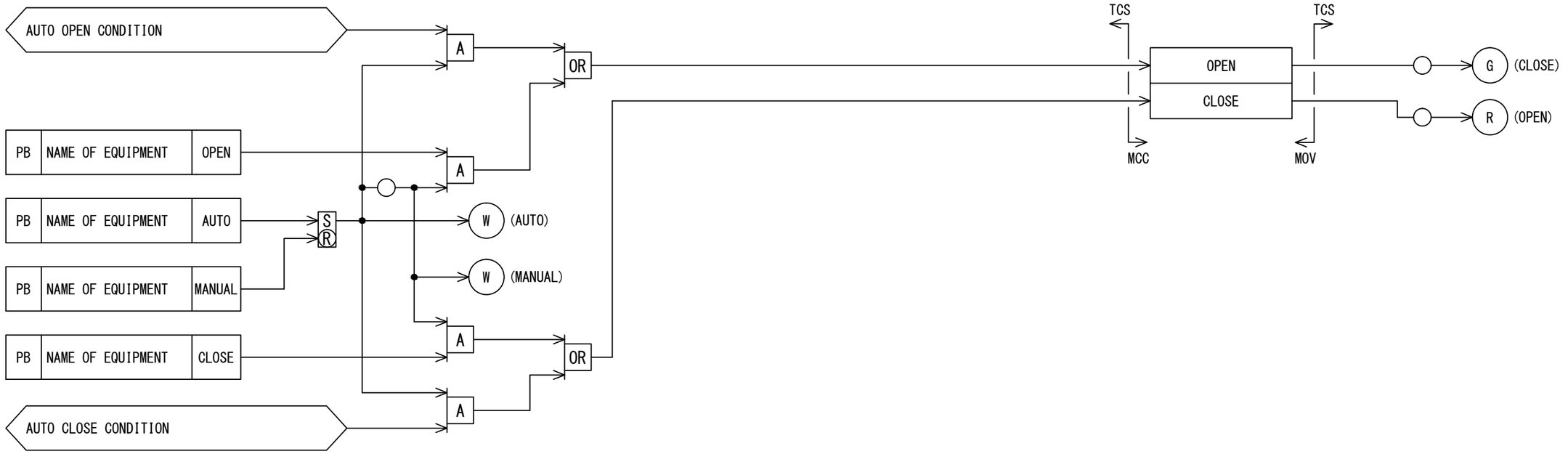
1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) OPEN FAULT
3. (NAME OF EQUIPMENT) CLOSE FAULT
4. (NAME OF EQUIPMENT) TORQUE SWITCH OPEN
5. (NAME OF EQUIPMENT) TORQUE SWITCH CLOSE

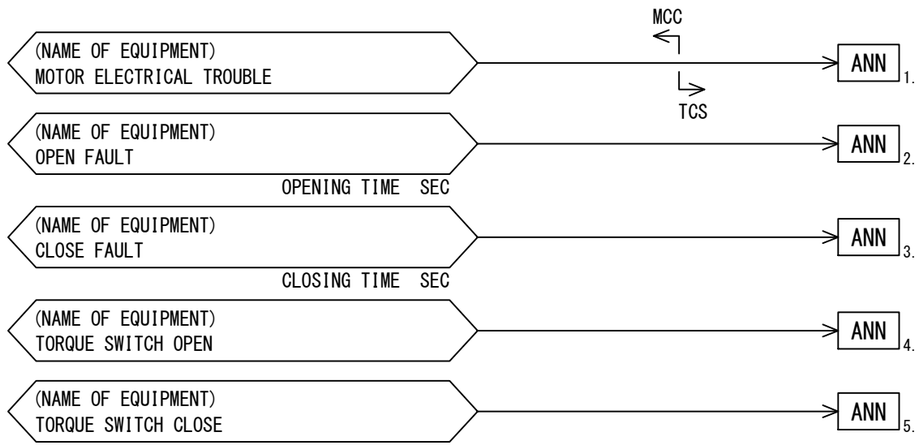
TYPE	POSITION SEAT (SEAL-IN)	PAGE 18
SERVICE	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (POSITION SEATING, SEAL-IN TYPE)	

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	18	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



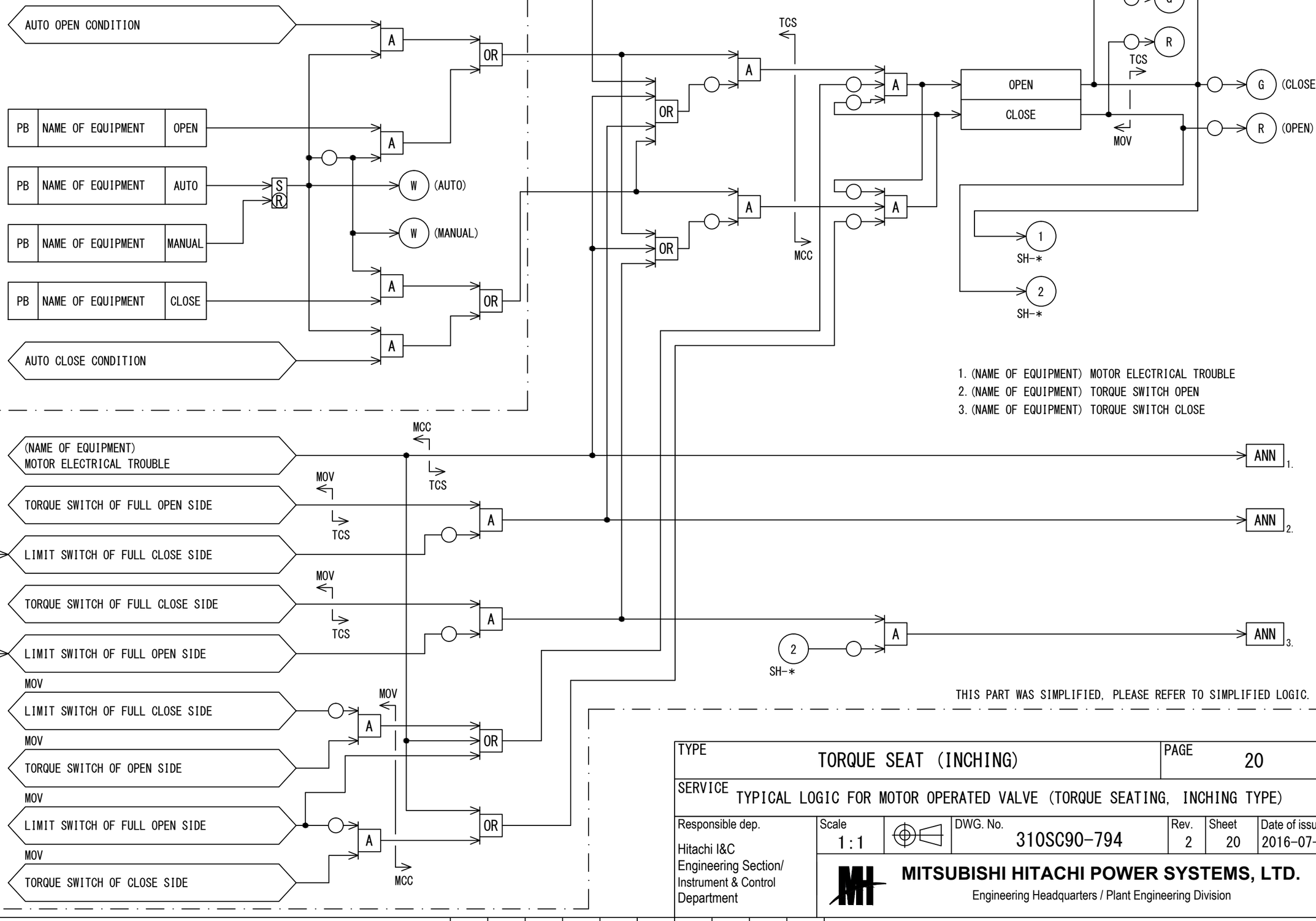


1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) OPEN FAULT
3. (NAME OF EQUIPMENT) CLOSE FAULT
4. (NAME OF EQUIPMENT) TORQUE SWITCH OPEN
5. (NAME OF EQUIPMENT) TORQUE SWITCH CLOSE

TYPE: POSITION SEAT (SEAL-IN)					PAGE: 19	
SERVICE: SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (POSITION SEATING, SEAL-IN TYPE)						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 19	Date of issue: 2016-07-18

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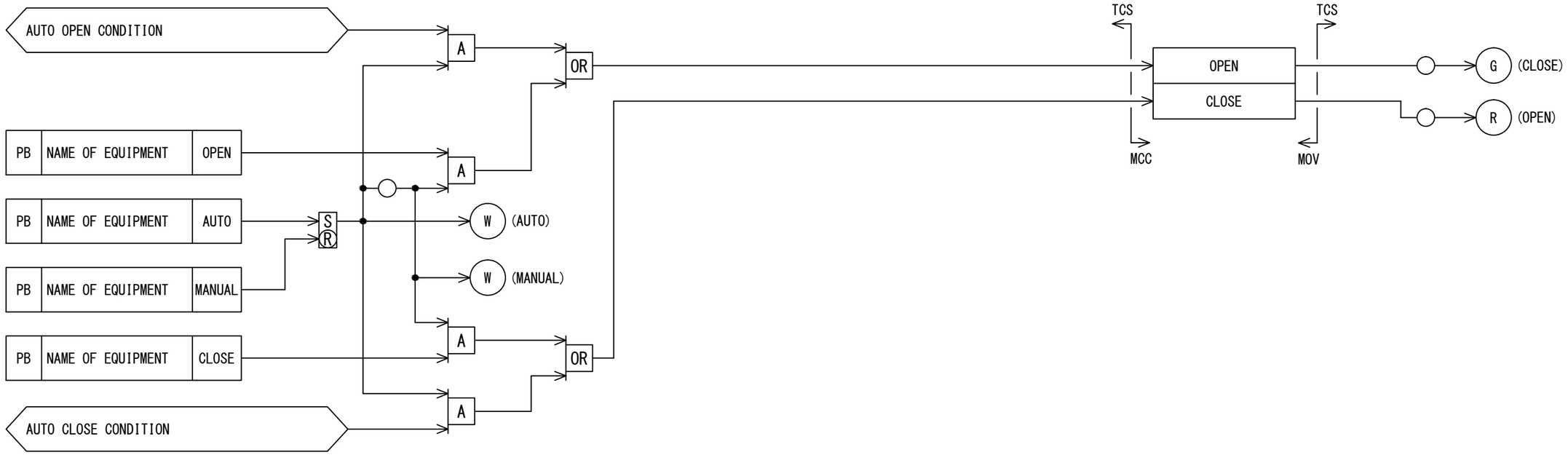
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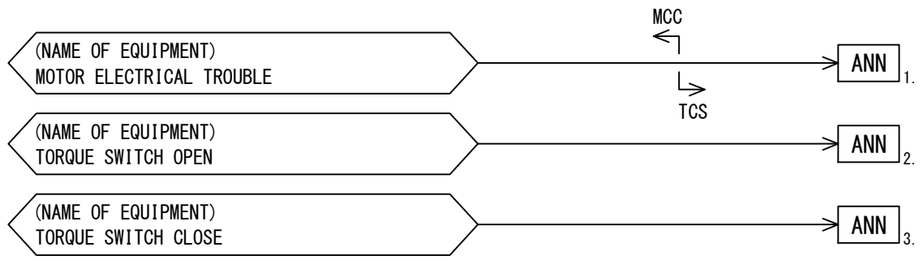


TYPE	TORQUE SEAT (INCHING)				PAGE	20
SERVICE	TYPICAL LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, INCHING TYPE)					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	20	2016-07-18

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DWG. No. 310SC90-794

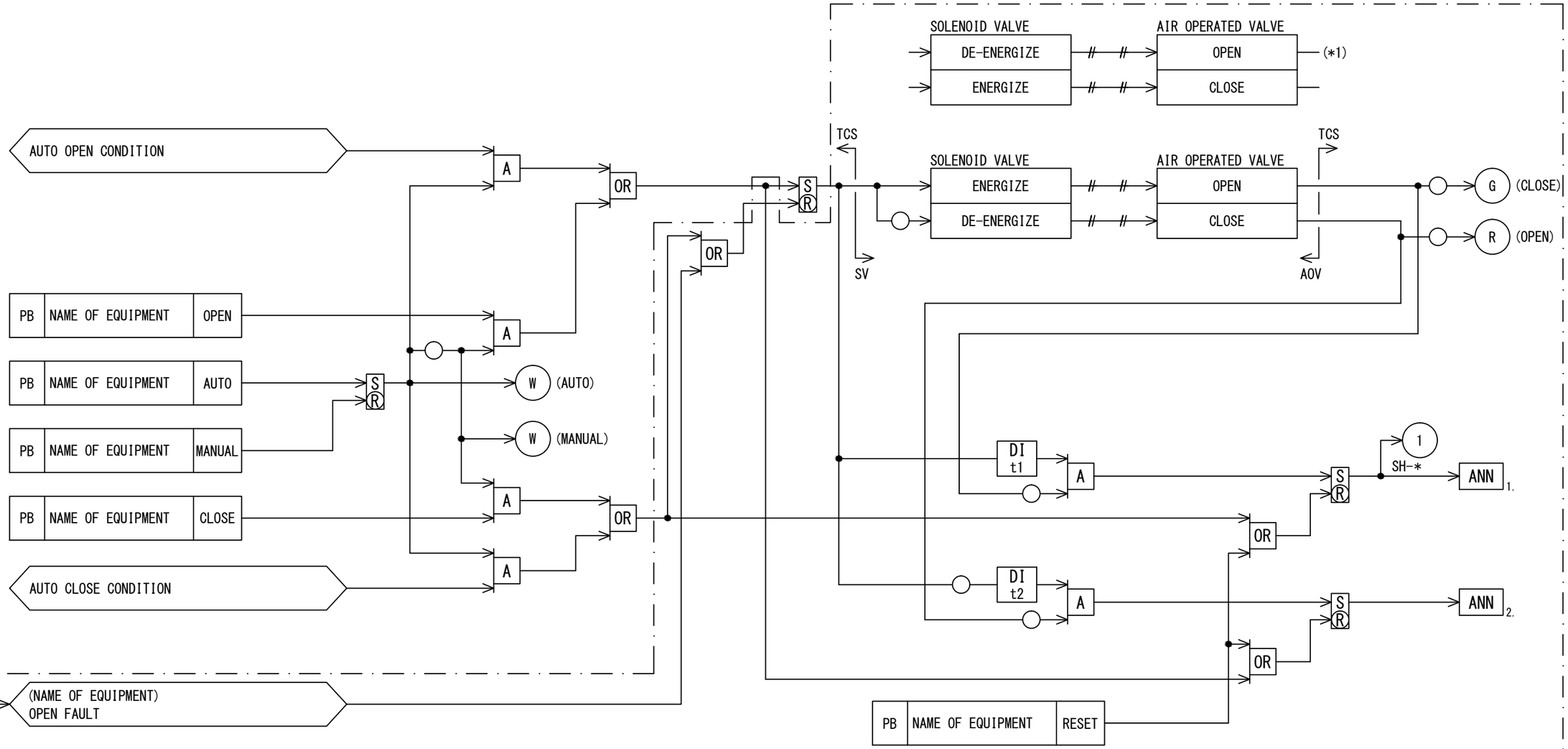




1. (NAME OF EQUIPMENT) MOTOR ELECTRICAL TROUBLE
2. (NAME OF EQUIPMENT) TORQUE SWITCH OPEN
3. (NAME OF EQUIPMENT) TORQUE SWITCH CLOSE

TYPE: TORQUE SEAT (INCHING)					PAGE: 21	
SERVICE: SIMPLIFIED LOGIC FOR MOTOR OPERATED VALVE (TORQUE SEATING, INCHING TYPE)						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 21	Date of issue: 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794



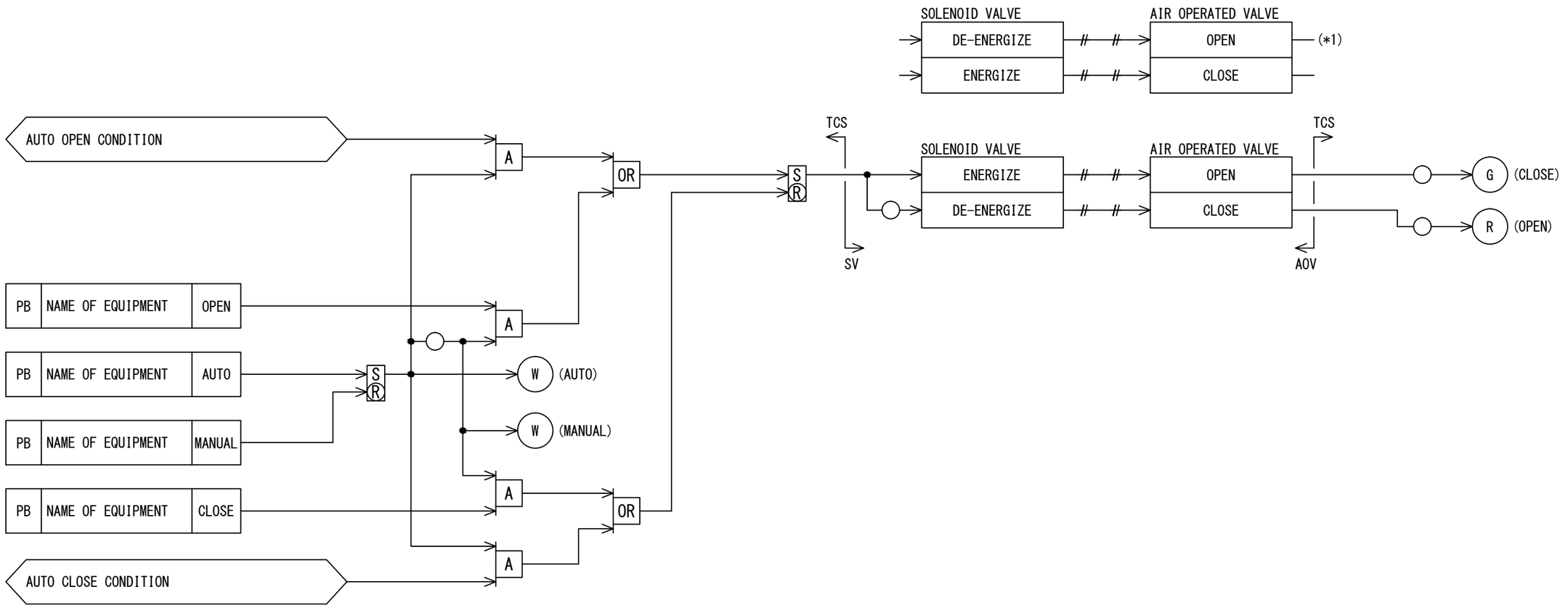
1. (NAME OF EQUIPMENT) OPEN FAULT
2. (NAME OF EQUIPMENT) CLOSE FAULT

THIS PART WAS SIMPLIFIED, PLEASE REFER TO SIMPLIFIED LOGIC.

NOTES:
(*1) TO REPLACE THE PART OF LOGIC IN CASE OF "ENERGIZE TO CLOSE" TYPE AIR OPERATED VALVE.
t1 : VALVE OPENING TIME+5sec
t2 : VALVE CLOSING TIME+5sec

TYPE –						PAGE 22	
SERVICE TYPICAL LOGIC FOR AIR OPERATED VALVE							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 22	Date of issue 2016-07-18	
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division							

DWG. No. 310SC90-794



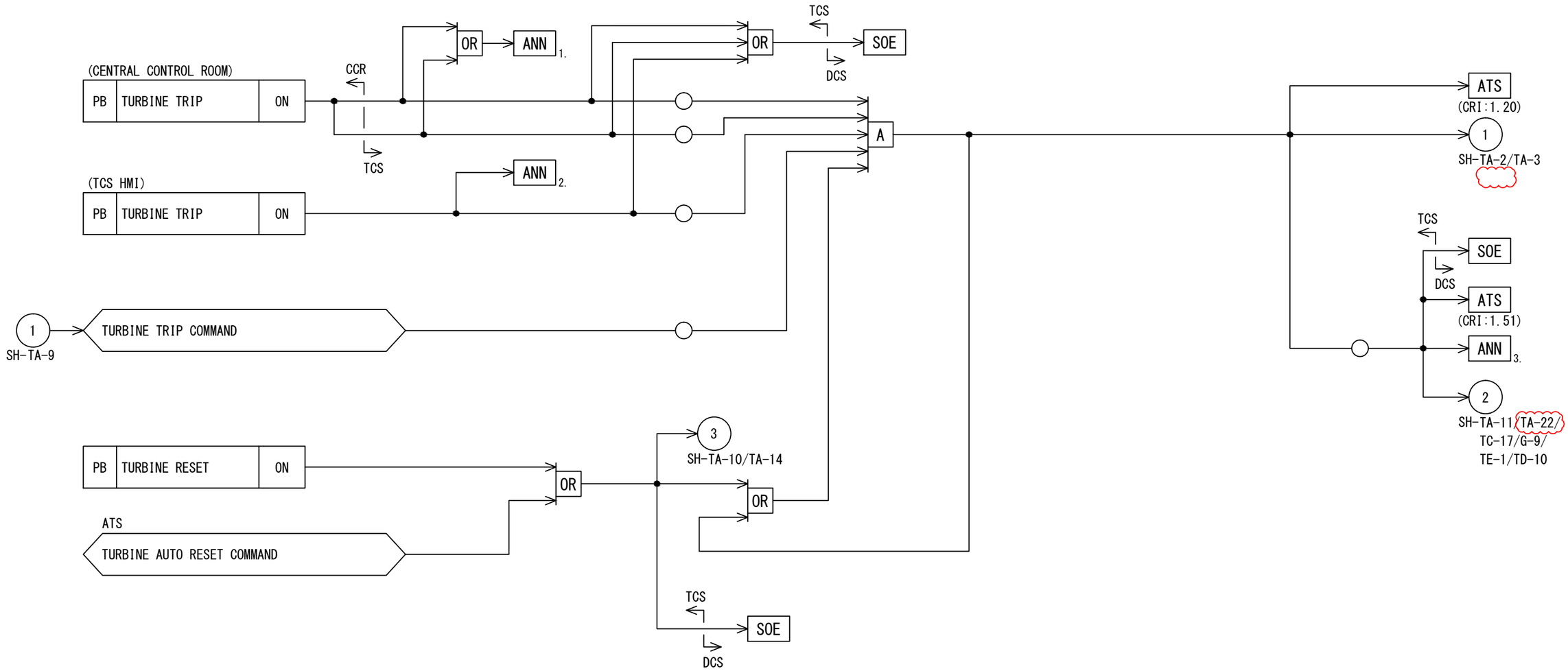
1. (NAME OF EQUIPMENT) OPEN FAULT
2. (NAME OF EQUIPMENT) CLOSE FAULT

NOTE:
(*1) TO REPLACE THE PART OF LOGIC IN CASE OF "ENERGIZE TO CLOSE" TYPE AIR OPERATED VALVE.

TYPE –					PAGE 23	
SERVICE SIMPLIFIED LOGIC FOR AIR OPERATED VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 23	Date of issue 2016-07-18

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Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

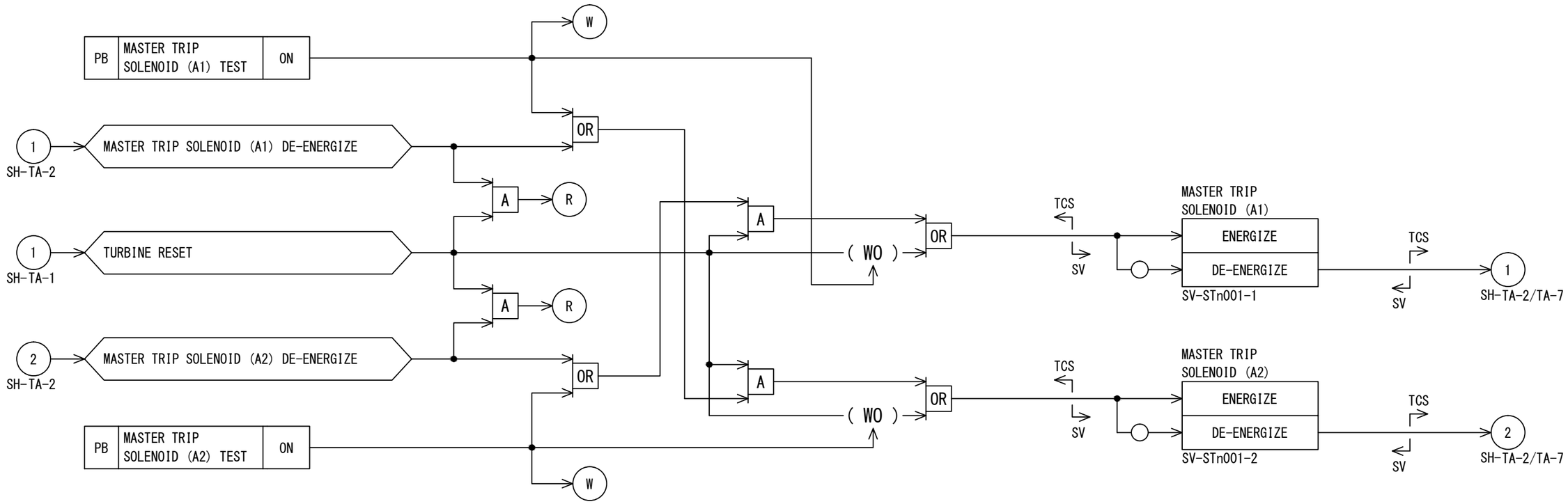


1. TURBINE MANUAL TRIP (CCR)
2. TURBINE MANUAL TRIP (TCS HMI)
3. TURBINE TRIP

TYPE	–				PAGE	TA-1	
SERVICE	TURBINE MASTER TRIP / MASTER RESET						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	24	2016-07-18	

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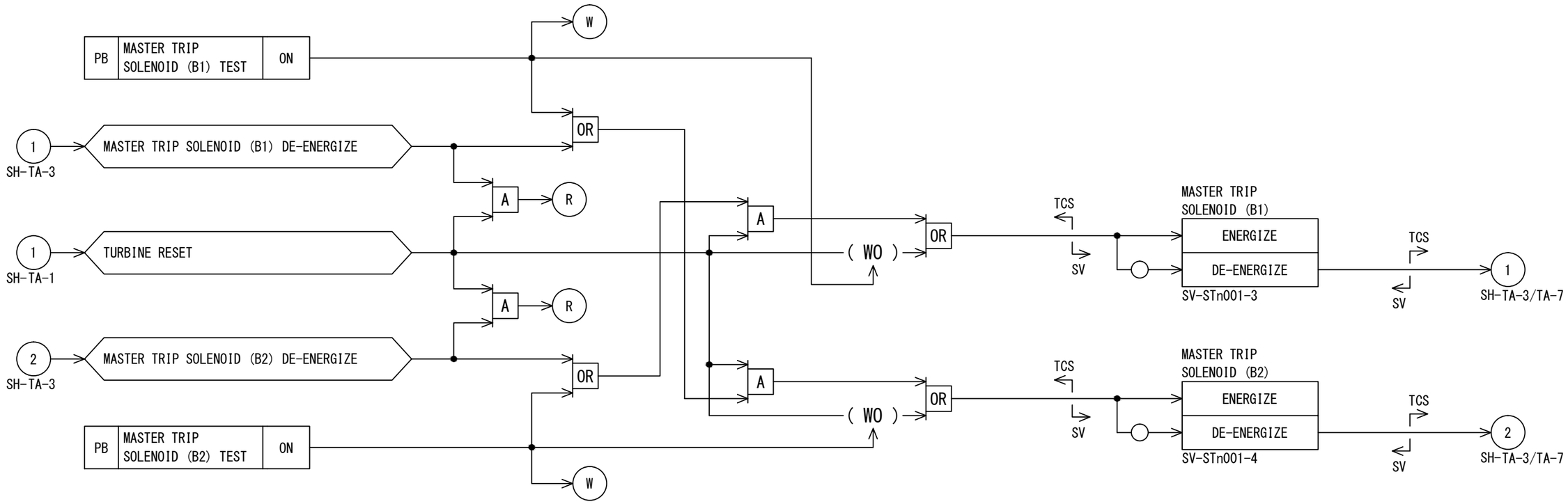
DWG. No. 310SC90-794



TYPE –					PAGE TA-2	
SERVICE MASTER TRIP SOLENOID (A1),(A2) TEST						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 25	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



TYPE –					PAGE TA-3	
SERVICE MASTER TRIP SOLENOID (B1), (B2) TEST						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 26	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794

ATS
52G OPEN COMMAND
FROM ATS SHUTDOWN SEQUENCE

2 SH-TA-31 MSV(A) FULL CLOSE
2 SH-TA-32 MSV(B) FULL CLOSE
7 SH-TA-37 RSV(A) FULL CLOSE
4 SH-TA-37 ICV(A) FULL CLOSE
7 SH-TA-38 RSV(B) FULL CLOSE
4 SH-TA-38 ICV(B) FULL CLOSE
2 SH-TA-33 CV(A) FULL CLOSE
2 SH-TA-34 CV(B) FULL CLOSE
TCS MAJOR TROUBLE
11 SH-G-9 (310SD03-749) 52G CLOSED

A
OR
A
A
A
DI 3s
A
OR
OR
OR
OR
OR

TCS
TURBINE TRIP
DCS
GPP
1 SH-G-9 (310SD03-749)
2 SH-G-9 (310SD03-749)
TCS

ATS (CRI:1.37)

TCS
FOR BOILER REHEATER PROTECTION
FOR BOILER REHEATER PROTECTION
FOR BOILER REHEATER PROTECTION
FOR BOILER REHEATER PROTECTION
HP-CV FULL CLOSE (BOTH (A) & (B))
DCS

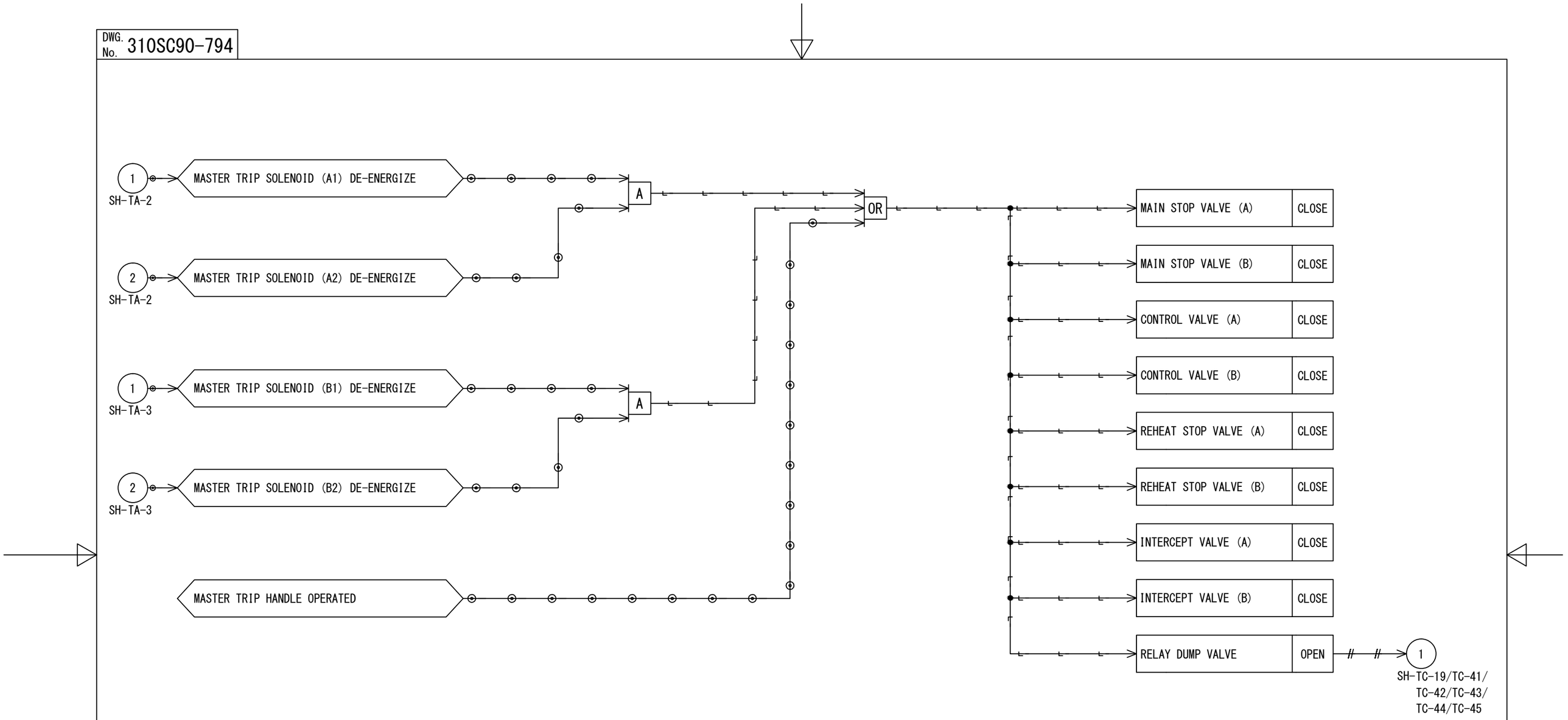
TCS
FOR HOUSE LOAD OPERATION
DCS

3 SH-TD-10

TYPE –					PAGE TA-6	
SERVICE MAIN STOP VALVE FULL CLOSE (FOR BOILER PROTECTION & ELECTRICAL TRIP)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 27	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



TYPE –						PAGE TA-7	
SERVICE MECHANICAL LOGIC							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 28	Date of issue 2016-07-18	
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division							

DWG. No. 310SC90-794

1 SH-TA-10 → CONDENSER VACUUM VERY LOW TRIP
1 SH-TA-11 → MAIN STEAM TEMP. VERY LOW TRIP
1 SH-TA-12 → THRUST BEARING FAILURE (FRONT) ABNORMAL TRIP
2 SH-TA-12 → THRUST BEARING FAILURE (REAR) ABNORMAL TRIP
1 SH-TA-13 → CONTROL OIL PRESS. VERY LOW TRIP
2 SH-TA-13 → BEARING OIL PRESS. VERY LOW TRIP
1 SH-TA-14 → LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP
4 SH-TA-14 → TURBINE TRIP OIL PRESS. VERY LOW TRIP
1 SH-TA-15 → TURBINE SPEED (TURBINE BACK-UP OVERSPEED) TRIP
2 SH-TA-15 → TURBINE SPEED (TURBINE OVERSPEED) TRIP
1 SH-TA-17 → NO.1 TURBINE SHAFT VIBRATION VERY HIGH TRIP
2 SH-TA-17 → NO.2 TURBINE SHAFT VIBRATION VERY HIGH TRIP
3 SH-TA-17 → NO.3 GEN. SHAFT VIBRATION VERY HIGH TRIP
4 SH-TA-17 → NO.4 GEN. SHAFT VIBRATION VERY HIGH TRIP
1 SH-TA-20 → HP TURBINE EXHAUST STEAM TEMP. VERY HIGH TRIP
1 SH-TA-21 → HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. VERY HIGH TRIP
1 SH-TA-22 → HP TURBINE EXHAUST STEAM PRESS. VERY HIGH TRIP

→ OR

DCS
BOILER MFT 1
BOILER MFT 2
TCS
TCS MAJOR TROUBLE (*2)
TCS → ANN 1.
TCS → SOE → DCS

9 SH-G-9 (310SD03-749) → GEN./UNIT TRANS. PROTECTION A & GEN. LOCKOUT RELAY A
10 SH-G-9 (310SD03-749) → GEN./UNIT TRANS. PROTECTION B & GEN. LOCKOUT RELAY B

A, A → OR → ANN 2. (*1)

OR → 1 SH-TA-1

ATS
TURBINE SHUTDOWN COMMAND

OR → ATS (CRI:1.41)

1. TCS MAJOR TROUBLE
2. BOILER MFT 1 AND 2 SYN.

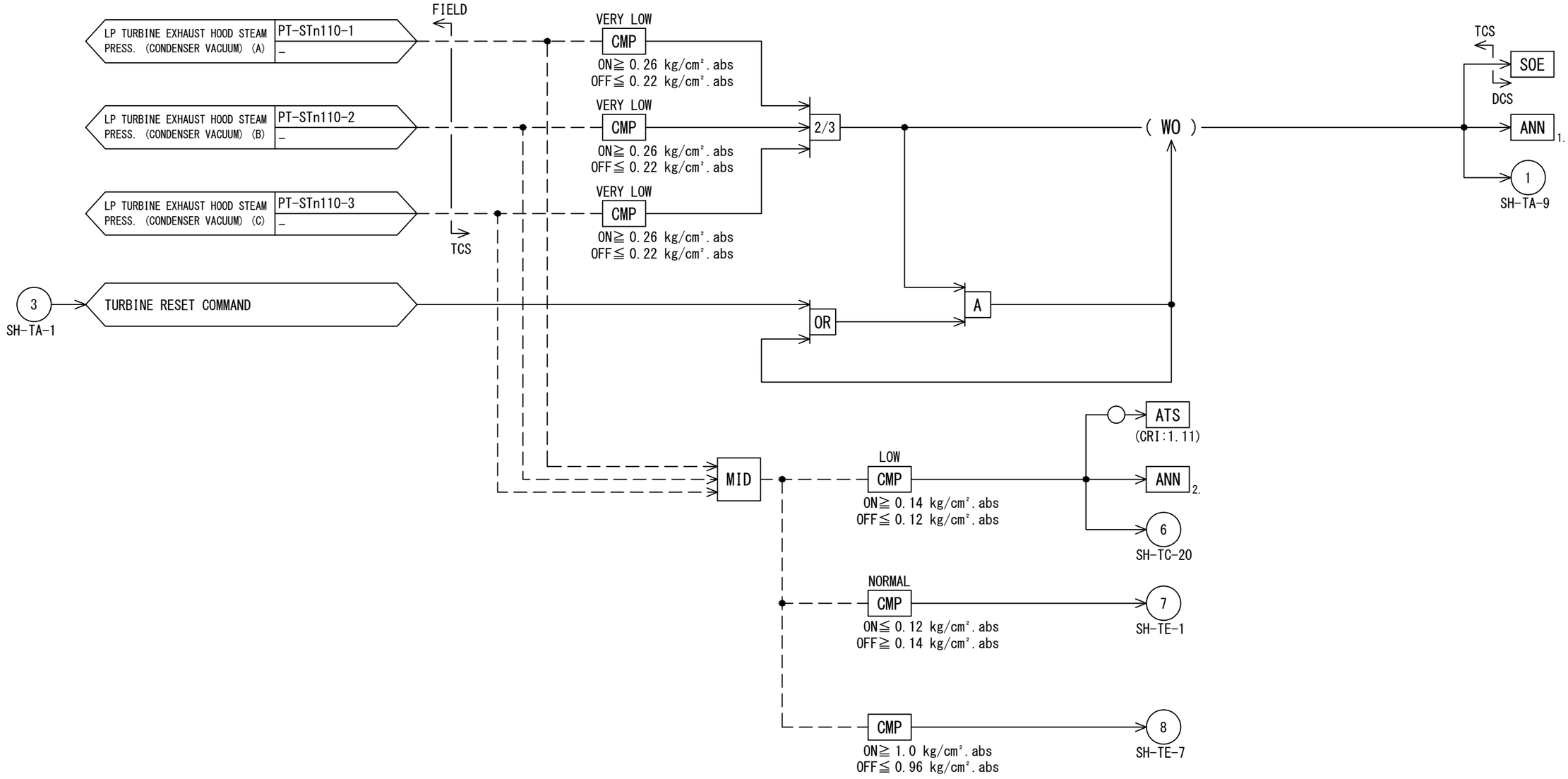
NOTES:
(*1) THIS ANN_2 LOGIC IS INDICATED BECAUSE THE NAME OF ANN WAS CHANGED FROM "TCS MINOR TROUBLE" ACCORDING TO FHI COMMENT AT WITNESS TEST OF TCS (2014-10-30).
(*2) THE DETAILS OF TCS MAJOR TROUBLE ARE DESCRIBED IN FIG.4 OF THE DOCUMENT "BASIC CONFIGURATION OF TURBINE CONTROL SYSTEM" (TC-13-100).

TYPE	–	PAGE	TA-9
SERVICE	TRIP LOGIC (1)(TOTAL)		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	29	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. CONDENSER VACUUM VERY LOW TRIP
2. CONDENSER VACUUM LOW

TYPE –					PAGE TA-10	
SERVICE TRIP LOGIC (2) (CONDENSER VACUUM LOW TRIP)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 30	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

2 SH-TA-1

TURBINE TRIP

MID

LOW CMP ON≦ 290° C OFF≧ 295° C

A

OR

ANN 2.

LOW (*2) CMP ON≦ VARIABLE OFF≧ VARIABLE

A

MID

FG (ALARM)

3 SH-TA-21

FIELD

MSV INLET STEAM TEMP. (A) TE-MSn902-1 –

VERY LOW (*1) CMP ON≦ VARIABLE OFF≧ VARIABLE

HP TURBINE FIRST STAGE STEAM PRESS. (A) PT-STn001-1 –

FG (TRIP)

TCS

4 SH-TA-21

A

GENERATOR LOAD (*3)

CMP ON≧ 20% OFF≦ 17%

S R

DT 3s

7 SH-TA-21

FIELD

MSV INLET STEAM TEMP. (B) TE-MSn902-2 –

VERY LOW (*1) CMP ON≦ VARIABLE OFF≧ VARIABLE

A

2/3

TCS

SOE

DCS

ANN 1.

HP TURBINE FIRST STAGE STEAM PRESS. (B) PT-STn001-2 –

FG (TRIP)

1 SH-TA-9

5 SH-TA-21

MSV INLET STEAM TEMP. (C) TE-MSn902-3 –

VERY LOW (*1) CMP ON≦ VARIABLE OFF≧ VARIABLE

A

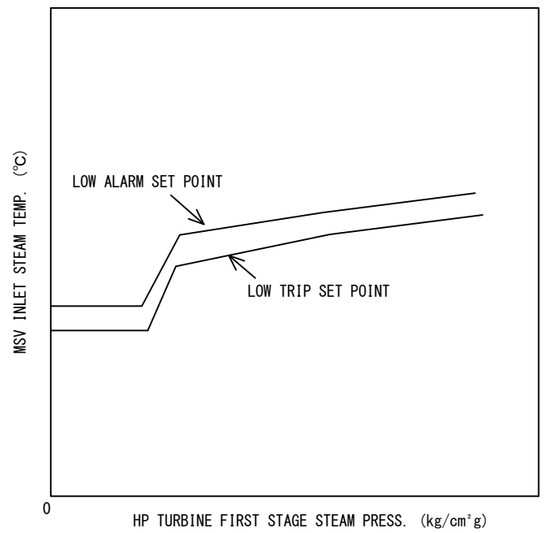
HP TURBINE FIRST STAGE STEAM PRESS. (C) PT-STn001-3 –

FG (TRIP)

TCS

6 SH-TA-21

VARIABLE SET POINT



HP TURBINE FIRST STAGE STEAM PRESS (kg/cm²g)	ANN	TRIP
0	410.3	395.3
33.4	410.3	395.3
42.8	432.0	417.0
55.2	434.6	419.6
61.7	436.6	421.6
72.2	438.4	423.4
88.0	440.5	425.5
104.2	442.2	427.2
150.0	447.0	432.0

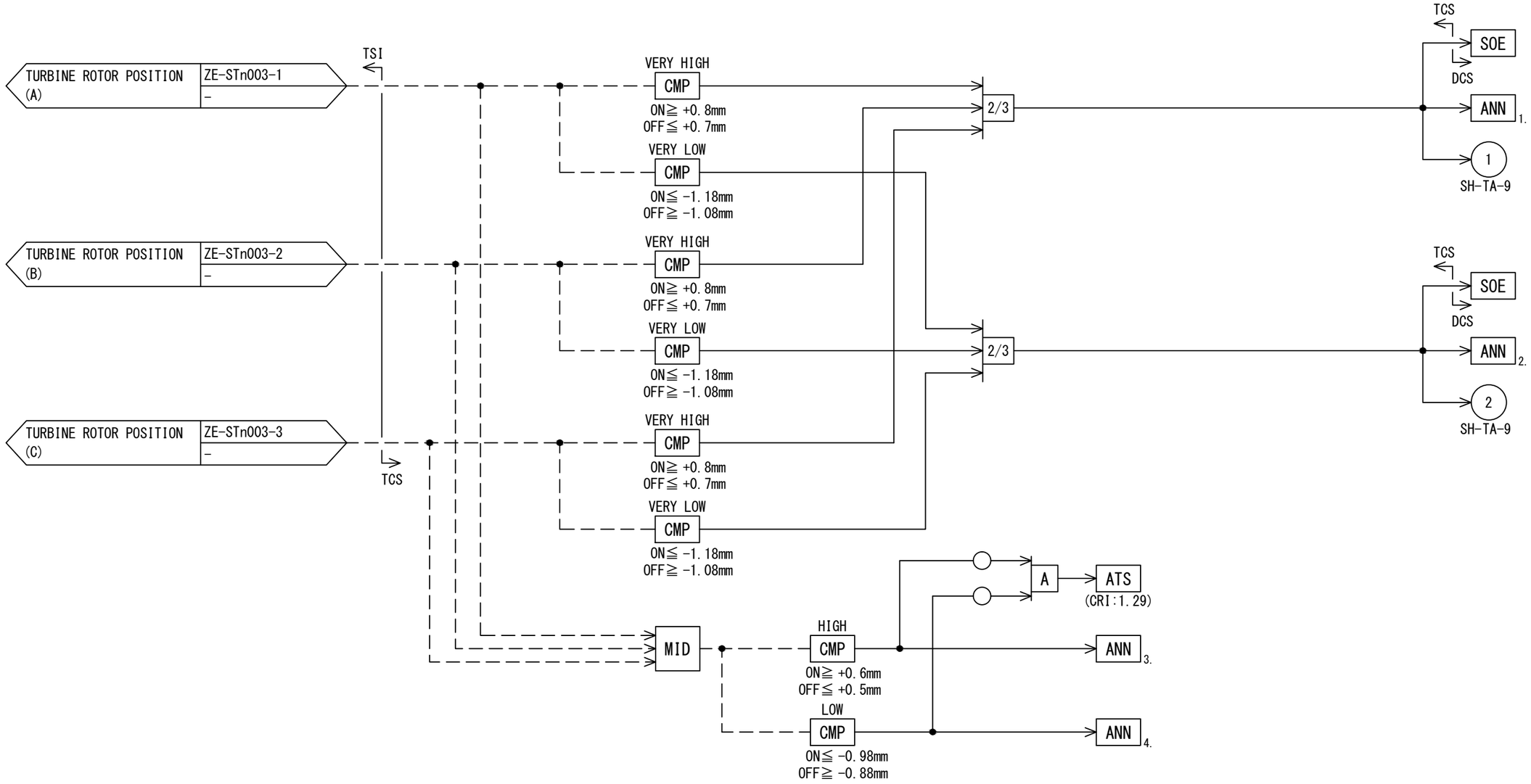
1. MAIN STEAM TEMP. VERY LOW
2. MAIN STEAM TEMP. LOW

NOTES:
(*1) OUTPUT IS TRUE, IF MSV INLET STEAM TEMP. IS VERY LOW TRIP LEVEL
(*2) OUTPUT IS TRUE, IF MSV INLET STEAM TEMP. IS LOW ALARM LEVEL
(*3) FOR THE LOAD SIGNAL, PLEASE REFER TO ONE LINE DIAGRAM GENERATOR MAIN CIRCUIT (310SD03-746)

TYPE –					PAGE TA-11	
SERVICE TRIP LOGIC (3) (MAIN STEAM TEMP. LOW TRIP)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 31	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

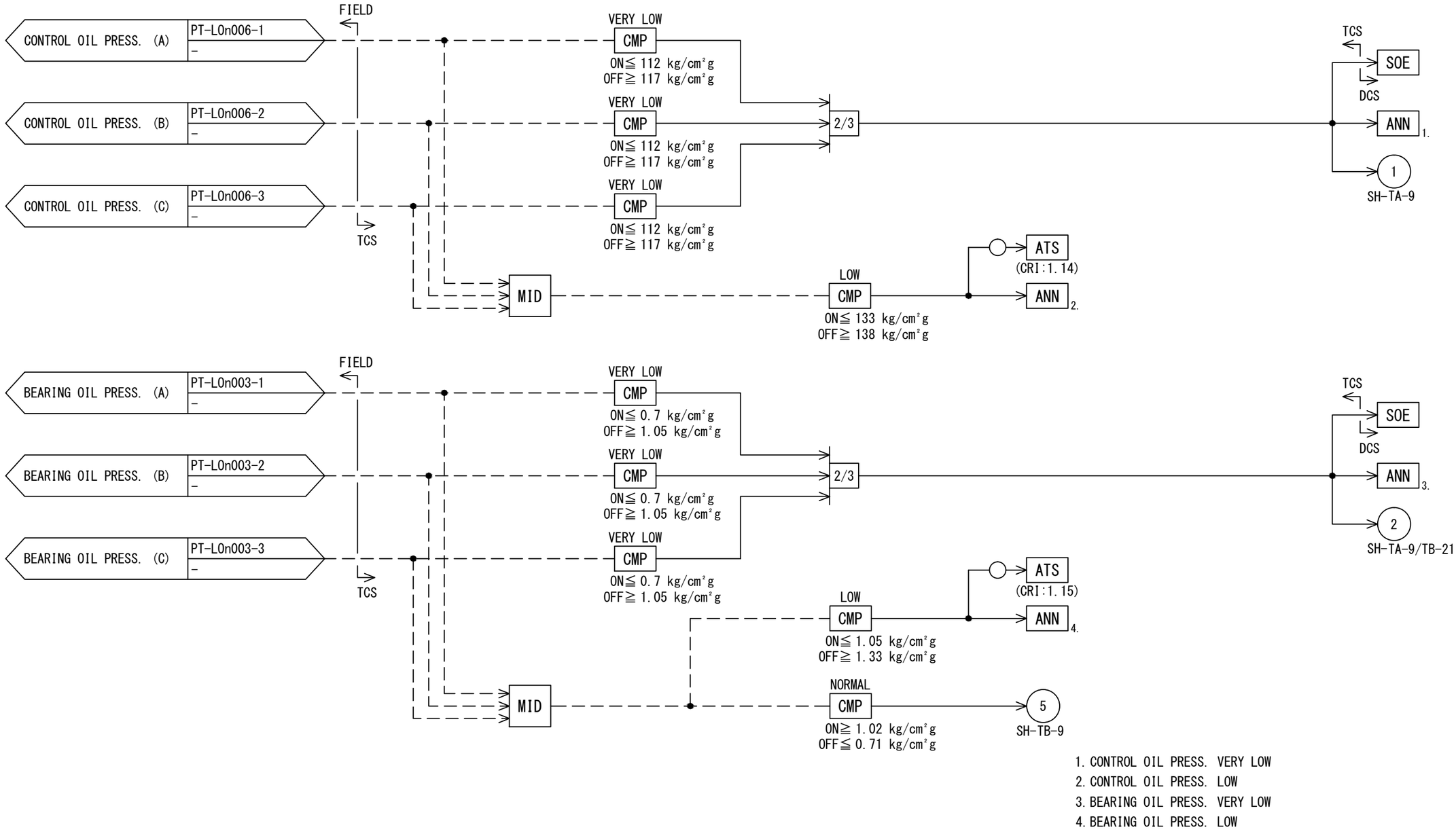


1. THRUST BEARING FAILURE (FRONT) ABNORMAL TRIP
2. THRUST BEARING FAILURE (REAR) ABNORMAL TRIP
3. THRUST BEARING FAILURE (FRONT) ABNORMAL
4. THRUST BEARING FAILURE (REAR) ABNORMAL

TYPE –						PAGE TA-12	
SERVICE TRIP LOGIC (4)(THRUST BEARING FAILURE TRIP)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 32	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



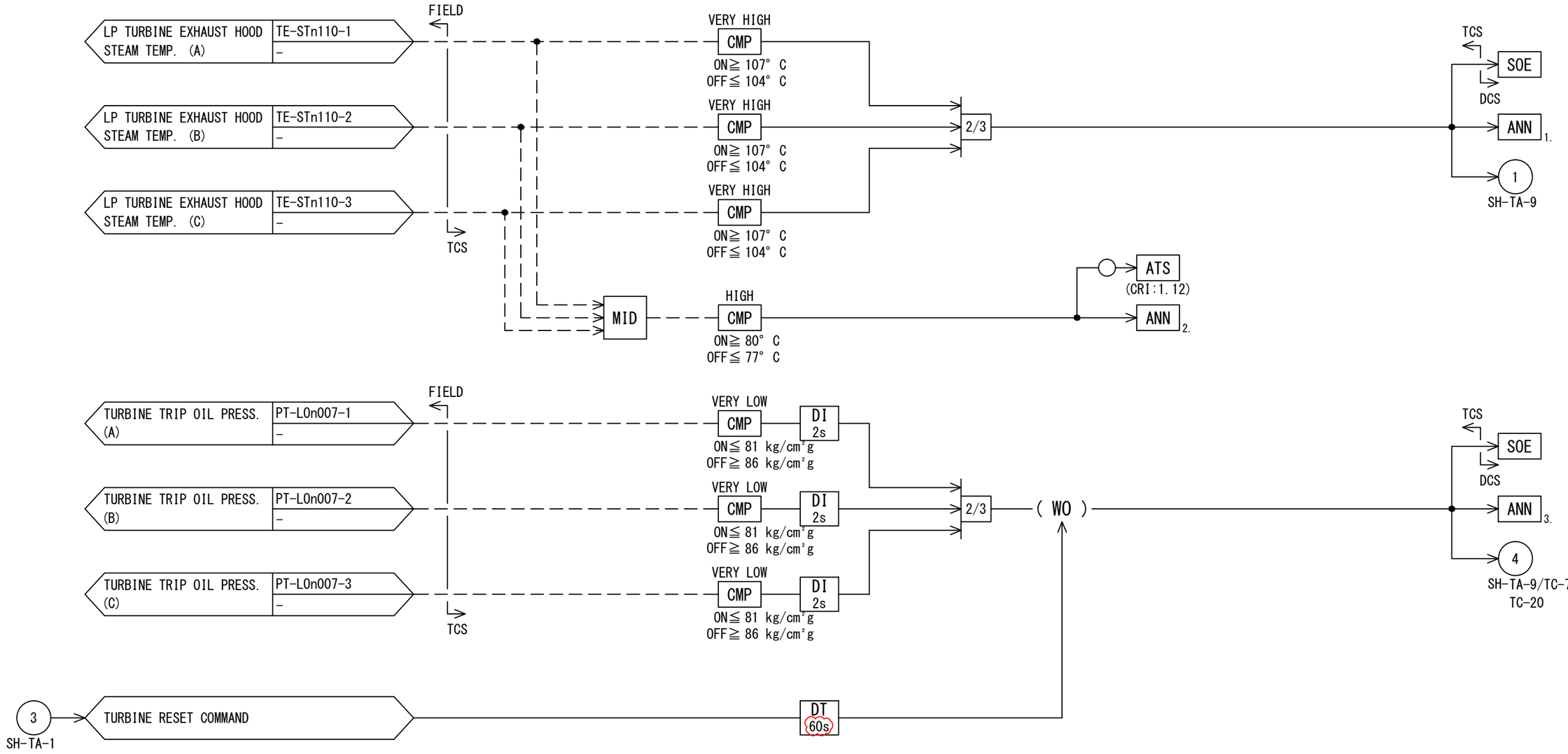
1. CONTROL OIL PRESS. VERY LOW
2. CONTROL OIL PRESS. LOW
3. BEARING OIL PRESS. VERY LOW
4. BEARING OIL PRESS. LOW

TYPE	–	PAGE	TA-13
SERVICE	TRIP LOGIC (5)(CONTROL OIL PRESS. VERY LOW TRIP/ BEARING OIL PRESS. VERY LOW TRIP)		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	33	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

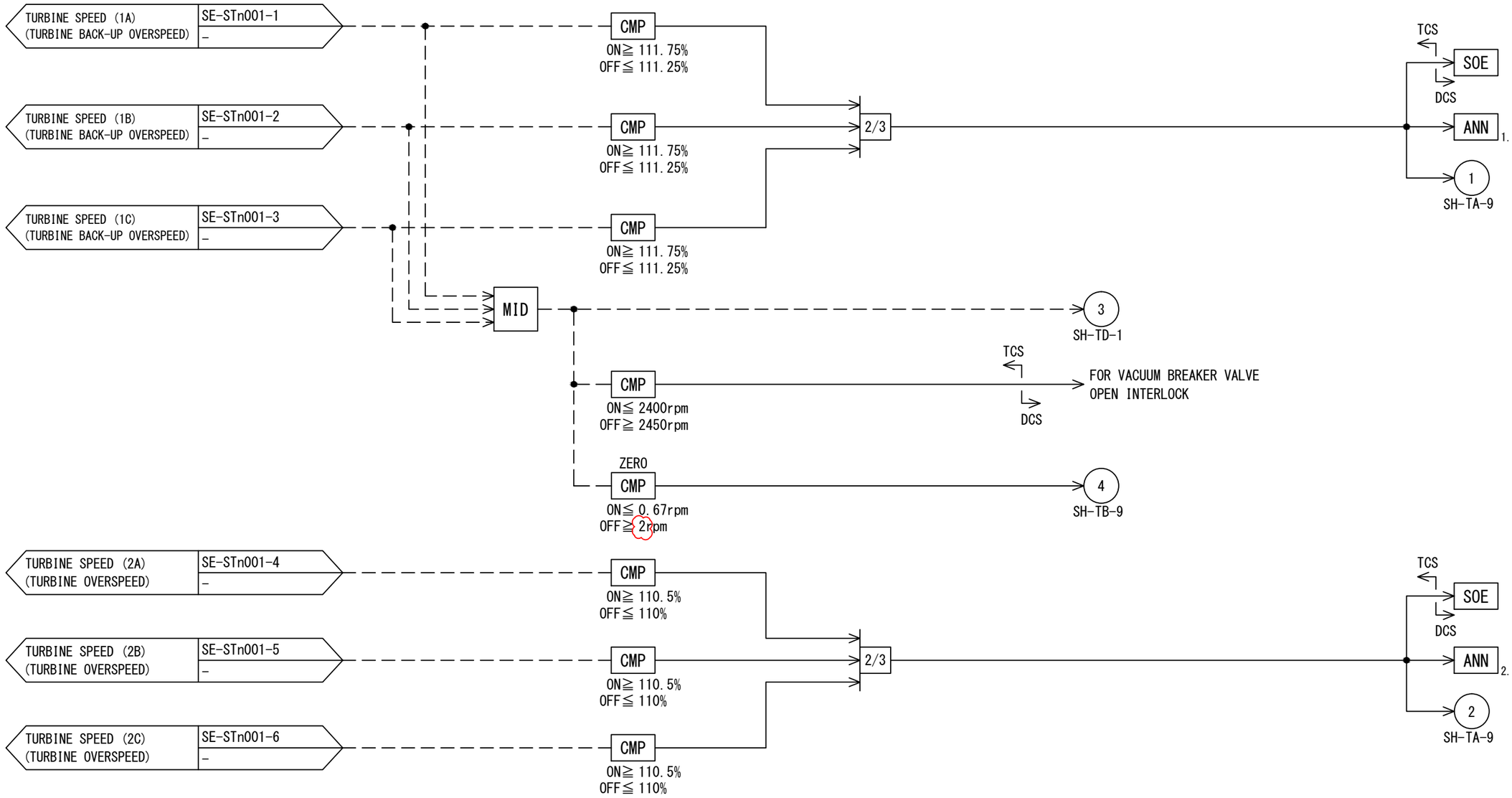


1. LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH
2. LP TURBINE EXHAUST HOOD STEAM TEMP. HIGH
3. TURBINE TRIP OIL PRESS. VERY LOW

TYPE	–				PAGE	TA-14
SERVICE	TRIP LOGIC (6) (LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP/TURBINE TRIP OIL PRESS. LOW TRIP)					
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 34	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. TURBINE SPEED (TURBINE BACK-UP OVERSPEED)
2. TURBINE SPEED (TURBINE OVERSPEED)

TYPE –					PAGE TA-15	
SERVICE TRIP LOGIC (7) (OVERSPEED TRIP/BACK-UP OVERSPEED TRIP)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 35	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

1 SH-TA-18 — NO.1 TURBINE SHAFT VIBRATION (X) HIGH
2 SH-TA-18 — NO.1 TURBINE SHAFT VIBRATION (Y) HIGH

(*2)
3 SH-TA-18 — NO.2 TURBINE SHAFT VIBRATION (X) HIGH
4 SH-TA-18 — NO.2 TURBINE SHAFT VIBRATION (Y) HIGH
5 SH-TA-18 — NO.3 GENERATOR SHAFT VIBRATION (X) HIGH
6 SH-TA-18 — NO.3 GENERATOR SHAFT VIBRATION (Y) HIGH
7 SH-TA-18 — NO.4 GENERATOR SHAFT VIBRATION (X) HIGH
8 SH-TA-18 — NO.4 GENERATOR SHAFT VIBRATION (Y) HIGH

(*1)
NO.1 TURBINE SHAFT VIBRATION (X) VERY HIGH — VE-STn001-1 — ON≧0.200mm
NO.1 TURBINE SHAFT VIBRATION (Y) VERY HIGH — VE-STn001-2 — ON≧0.200mm
NO.2 TURBINE SHAFT VIBRATION (X) VERY HIGH — VE-STn002-1 — ON≧0.200mm
NO.2 TURBINE SHAFT VIBRATION (Y) VERY HIGH — VE-STn002-2 — ON≧0.200mm
NO.3 GENERATOR SHAFT VIBRATION (X) VERY HIGH — VE-GEn001-1 — ON≧0.200mm
NO.3 GENERATOR SHAFT VIBRATION (Y) VERY HIGH — VE-GEn001-2 — ON≧0.200mm
NO.4 GENERATOR SHAFT VIBRATION (X) VERY HIGH — VE-GEn002-1 — ON≧0.200mm
NO.4 GENERATOR SHAFT VIBRATION (Y) VERY HIGH — VE-GEn002-2 — ON≧0.200mm

OR → A (TSI) → DI 3s → TCS SOE / DCS / ANN → 1 SH-TA-9

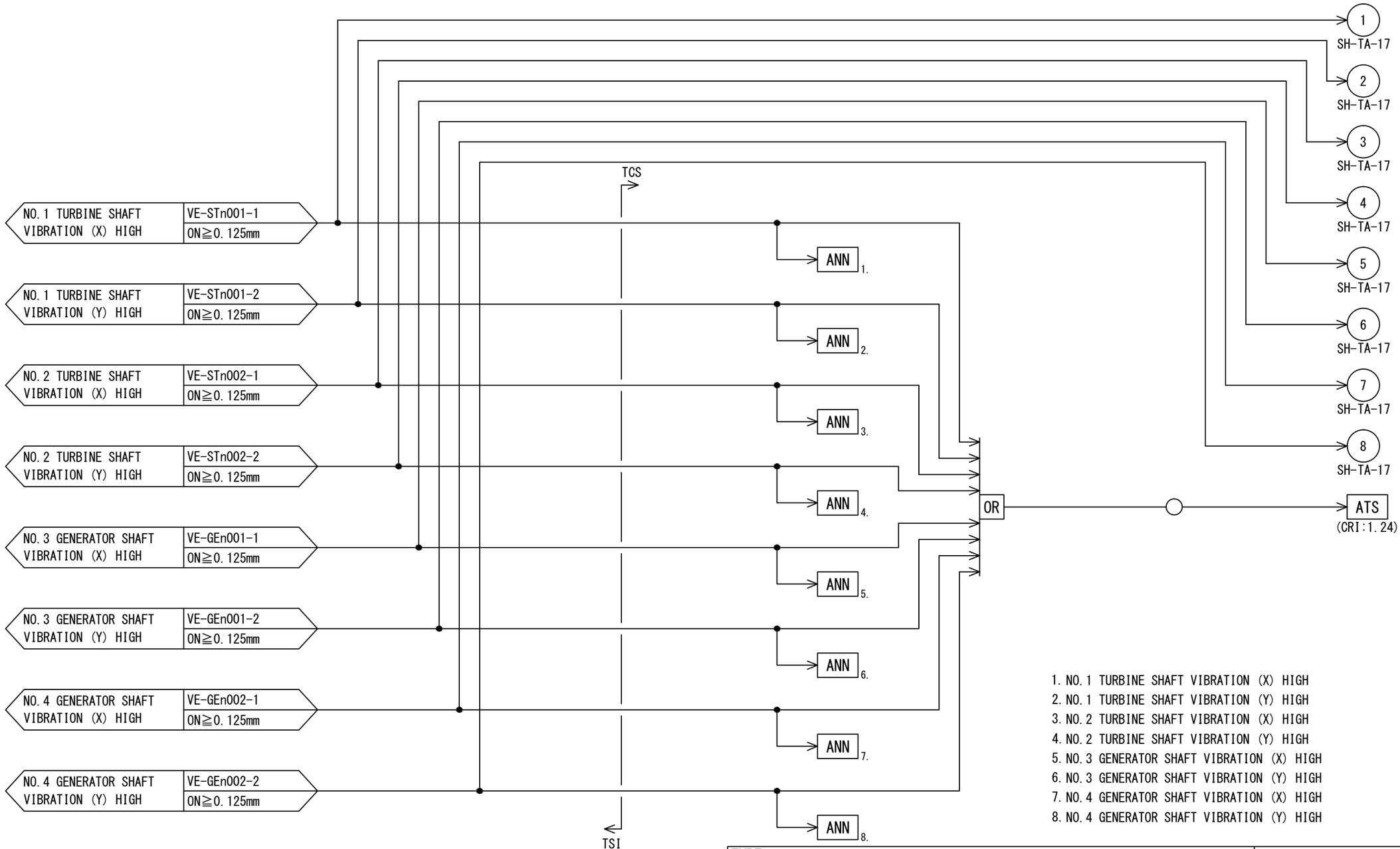
1. NO.1 TURBINE SHAFT VIBRATION VERY HIGH
2. NO.2 TURBINE SHAFT VIBRATION VERY HIGH
3. NO.3 GENERATOR SHAFT VIBRATION VERY HIGH
4. NO.4 GENERATOR SHAFT VIBRATION VERY HIGH

NOTES:
"NO.1 TURBINE SHAFT VIBRATION VERY HIGH" IS ACTUATED AS THE FOLLOWING 2 CONDITIONS (*1,*2) BOTH SATISFIED;
(*1) NO.1 TURBINE SHAFT VIBRATION X OR Y VERY HIGH IS SATISFIED.
(*2) ANY OTHER SHAFT VIBRATION HIGH EXCEPT FOR NO.1 TURBINE SHAFT IS SATISFIED.
TRIP SIGNAL OF NO.2 TURBINE AND NO.3,4 GENERATOR BEARING VIBRATION ARE SIMILAR TO THAT OF NO.1 TURB. BRG. VIBRATION.

TYPE	–					PAGE	TA-17
SERVICE	TRIP LOGIC (8)(BEARING VIBRATION VERY HIGH TRIP (1))						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	36	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

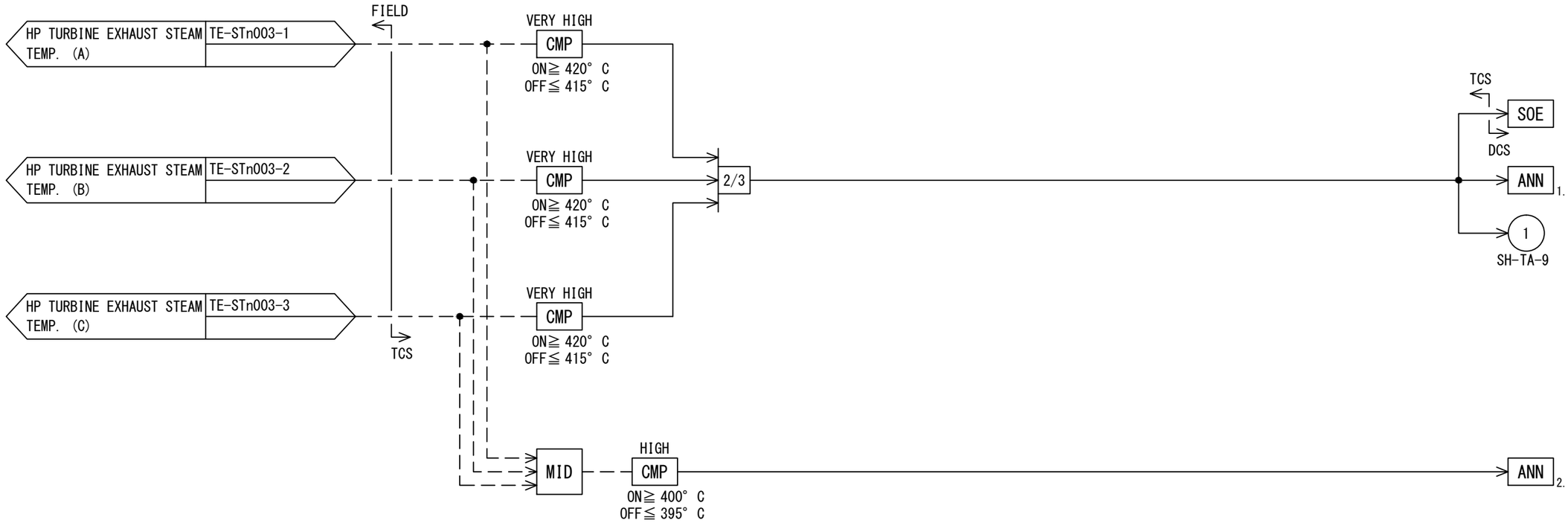


1. NO.1 TURBINE SHAFT VIBRATION (X) HIGH
2. NO.1 TURBINE SHAFT VIBRATION (Y) HIGH
3. NO.2 TURBINE SHAFT VIBRATION (X) HIGH
4. NO.2 TURBINE SHAFT VIBRATION (Y) HIGH
5. NO.3 GENERATOR SHAFT VIBRATION (X) HIGH
6. NO.3 GENERATOR SHAFT VIBRATION (Y) HIGH
7. NO.4 GENERATOR SHAFT VIBRATION (X) HIGH
8. NO.4 GENERATOR SHAFT VIBRATION (Y) HIGH

TYPE –					PAGE TA-18	
SERVICE TRIP LOGIC (9)(BEARING VIBRATION VERY HIGH TRIP (2))						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 37	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

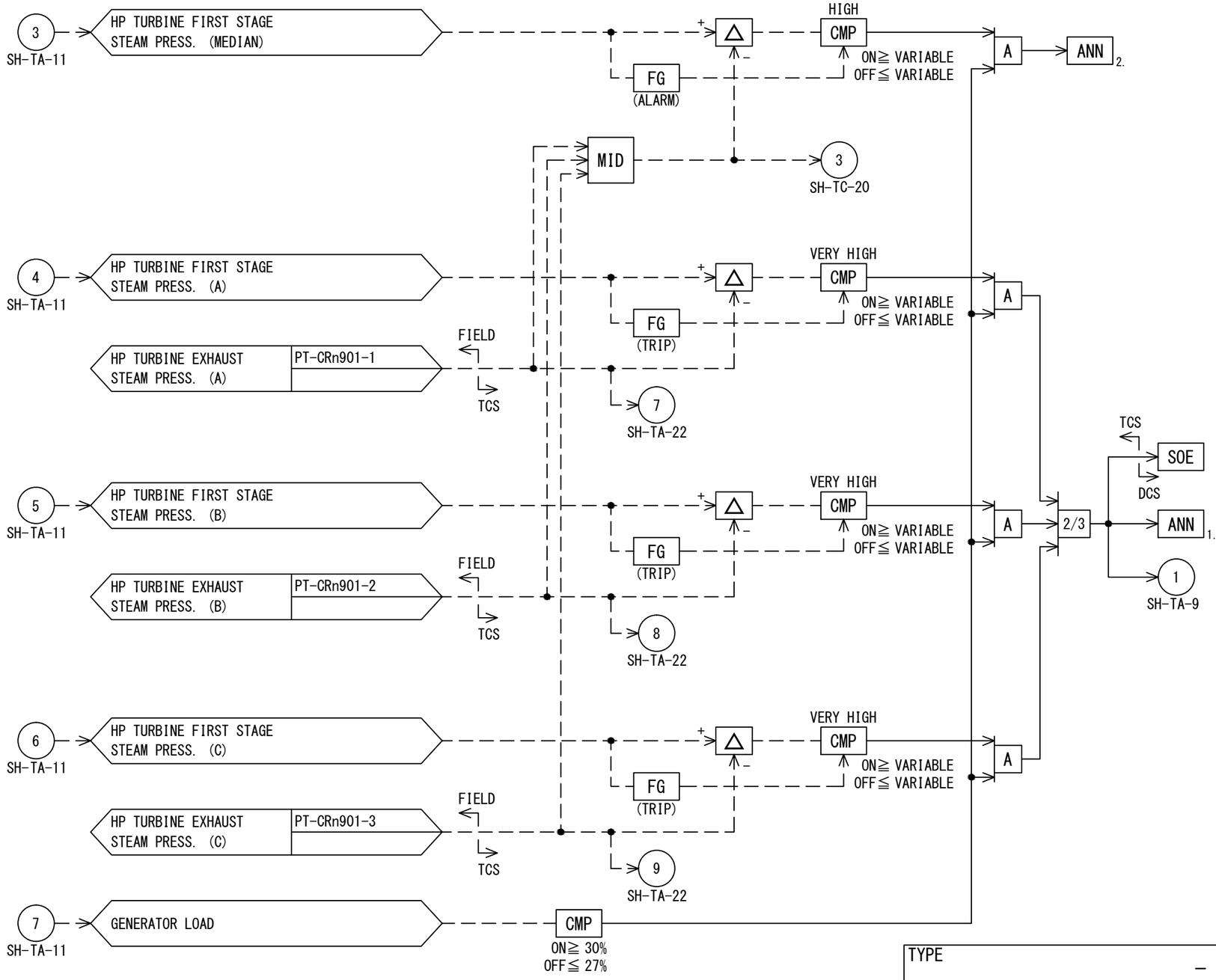


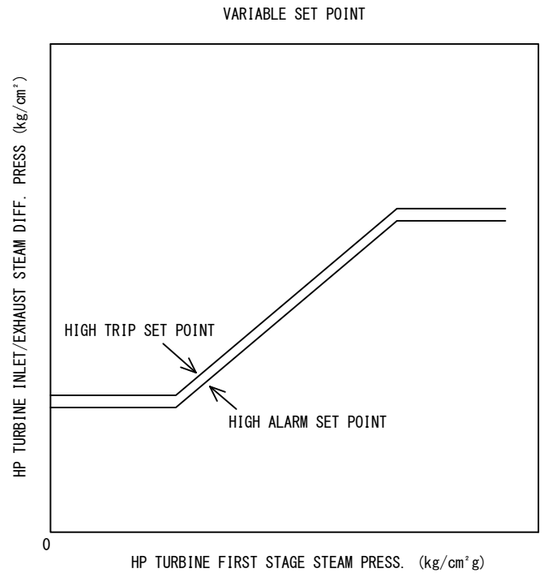
1. HP TURBINE EXHAUST STEAM TEMP. VERY HIGH
2. HP TURBINE EXHAUST STEAM TEMP. HIGH

TYPE –						PAGE TA-20	
SERVICE TRIP LOGIC (10) (HP TURBINE EXHAUST STEAM TEMP. VERY HIGH TRIP)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 38	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





HP TURBINE FIRST STAGE STEAM PRESS (kg/cm²g)	ANN	TRIP
0	24.72	26.72
31.9	24.72	26.72
51.5	37.02	39.02
72.2	50.22	52.22
104.2	70.92	72.92
107.4	72.92	74.92
111.1	75.42	77.42
150.0	75.42	77.42

1. HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. VERY HIGH
2. HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. HIGH

TYPE	–	PAGE	TA-21
SERVICE	TRIP LOGIC (11) (HP TURBINE INLET/ EXHAUST STEAM DIFF. PRESS. VERY HIGH TRIP)		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	39	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

7
SH-TA-21
HP TURBINE EXHAUST STEAM PRESS. (A)

8
SH-TA-21
HP TURBINE EXHAUST STEAM PRESS. (B)

9
SH-TA-21
HP TURBINE EXHAUST STEAM PRESS. (C)

VERY HIGH
CMP
ON≧ 44.9 kg/cm²g
OFF≦ 42.9 kg/cm²g

VERY HIGH
CMP
ON≧ 44.9 kg/cm²g
OFF≦ 42.9 kg/cm²g

VERY HIGH
CMP
ON≧ 44.9 kg/cm²g
OFF≦ 42.9 kg/cm²g

2/3

TCS
SOE
DCS
ANN 1.
1
SH-TA-9

MID

HIGH
CMP
ON≧ 42.7 kg/cm²g
OFF≦ 40.7 kg/cm²g

A

ANN 2.

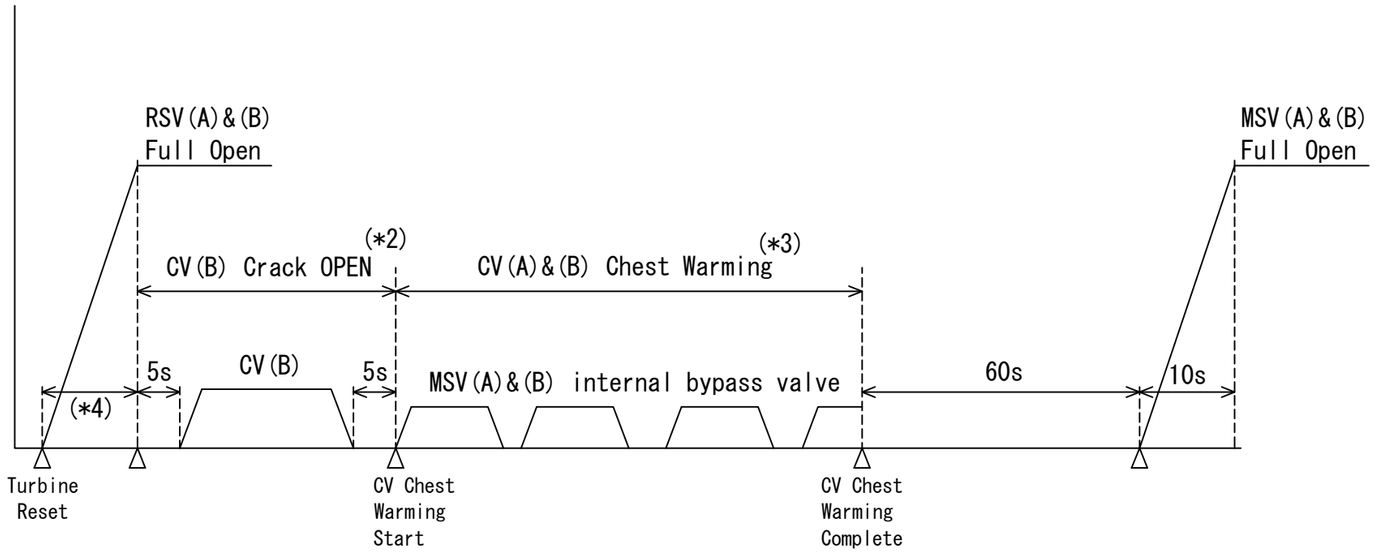
2
SH-TA-1
TURBINE TRIP

1. HP TURBINE EXHAUST STEAM PRESS. VERY HIGH
2. HP TURBINE EXHAUST STEAM PRESS. HIGH

TYPE –					PAGE TA-22	
SERVICE TRIP LOGIC (12) (HP TURBINE EXHAUST STEAM PRESS. VERY HIGH TRIP)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 40	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

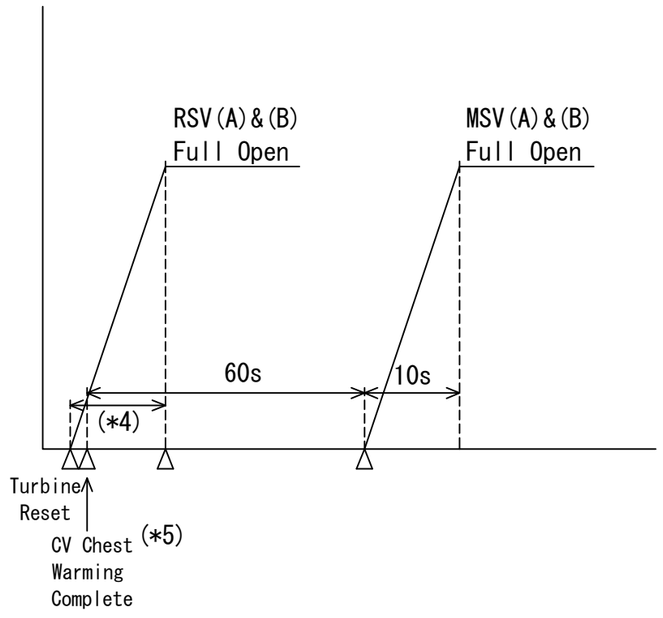
The following figure indicated the MSV(A)&(B) / RSV(A)&(B) Open procedure when the steam turbine is reset for turbine start-up.

Case 1.
CV(B) Crack open sequence & CV Chest warming required.



ADDED

Case 2.
CV(B) Crack open sequence & CV Chest warming not required.



NOTE:
*1): THE SEQUENTIAL OPERATION IS TO PREVENT THE CONTROL OIL PRESSURE FROM EXTREMELY DROPPING OUT. FOR THE DETAIL MECHANISM OF MSV(A)&(B) / RSV(A)&(B), PLEASE REFER TO "DIAGRAM OF STEAM TURBINE CONTROL MECHANISM (DWG. 310SD14-355)".

*2): FOR CV(B) CRACK OPEN, PLEASE REFER TO "ATS CRITERIA LIST AND CRITERIA TABLE (DWG. 310SC90-796)".

*3): FOR CV(A)&(B) CHEST WARMING, PLEASE REFER TO "ATS UNIT SEQUNCE CONTROL PROGRAM (DWG. 310SC90-795)".

*4): THE OPENING TIME OF RSV(A)&(B) FULL OPEN IS DESIGNED AS 10-40S. THE ACTUAL OPENING TIME DEPENDS ON THE ACTUAL RSV MECHANICAL OPERATING CONDITIONS.

*5): MSV(A)&(B) FULL OPEN REQUIRES CV(A)&(B) WARMING COMPLETE.

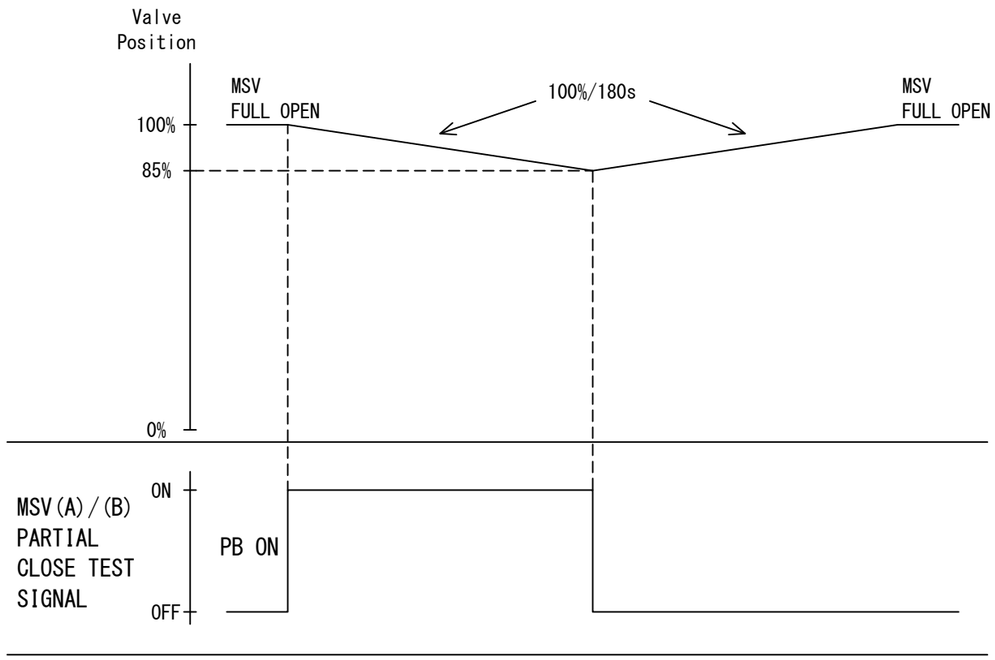
TYPE	–				PAGE	TA-29
SERVICE	MSV(A)&(B) / RSV(A)&(B) OPENING PROCEDURE					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	41	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

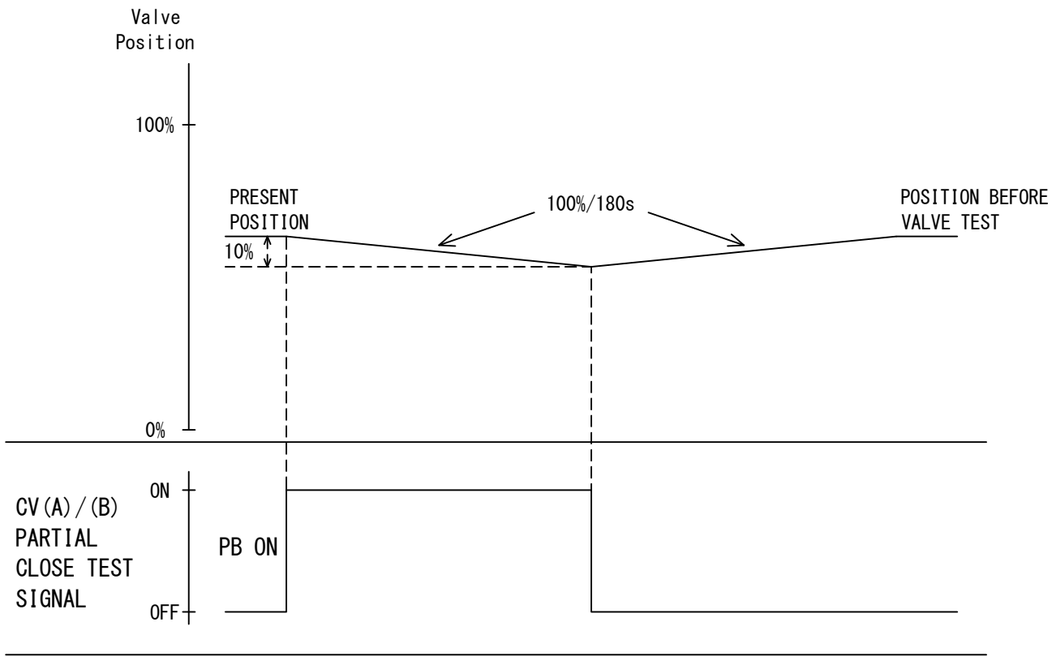
MSV/CV PARTIAL CLOSE TEST

MSV(A)/(B) PARTIAL TEST CHART



PLEASE NOTE THAT MSV(A)/(B) CLOSING RATE SHALL BE 100%/180S OR LESS DURING THE PARTIAL VALVE TEST TO AVOID THE MSP ABNORMAL INCREASE.

CV(A)/(B) PARTIAL TEST CHART

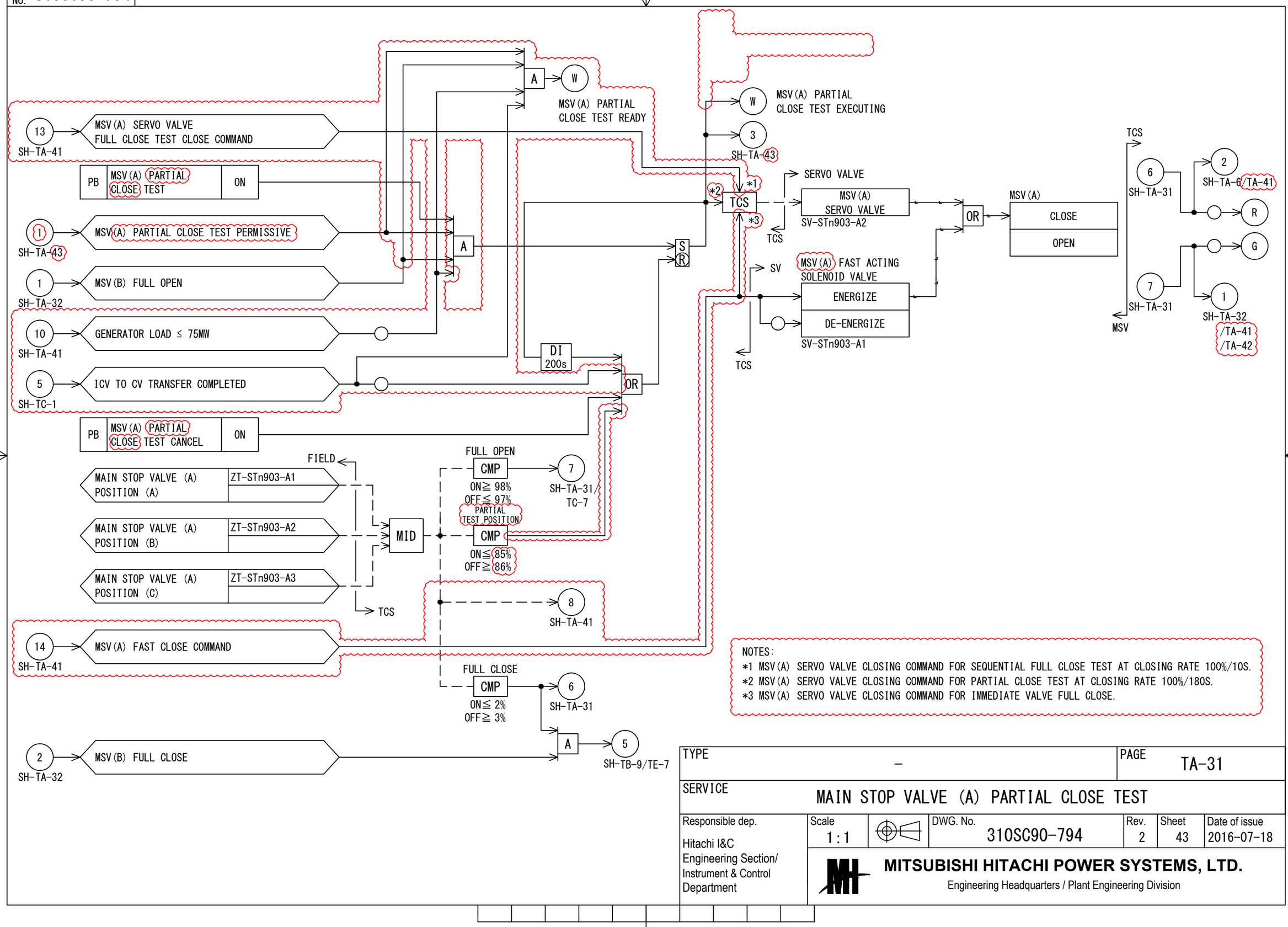


PLEASE NOTE THAT CV(A)/(B) CLOSING RATE SHALL BE 100%/180S OR LESS DURING THE PARTIAL VALVE TEST TO AVOID THE MSP ABNORMAL INCREASE.

ADDED

TYPE –					PAGE TA-30	
SERVICE MSV (A)/(B), CV (A)/(B) PARTIAL CLOSE TEST CHART						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 42	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794



NOTES:
*1 MSV(A) SERVO VALVE CLOSING COMMAND FOR SEQUENTIAL FULL CLOSE TEST AT CLOSING RATE 100%/10S.
*2 MSV(A) SERVO VALVE CLOSING COMMAND FOR PARTIAL CLOSE TEST AT CLOSING RATE 100%/180S.
*3 MSV(A) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.

TYPE	–					PAGE	TA-31
SERVICE	MAIN STOP VALVE (A) PARTIAL CLOSE TEST						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-794	2	43	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

13 SH-TA-42 — MSV(B) SERVO VALVE FULL CLOSE TEST CLOSE COMMAND

PB | MSV(B) PARTIAL CLOSE TEST | ON

2 SH-TA-43 — MSV(B) PARTIAL CLOSE TEST PERMISSIVE

1 SH-TA-31 — MSV(A) FULL OPEN

10 SH-TA-41 — GENERATOR LOAD ≤ 75MW

5 SH-TC-1 — ICV TO CV TRANSFER COMPLETED

PB | MSV(B) PARTIAL CLOSE TEST CANCEL | ON

A → W — MSV(B) PARTIAL CLOSE TEST READY

A

DI 200s

OR

S / R

W — MSV(B) PARTIAL CLOSE TEST EXECUTING

3 SH-TA-43

*1 *2 *3

TCS

SERVO VALVE

MSV(B) SERVO VALVE SV-STn903-B2

TCS

SV

MSV(B) FAST ACTING SOLENOID VALVE

ENERGIZE / DE-ENERGIZE

SV-STn903-B1

TCS

OR

MSV(B) CLOSE / OPEN

TCS

4 SH-TA-32 — 2 SH-TA-6/TA-31 /TA-42 — R

G

5 SH-TA-32 — 1 SH-TA-31 /TA-32 /TA-41 /TA-42

MSV

FIELD

MAIN STOP VALVE (B) POSITION (A) — ZT-STn903-B1

MAIN STOP VALVE (B) POSITION (B) — ZT-STn903-B2

MAIN STOP VALVE (B) POSITION (C) — ZT-STn903-B3

MID

TCS

FULL OPEN — CMP — ON≧ 98% OFF≦ 97% — 5 SH-TA-32/TC-7

PARTIAL TEST POSITION — CMP — ON≦ 85% OFF≧ 86%

8 SH-TA-42

FULL CLOSE — CMP — ON≦ 2% OFF≧ 3% — 4 SH-TA-32

14 SH-TA-42 — MSV(B) FAST CLOSE COMMAND

1 SH-TA-31 — MSV(A) FULL OPEN

1 SH-TA-32 — MSV(B) FULL OPEN

A → ATS (CRI:1.21)

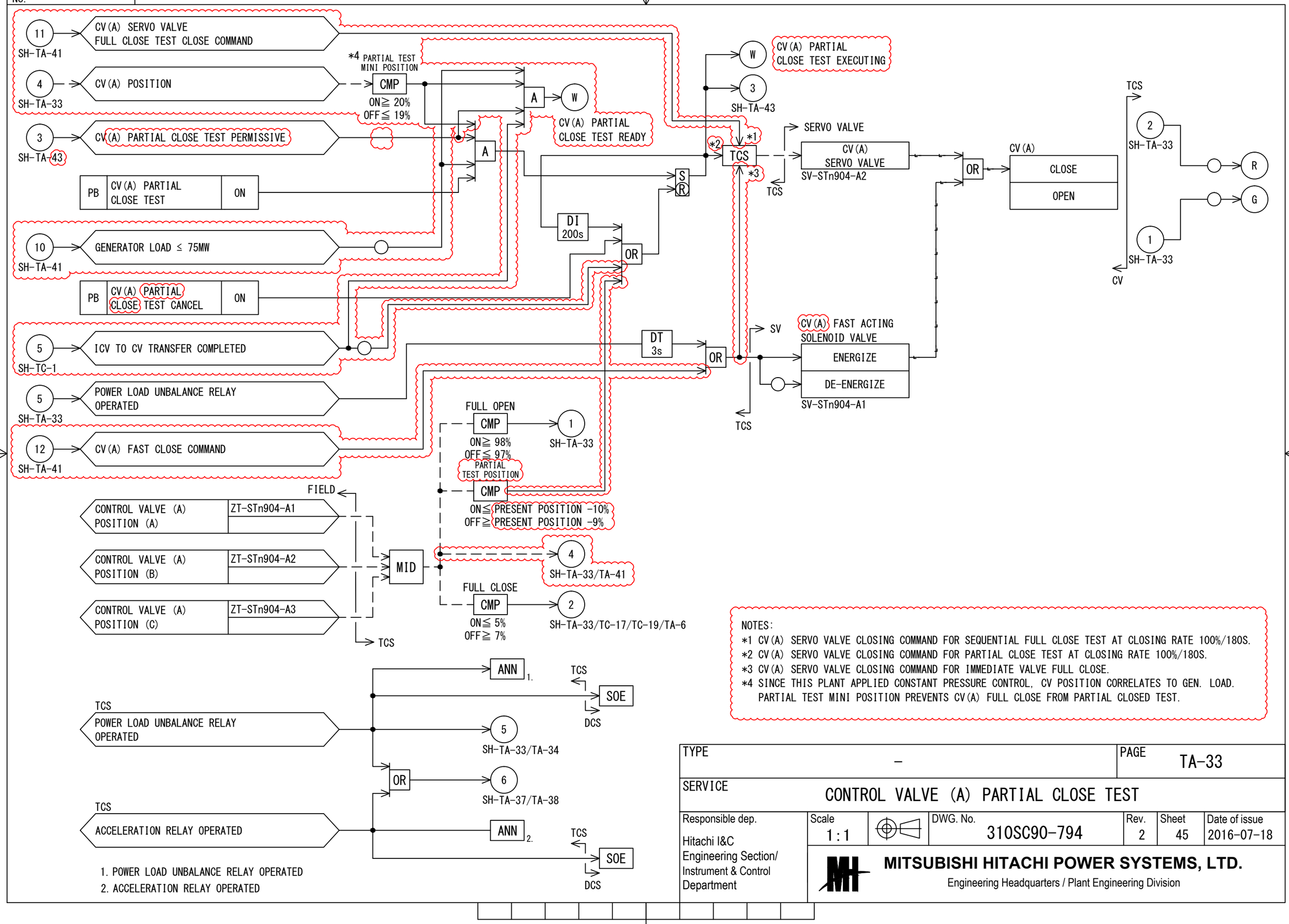
NOTES:
*1 MSV(B) SERVO VALVE CLOSING COMMAND FOR SEQUENTIAL FULL CLOSE TEST AT CLOSING RATE 100%/10S.
*2 MSV(B) SERVO VALVE CLOSING COMMAND FOR PARTIAL CLOSE TEST AT CLOSING RATE 100%/180S.
*3 MSV(B) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.

TYPE	–	PAGE	TA-32
SERVICE	MAIN STOP VALVE (B) PARTIAL CLOSE TEST		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	44	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



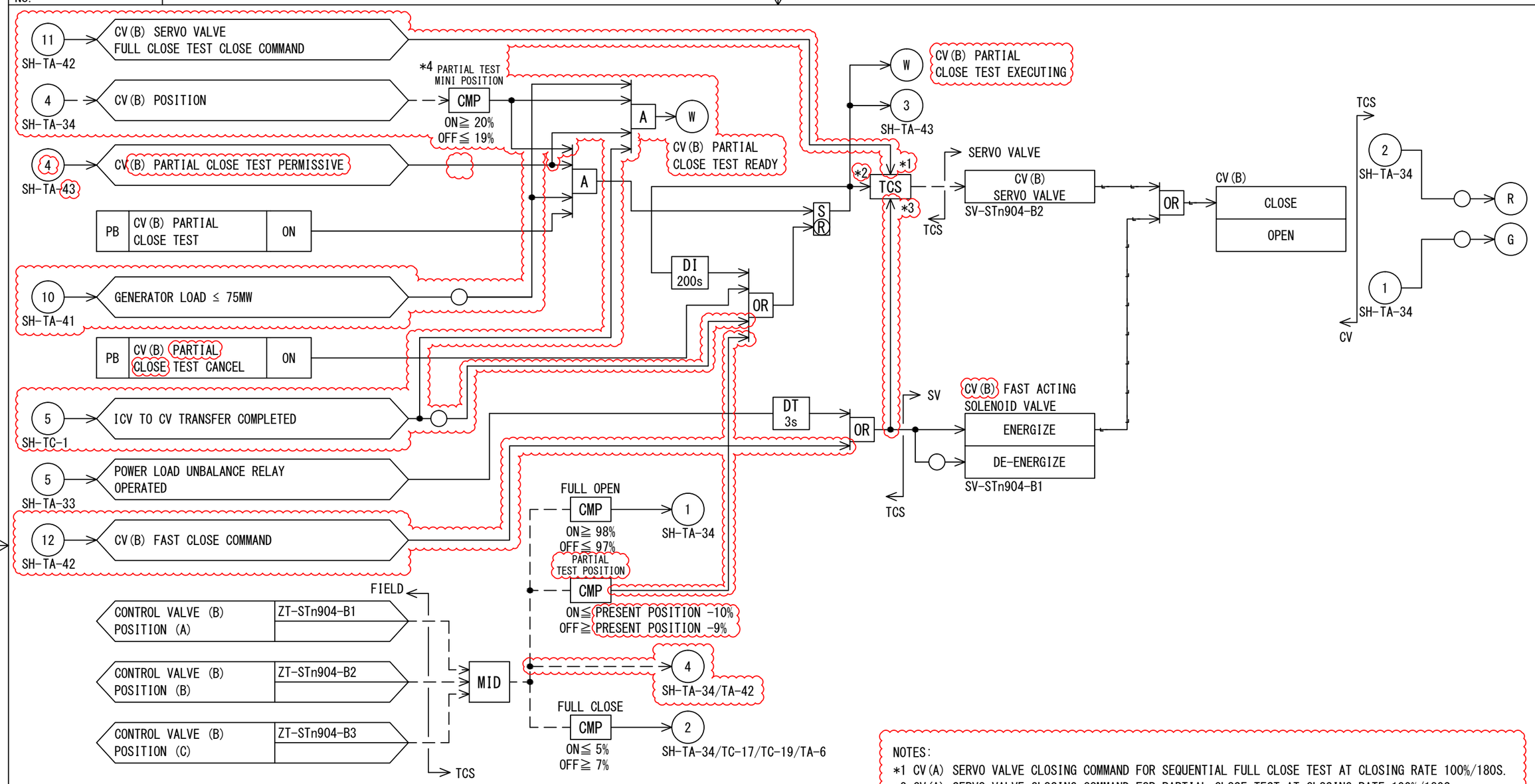
NOTES:
*1 CV(A) SERVO VALVE CLOSING COMMAND FOR SEQUENTIAL FULL CLOSE TEST AT CLOSING RATE 100%/180S.
*2 CV(A) SERVO VALVE CLOSING COMMAND FOR PARTIAL CLOSE TEST AT CLOSING RATE 100%/180S.
*3 CV(A) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.
*4 SINCE THIS PLANT APPLIED CONSTANT PRESSURE CONTROL, CV POSITION CORRELATES TO GEN. LOAD. PARTIAL TEST MINI POSITION PREVENTS CV(A) FULL CLOSE FROM PARTIAL CLOSED TEST.

TYPE	–	PAGE	TA-33
SERVICE	CONTROL VALVE (A) PARTIAL CLOSE TEST		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	45	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



NOTES:
*1 CV(A) SERVO VALVE CLOSING COMMAND FOR SEQUENTIAL FULL CLOSE TEST AT CLOSING RATE 100%/180S.
*2 CV(A) SERVO VALVE CLOSING COMMAND FOR PARTIAL CLOSE TEST AT CLOSING RATE 100%/180S.
*3 CV(A) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.
*4 SINCE THIS PLANT APPLIED CONSTANT PRESSURE CONTROL, CV POSITION CORRELATES TO GEN. LOAD. PARTIAL TEST MINI POSITION PREVENTS CV(A) FULL CLOSE FROM PARTIAL CLOSED TEST.

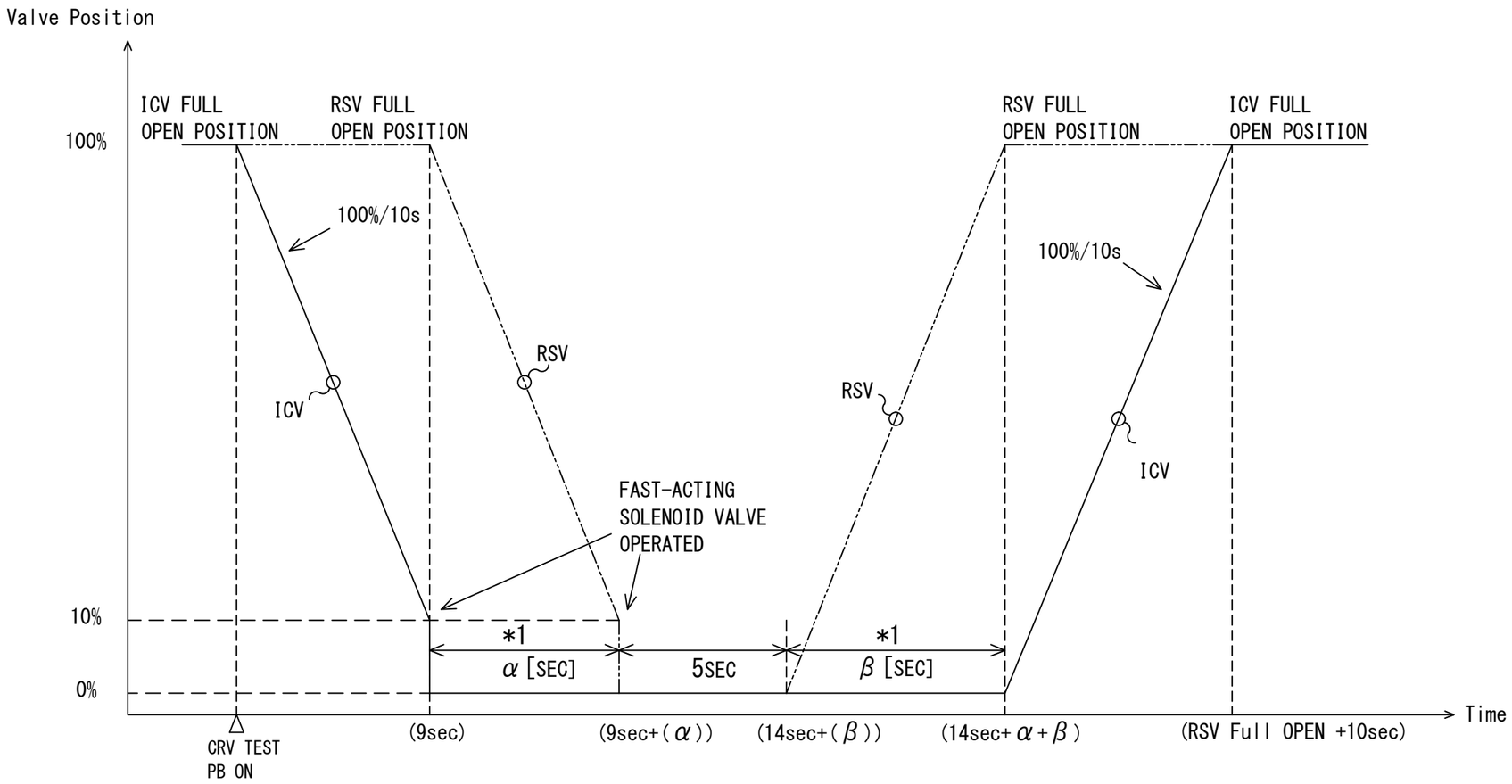
TYPE –					PAGE TA-34	
SERVICE CONTROL VALVE (B) PARTIAL CLOSE TEST						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 46	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

CRV (A)/(B) CLOSE TEST CHART

ADDED



ONCE ICV POSITION REACHED 10% OR LESS, ICV SERVO VALVE SHALL BE IMMEDIATELY CHANGED TO FULL CLOSE POSITION AS WELL AS PLU/ACCELERATION RELAY ACTIVATION.

NOTE:
*1): THE ACTUAL RSV(A)/(B)OPENING/CLOSING TIME DEPENDS ON THE ACTUAL REHEAT STEAM PRESSURE.
$\alpha+\beta$ IS DESIGNED AS 10s-40s.

TYPE –					PAGE TA-36	
SERVICE CRV (A)/(B) CLOSE TEST CHART						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 47	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794

NOTES:
*1 ICV(A) SERVO VALVE CLOSING COMMAND FOR CRV TEST AT 100%/10S.
*2 ICV(A) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.

FIELD
INTERCEPT VALVE (A) POSITION (A) — ZT-STn905-A4
INTERCEPT VALVE (A) POSITION (B) — ZT-STn905-A5
INTERCEPT VALVE (A) POSITION (C) — ZT-STn905-A6
MID
TCS

TEST POSITION
CMP
ON≦ 10%
OFF≧ 11%
8
SH-TA-37

FULL OPEN POSITION
CMP
ON≧ 98%
OFF≦ 97%
G

FULL CLOSE POSITION
CMP
ON≦ 2%
OFF≧ 3%
4
SH-TA-6/TA-38
R

PB | CRV (A) TEST | ON

7 SH-TA-43 — CRV(A) FULL CLOSE TEST PERMISSIVE
1 SH-TA-38 — RSV(B) FULL OPEN
1 SH-TA-37 — RSV(A) FULL OPEN
8 SH-TA-37 — ICV(A) TEST POSITION
6 SH-TA-33 — ICV FAST CLOSE COMMAND
5 SH-TC-1 — ICV TO CV TRANSFER COMPLETED
2 SH-TA-37 — RSV(A) FULL CLOSE

A
S/R
A
W
CRV(A) CLOSE TEST READY
A
DT 2s
A
DT 8s
(WO)
OR
DT 1s
OR
*1
TCS
*2
TCS
SERVO VALVE
ICV(A) SERVO VALVE
SV-STn905-A2

DI 5s
DI 60s
OR
W
5
SH-TA-43
DT 1s

SV
INTERCEPT VALVE (A) FAST ACTING SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn905-A1
TCS
OR
ICV(A)
CLOSE
CONTROL
OPEN

SV
REHEAT STOP VALVE (A) TEST SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn906-A2
TCS

SV
REHEAT STOP VALVE (A) FAST ACTING SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn906-A1
TCS

OR
RSV(A)
CLOSE
OPEN

7 SH-TA-37 — 2 SH-TA-37 — R
G
6 SH-TA-37 — 1 SH-TA-37 /TA-38

PB | CRV (A) TEST CANCEL | ON

FIELD
REHEAT STOP VALVE (A) POSITION (A) — ZT-STn905-A1
REHEAT STOP VALVE (A) POSITION (B) — ZT-STn905-A2
REHEAT STOP VALVE (A) POSITION (C) — ZT-STn905-A3
MID
TCS

TEST POSITION
CMP
ON≦ 10%
OFF≧ 11%

FULL OPEN POSITION
CMP
ON≧ 98%
OFF≦ 97%
6
SH-TA-37

FULL CLOSE POSITION
CMP
ON≦ 2%
OFF≧ 3%
7
SH-TA-37/TA-6

1 SH-TA-37 — RSV(A) FULL OPEN
1 SH-TA-38 — RSV(B) FULL OPEN
A
ATS
(CRI:1.22)

TYPE	–	PAGE	TA-37
SERVICE	COMBINED REHEAT VALVE (A) TEST		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	48	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

NOTES:
*1 ICV(B) SERVO VALVE CLOSING COMMAND FOR CRV TEST AT 100%/10S.
*2 ICV(B) SERVO VALVE CLOSING COMMAND FOR IMMEDIATE VALVE FULL CLOSE.

FIELD
INTERCEPT VALVE (B) POSITION (A) — ZT-STn905-B4
INTERCEPT VALVE (B) POSITION (B) — ZT-STn905-B5
INTERCEPT VALVE (B) POSITION (C) — ZT-STn905-B6
MID
TCS

TEST POSITION
CMP
ON≦ 10%
OFF≧ 11%
8
SH-TA-38

FULL OPEN POSITION
CMP
ON≧ 98%
OFF≦ 97%
G

FULL CLOSE POSITION
CMP
ON≦ 2%
OFF≧ 3%
4
SH-TA-6/TA-38
R

PB | CRV (B) TEST | ON

8 SH-TA-43 — CRV(B) FULL CLOSE TEST PERMISSIVE
1 SH-TA-37 — RSV(A) FULL OPEN
1 SH-TA-38 — RSV(B) FULL OPEN
8 SH-TA-38 — ICV(B) TEST POSITION
6 SH-TA-33 — ICV FAST CLOSE COMMAND
5 SH-TC-1 — ICV TO CV TRANSFER COMPLETED
2 SH-TA-38 — RSV(B) FULL CLOSE

A
S
R
A
W
CRV(B) CLOSE TEST READY
A
DT 2s
A
DT 8s
(WO)
OR
DT 1s
OR
*1
TCS
*2
TCS
SERVO VALVE
ICV(B) SERVO VALVE
SV-STn905-B2

DT 2.5s
DI 5s
DI 60s
OR
W
5
SH-TA-43
DT 1s
A

SV
INTERCEPT VALVE (B) FAST ACTING SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn905-B1
TCS
OR
ICV(B)
CLOSE
CONTROL
OPEN

SV
REHEAT STOP VALVE (B) TEST SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn906-B2
TCS

SV
REHEAT STOP VALVE (B) FAST ACTING SOLENOID VALVE
ENERGIZE
DE-ENERGIZE
SV-STn906-B1
TCS

OR
RSV(B)
CLOSE
OPEN

7 SH-TA-38 — 2 SH-TA-38 — R
G
6 SH-TA-38 — 1 SH-TA-37 /TA-38

PB | CRV (B) TEST CANCEL | ON

FIELD
REHEAT STOP VALVE (B) POSITION (A) — ZT-STn905-B1
REHEAT STOP VALVE (B) POSITION (B) — ZT-STn905-B2
REHEAT STOP VALVE (B) POSITION (C) — ZT-STn905-B3
MID
TCS

TEST POSITION
CMP
ON≦ 10%
OFF≧ 11%

FULL OPEN POSITION
CMP
ON≧ 98%
OFF≦ 97%
6
SH-TA-38

FULL CLOSE POSITION
CMP
ON≦ 2%
OFF≧ 3%
7
SH-TA-38/TA-6

4 SH-TA-37 — ICV(A) FULL CLOSE
4 SH-TA-38 — ICV(B) FULL CLOSE
A
9
SH-TB-9

TYPE	–	PAGE	TA-38
SERVICE	COMBINED REHEAT VALVE (B) TEST		

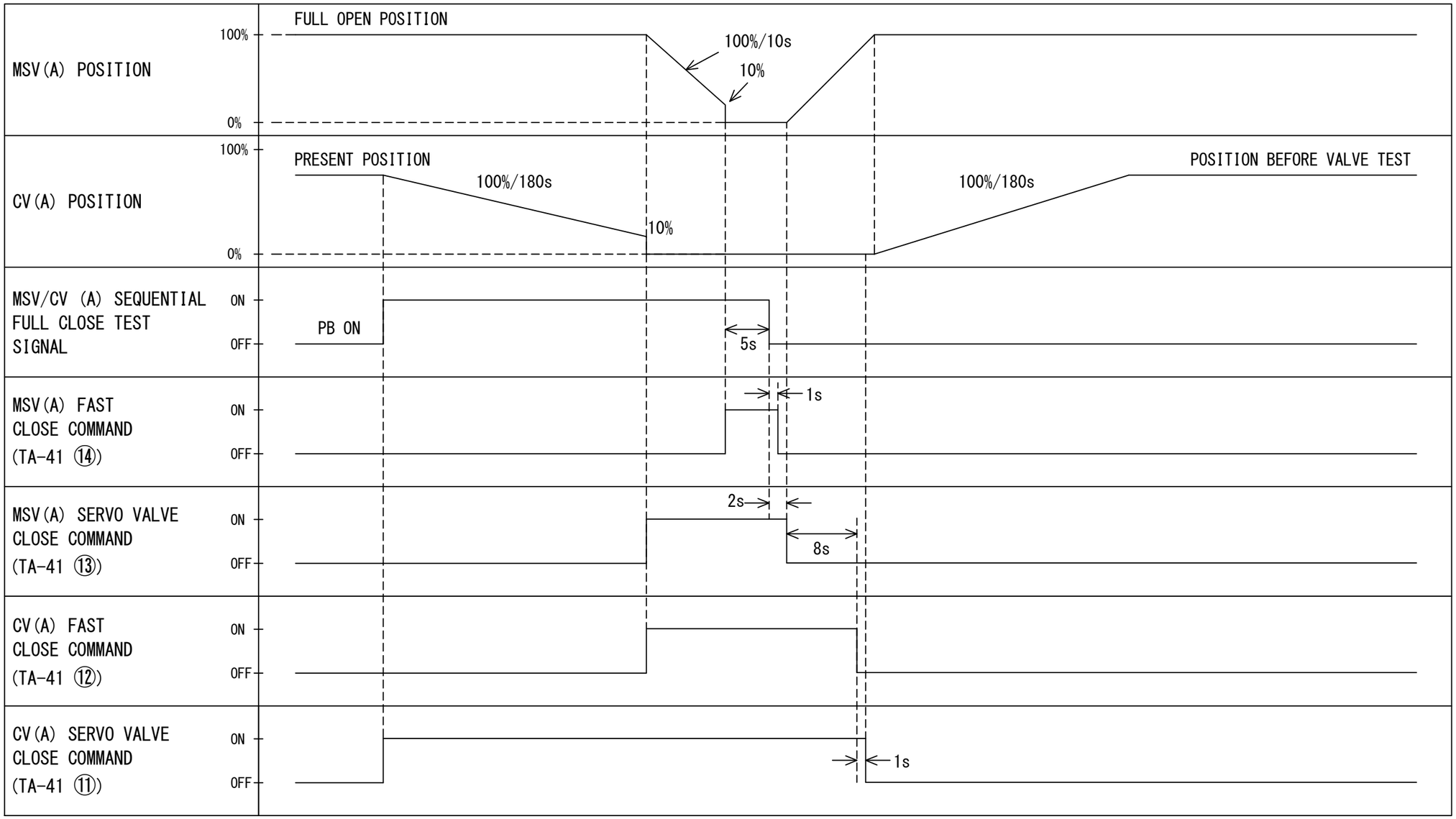
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	49	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

ADDED

MSV/CV (A)/(B) SEQUENTIAL FULL CLOSE TEST CHART



NOTES:
THE ABOVE TEST ALLOWS THE OPERATOR TO ACHIEVE BOTH MSV AND CV FULL CLOSE TEST AT ONE TIME.
PLEASE NOTE THAT THE ABOVE TEST SHALL ONLY BE OPERATED WHEN THE GENERATOR LOAD IS 75MW OR LESS TO AVOID THE ABNORMAL INCREASE OF THE MAIN STEAM PRESSURE (MSP).
FURTHERMORE THE MW VALUE (75MW) SHALL BE ADJUSTED AT SITE COMMISSIONING.

TYPE	–	PAGE	TA-40
SERVICE	MSV/CV (A)/(B) SEQUENTIAL FULL CLOSE TEST CHART		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	50	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

PB | MSV/CV(A) SEQUENTIAL FULL CLOSE TEST | ON

5 SH-TA-43 — MSV/CV(A) SEQUENTIAL FULL CLOSE TEST PERMISSIVE

1 SH-TA-32 — MSV(B) FULL OPEN

GENERATOR LOAD — *1 CMP ON≦ 75MW OFF≧ 76MW

5 SH-TA-41 — MSV/CV(A) FULL CLOSE TEST STATE

1 SH-TA-31 — MSV(A) FULL OPEN

4 SH-TA-33 — CV(A) POSITION — FAST CLOSE POSITION CMP ON≦ 10% OFF≧ 11%

A → W — MSV/CV (A) SEQUENTIAL FULL CLOSE TEST READY

A → S/R

10 — SH-TA-31/TA-32 /TA-33/TA-34 /TA-42

TCS — MSV/CV(A) SEQUENTIAL FULL CLOSE TEST EXECUTING — DCS

A

5 — SH-TA-41/TA-43

OR → CV(A) SERVO VALVE CLOSE COMMAND → 11 SH-TA-33

A → DT 2s → OR

DT 1s → OR

DT 8s — (WO) → OR → CV(A) FAST CLOSE COMMAND → 12 SH-TA-33

MSV(A) SERVO VALVE CLOSE COMMAND → 13 SH-TA-31

PB | MSV/CV(A) SEQUENTIAL FULL CLOSE TEST CANCEL | ON

2 SH-TA-31 — MSV(A) FULL CLOSE — DI 5s → OR

5 SH-TC-1 — ICV TO CV TRANSFER COMPLETE — DI 300s → OR

OR

DT 1s → A → MSV(A) FAST CLOSE COMMAND → 14 SH-TA-31

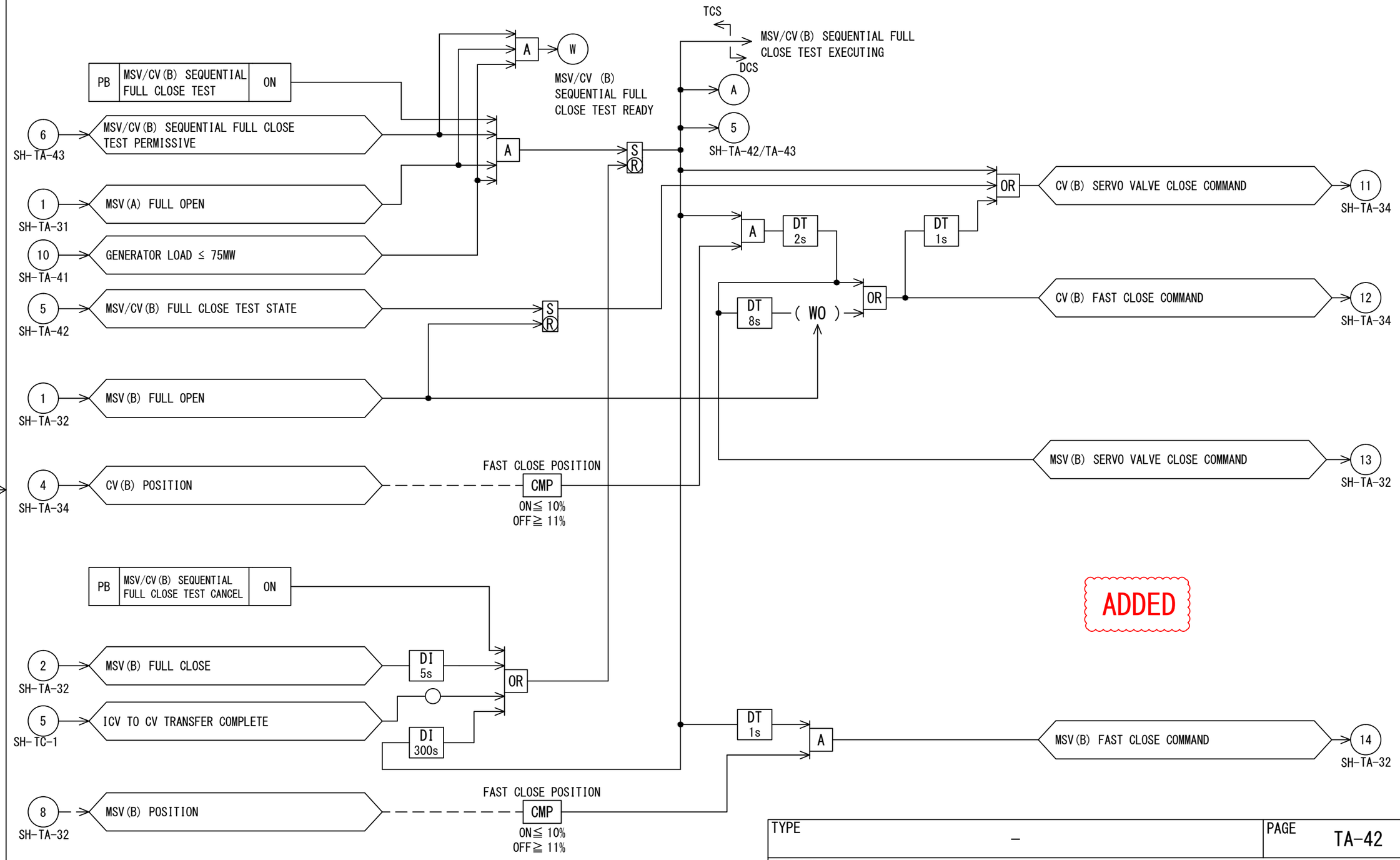
8 SH-TA-31 — MSV(A) POSITION — FAST CLOSE POSITION CMP ON≦ 10% OFF≧ 11%

ADDED

NOTES:
*1 Since the boiler maximum operating pressure is limited, this value should be adjusted at site according to MSP increasing rate.

TYPE –						PAGE TA-41		
SERVICE MSV/CV(A) SEQUENTIAL FULL CLOSE TEST								
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794		Rev. 2	Sheet 51	Date of issue 2016-07-18	
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division								

DWG. No. 310SC90-794



ADDED

TYPE –					PAGE TA-42	
SERVICE MSV/CV(B) SEQUENTIAL FULL CLOSE TEST						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 52	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

3 SH-TA-31 → MSV(A) PARTIAL CLOSE TEST STATE

3 SH-TA-32 → MSV(B) PARTIAL CLOSE TEST STATE

3 SH-TA-33 → CV(A) PARTIAL CLOSE TEST STATE

3 SH-TA-34 → CV(B) PARTIAL CLOSE TEST STATE

5 SH-TA-41 → MSV/CV (A) FULL CLOSE TEST STATE

5 SH-TA-42 → MSV/CV (B) FULL CLOSE TEST STATE

5 SH-TA-37 → CRV(A) TEST STATE

5 SH-TA-38 → CRV(B) TEST STATE

ADDED

OR

TCS

OR → 10MAY50EA003XB01 TURBINE VALVE TEST

DCS

MSV(A) PARTIAL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 1 SH-TA-31

MSV(B) PARTIAL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 2 SH-TA-32

CV(A) PARTIAL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 3 SH-TA-33

CV(B) PARTIAL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 4 SH-TA-34

MSV/CV(A) FULL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 5 SH-TA-41

MSV/CV(B) FULL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 6 SH-TA-42

CRV(A) FULL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 7 SH-TA-37

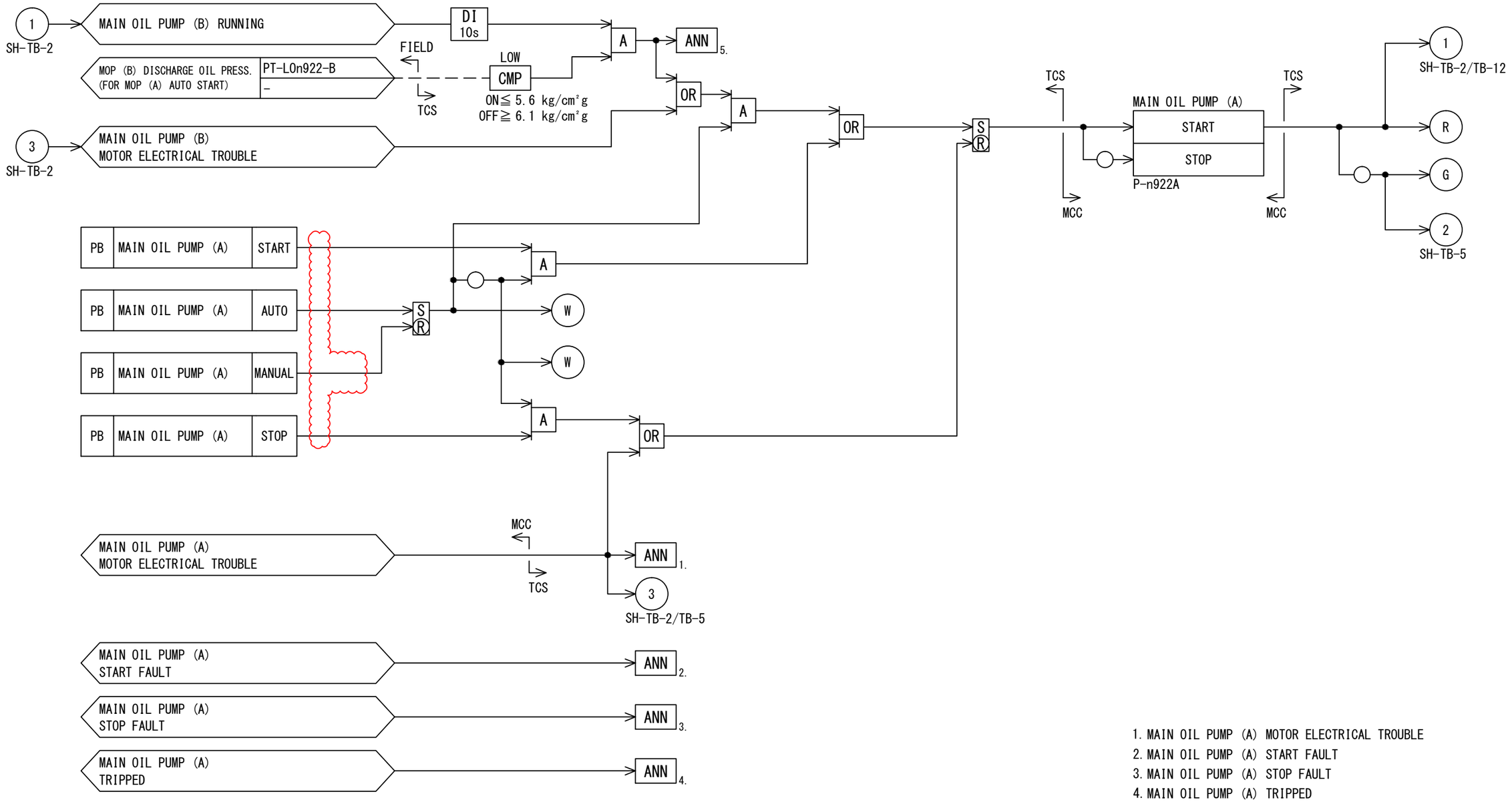
CRV(B) FULL CLOSE TEST PERMISSIVE (OTHER TESTS STATE) → 8 SH-TA-38

TYPE	–
PAGE	TA-43
SERVICE	MSV/CV (A)/(B) SEQUENTIAL FULL CLOSE TEST PERMISSIVE

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	53	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



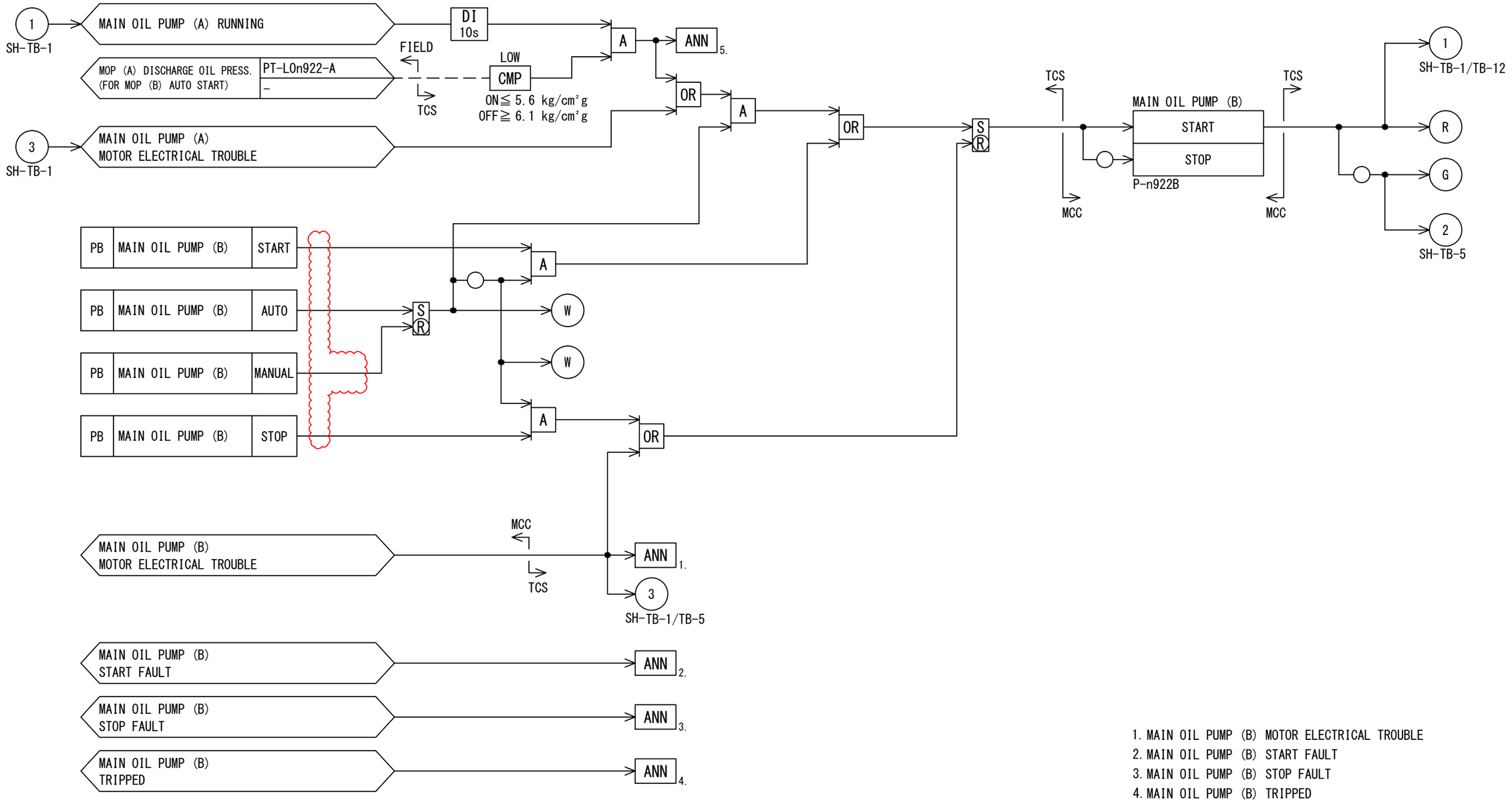
1. MAIN OIL PUMP (A) MOTOR ELECTRICAL TROUBLE
2. MAIN OIL PUMP (A) START FAULT
3. MAIN OIL PUMP (A) STOP FAULT
4. MAIN OIL PUMP (A) TRIPPED
5. MOP (B) DISCHARGE OIL PRESS. (FOR MOP (A) AUTO START) LOW

NOTE:
WHEN TCS MAJOR TROUBLE OCCURS, MOP WILL BE STARTED.

TYPE –						PAGE TB-1	
SERVICE MAIN OIL PUMP (A)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 54	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



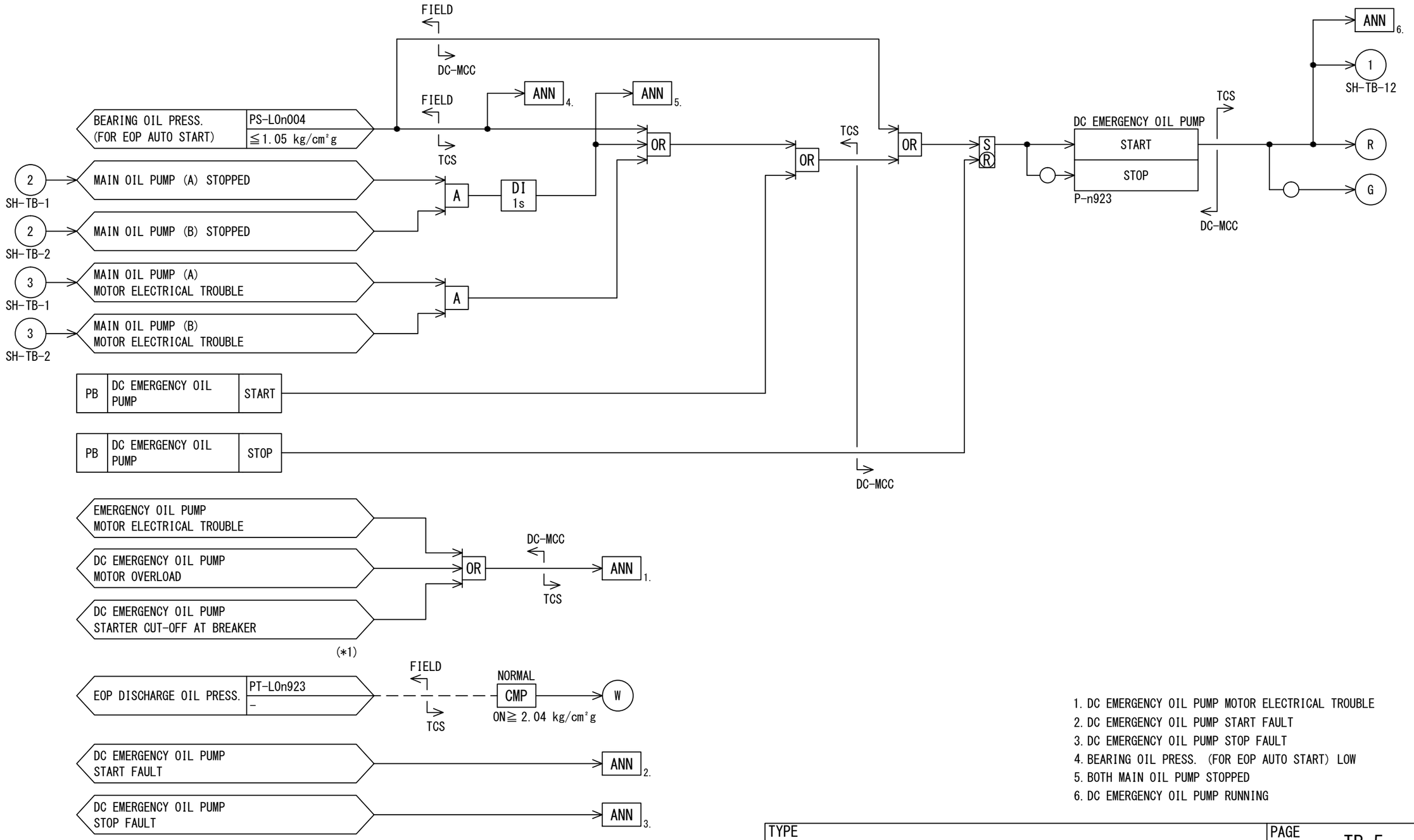
1. MAIN OIL PUMP (B) MOTOR ELECTRICAL TROUBLE
2. MAIN OIL PUMP (B) START FAULT
3. MAIN OIL PUMP (B) STOP FAULT
4. MAIN OIL PUMP (B) TRIPPED
5. MOP (A) DISCHARGE OIL PRESS. (FOR MOP (B) AUTO START) LOW

NOTE:
WHEN TCS MAJOR TROUBLE OCCURS, MOP WILL BE STARTED.

TYPE –					PAGE TB-2	
SERVICE MAIN OIL PUMP (B)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 55	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. DC EMERGENCY OIL PUMP MOTOR ELECTRICAL TROUBLE
2. DC EMERGENCY OIL PUMP START FAULT
3. DC EMERGENCY OIL PUMP STOP FAULT
4. BEARING OIL PRESS. (FOR EOP AUTO START) LOW
5. BOTH MAIN OIL PUMP STOPPED
6. DC EMERGENCY OIL PUMP RUNNING

NOTE:
(*1) THE POWER SOURCE OR THE CONTROL POWER SOURCE ARE UNABLE TO BE SUPPLIED BY DC-MCC.

TYPE –					PAGE TB-5	
SERVICE DC EMERGENCY OIL PUMP						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 56	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

CONDITION / DEVICE		REMARKS
TURBINE SPEED	ZERO SPEED ON (0.67rpm); TURNING 3 rpm; ZERO SPEED OFF (2rpm); 4 rpm	
	TURBINE TRIPPED; TURNING GEAR ENGAGE; TURBINE ACCELERATION; TURNING GEAR DIS-ENGAGE	
MSV & CV FULL CLOSE	ON / OFF	
TURBINE TURNING GEAR	ENGAGE / DIS-ENGAGE; 30s	
ZERO SPEED	ON / OFF	
TURNING GEAR MOTOR	RUNNING / STOPPED; 5s; 5s; 5s	
TURNING GEAR ENGAGE SOLENOID VALVE	ENERGIZE / DE-ENERGIZE; 30s	

TYPE	–					PAGE	TB-8
SERVICE	TURNING GEAR ENGAGE TIME CHART (AUTO MODE)						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	57	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

(LOCAL)

PB | TURNING GEAR MOTOR | JOG

FIELD → TCS

5 SH-TA-31 — BOTH MSV FULL CLOSE

9 SH-TA-38 — BOTH ICV FULL CLOSE

5 SH-TA-13 — BEARING OIL PRESS. NORMAL

HAND TURNING INTERLOCK POSITION | ZS-L0n003 | –

FIELD → TCS

PB | TURNING GEAR MOTOR | START

PB | TURNING GEAR MOTOR | AUTO

PB | TURNING GEAR MOTOR | MANUAL

PB | TURNING GEAR MOTOR | STOP

TURNING GEAR MOTOR ELECTRICAL TROUBLE → TCS → MCC

4 SH-TA-15 — TURBINE ZERO SPEED

4 SH-TB-10 — TURNING GEAR ENGAGED

1 SH-TB-9 — TURNING GEAR MOTOR RUNNING

4 SH-TB-10

7 SH-TB-10

6 SH-TB-10

5 SH-TB-10

8 SH-TB-10

DI 5s

DI 3s

DI 30s

DI 45s

A, OR, S/R

TURNING GEAR MOTOR
START
STOP

TCS → MCC

TCS → FIELD → R / 1 SH-TB-9/TB-10 / R / G

ANN 1.

TURNING GEAR MOTOR START FAULT → ANN 2.

TURNING GEAR MOTOR STOP FAULT → ANN 3.

TURNING GEAR MOTOR TRIPPED → ANN 4.

ANN 5.

ATS (CRI:1.19)

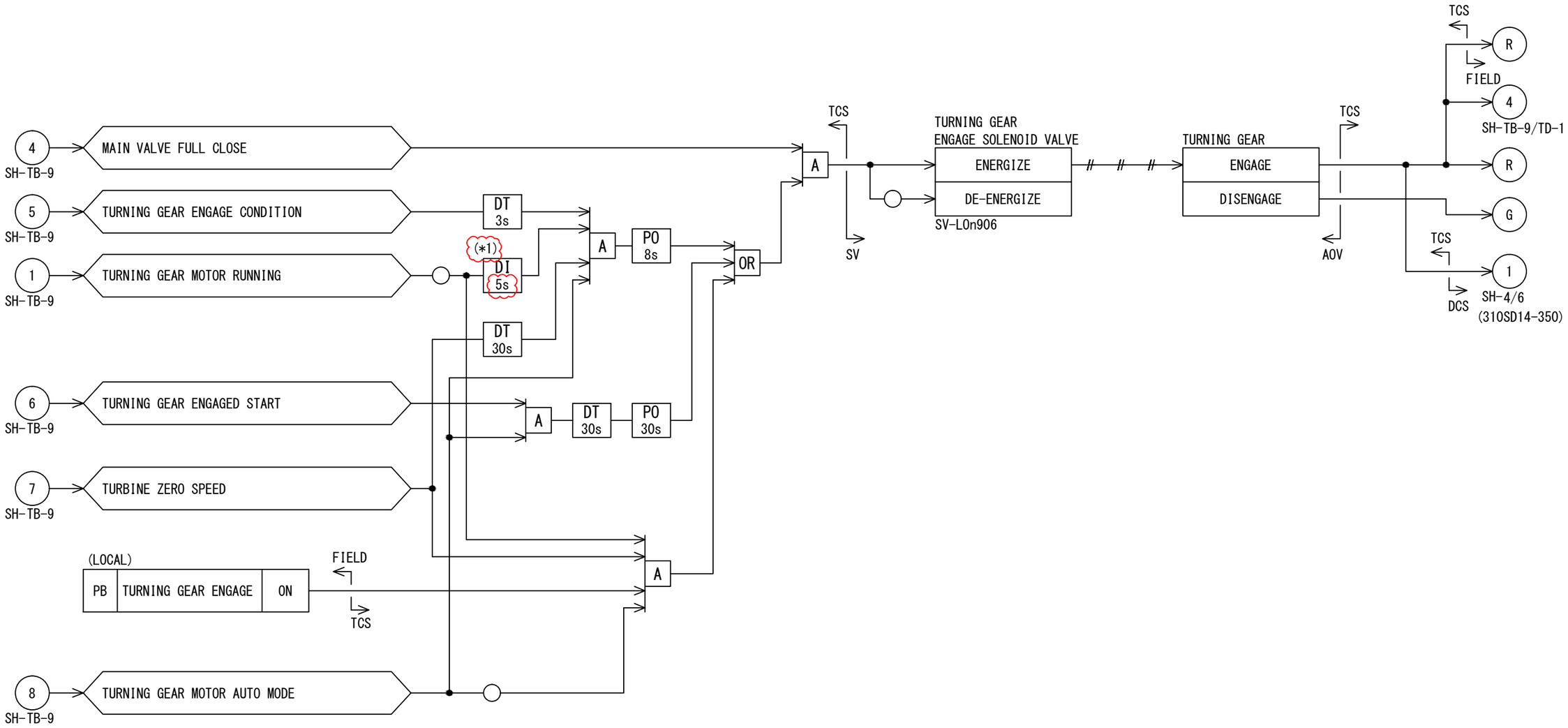
14 SH-TE-1/TE-6

1. TURNING GEAR MOTOR ELECTRICAL TROUBLE
2. TURNING GEAR MOTOR START FAULT
3. TURNING GEAR MOTOR STOP FAULT
4. TURNING GEAR MOTOR TRIPPED
5. TURNING FAILURE

TYPE –						PAGE TB-9	
SERVICE TURNING GEAR MOTOR							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 58	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

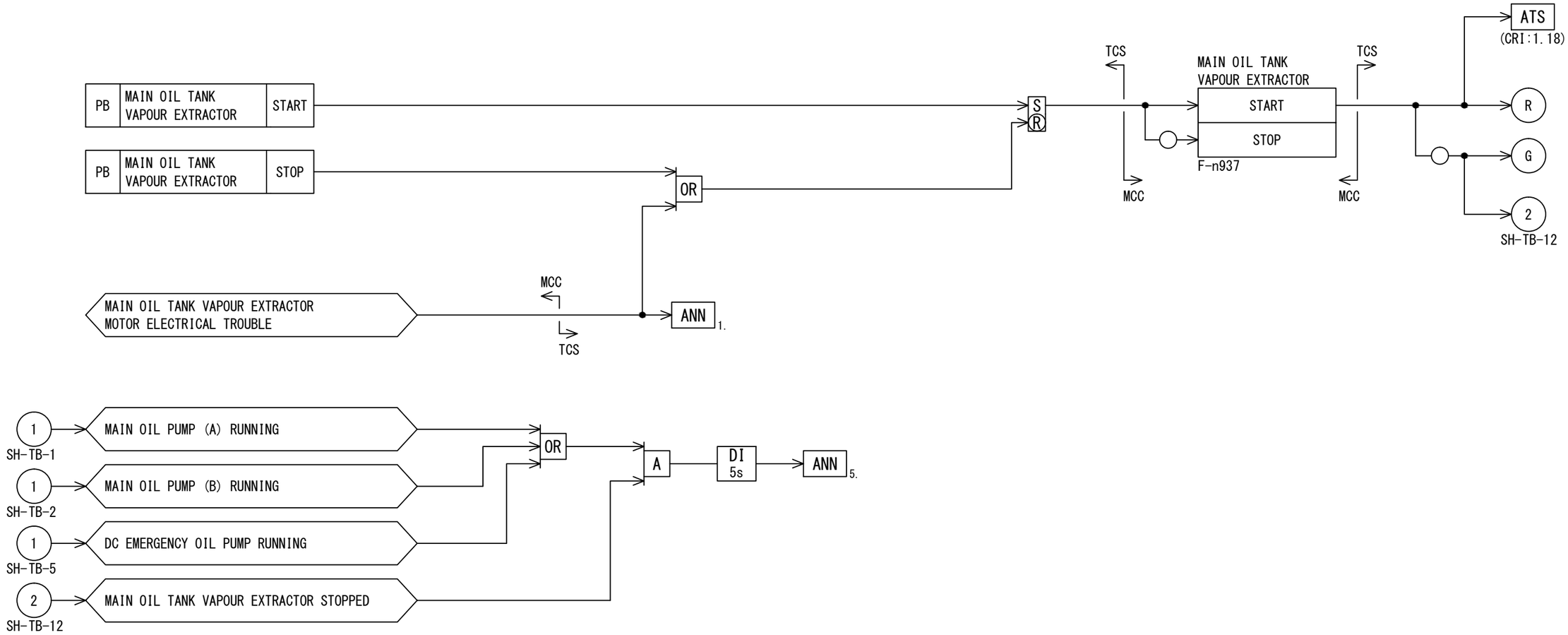


NOTE:
(*1) THIS VALUE SHOULD BE ADJUSTED AT SITE COMMISSIONING.
THE TURNING GEAR ON SIGNAL IS REQUIRED TO OUTPUT BEFORE THE COMPLETE STOP OF TURNING GEAR MOTOR.
SINCE THE MECHANICAL CHARACTERISTIC OF EACH MOTOR MIGHT BE SLIGHTLY DIFFERENT, THE TIMER NEEDS TO BE DETERMINED AT SITE COMMISSIONING.

TYPE	–					PAGE	TB-10
SERVICE	TURNING GEAR ENGAGE SOLENOID						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	59	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

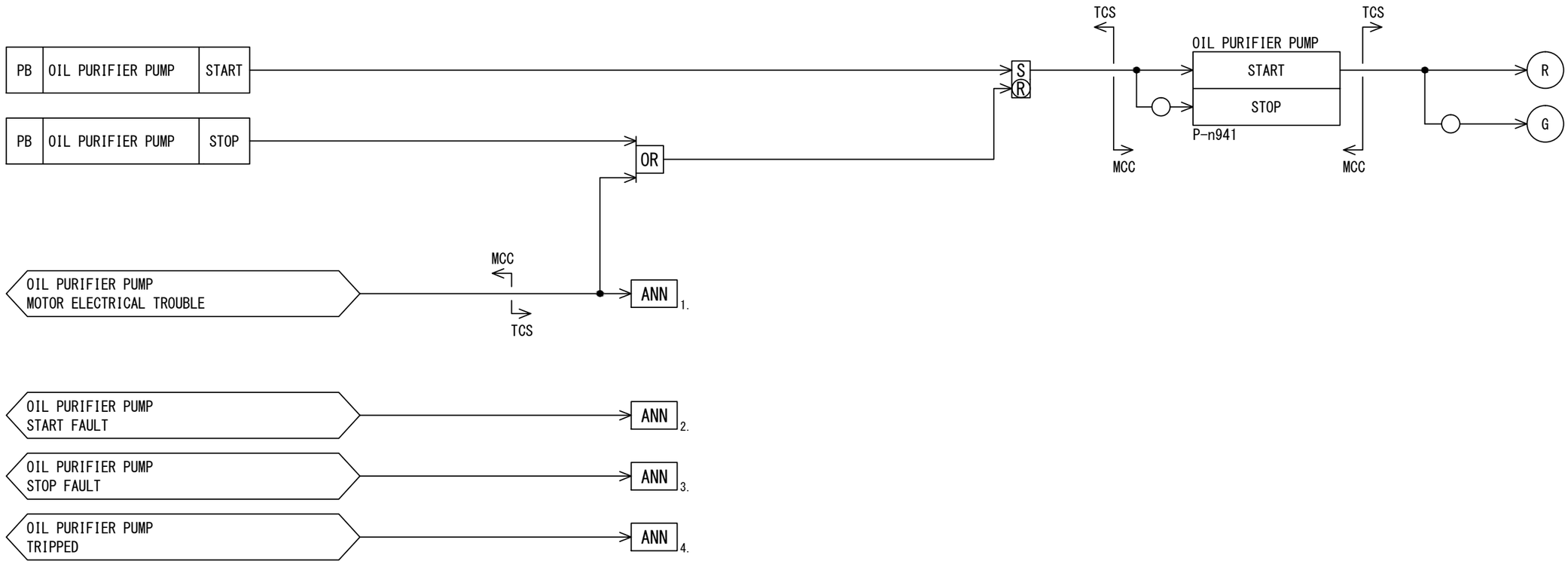


1. MAIN OIL TANK VAPOUR EXTRACTOR MOTOR ELECTRICAL TROUBLE
2. MAIN OIL TANK VAPOUR EXTRACTOR START FAULT
3. MAIN OIL TANK VAPOUR EXTRACTOR STOP FAULT
4. MAIN OIL TANK VAPOUR EXTRACTOR TRIPPED
5. MAIN OIL TANK VAPOUR EXTRACTOR STOPPED

TYPE –					PAGE TB-12	
SERVICE MAIN OIL TANK VAPOUR EXTRACTOR						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 60	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



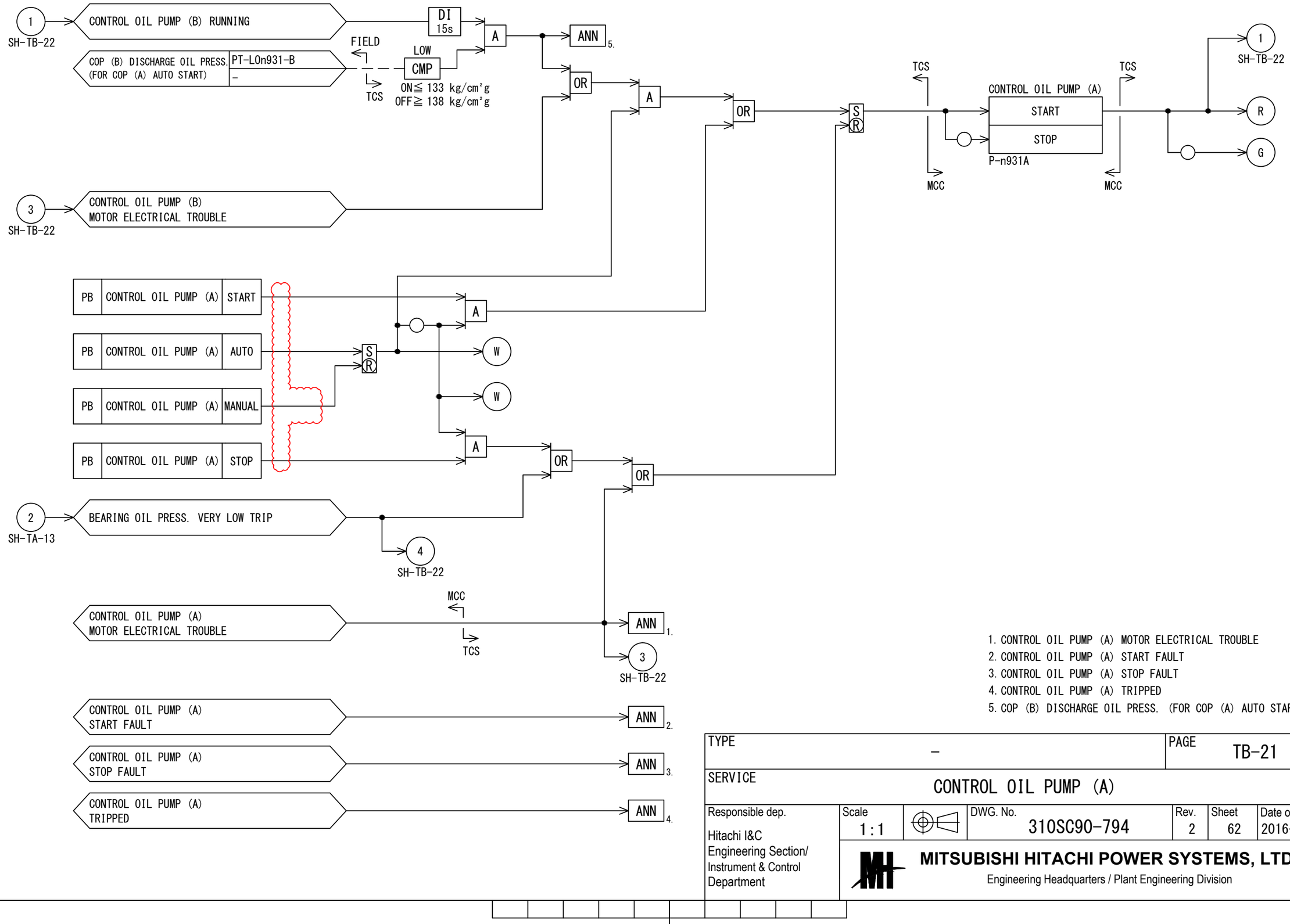
1. OIL PURIFIER PUMP MOTOR ELECTRICAL TROUBLE
2. OIL PURIFIER PUMP START FAULT
3. OIL PURIFIER PUMP STOP FAULT
4. OIL PURIFIER PUMP TRIPPED

TYPE	–	PAGE	TB-15
SERVICE	OIL PURIFIER PUMP		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	61	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

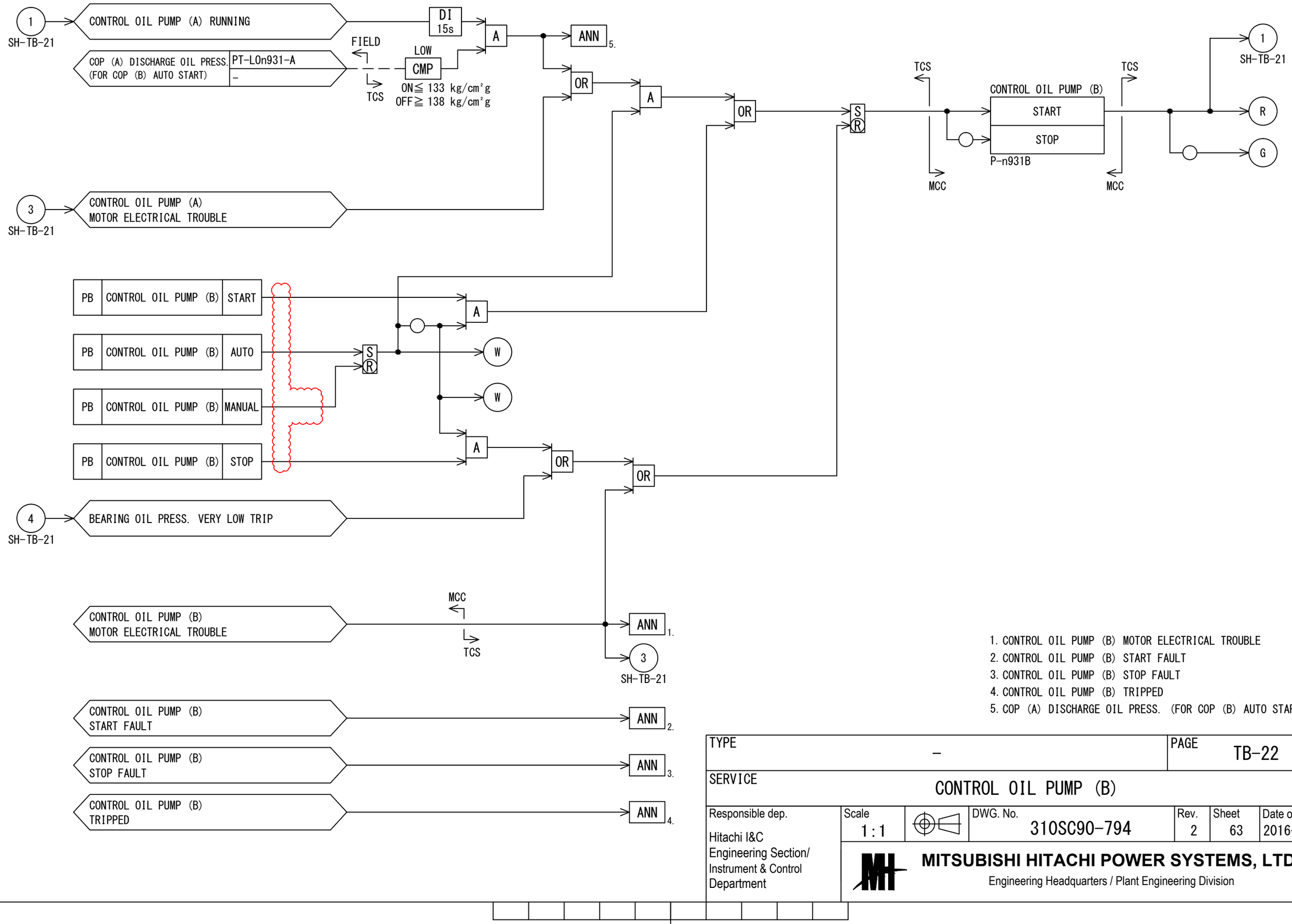


1. CONTROL OIL PUMP (A) MOTOR ELECTRICAL TROUBLE
2. CONTROL OIL PUMP (A) START FAULT
3. CONTROL OIL PUMP (A) STOP FAULT
4. CONTROL OIL PUMP (A) TRIPPED
5. COP (B) DISCHARGE OIL PRESS. (FOR COP (A) AUTO START) LOW

TYPE –						PAGE TB-21	
SERVICE CONTROL OIL PUMP (A)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794		Rev. 2	Sheet 62	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. CONTROL OIL PUMP (B) MOTOR ELECTRICAL TROUBLE
2. CONTROL OIL PUMP (B) START FAULT
3. CONTROL OIL PUMP (B) STOP FAULT
4. CONTROL OIL PUMP (B) TRIPPED
5. COP (A) DISCHARGE OIL PRESS. (FOR COP (B) AUTO START) LOW

TYPE –						PAGE TB-22	
SERVICE CONTROL OIL PUMP (B)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 63	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

4
SH-TC-2

52G OPEN

PB | MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE | OPEN

PB | MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE | AUTO

PB | MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE | MANUAL

PB | MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE | CLOSE

S
R

A

A

A

A

W

W

OR

OR

PO
3s

PO
3s

TCS

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE

OPEN

CLOSE

MV-STn002A

TCS

TCS

MCC

MOV

1
SH-TC-1

G

R

ICV TO CV TRANSFER COMPLETED

TCS

DCS

SH-

5
SH-TC-2/TC-5/TC-7/TC-8/TA-31/TA-32/TA-33/
TA-34/TA-37/TA-38/TA-41/TA-42

1
SH-TC-1
MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
FULL OPEN

1
SH-TC-2
MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE
FULL OPEN

1
SH-TC-7
CV(A) ABOVE SEAT SECONDARY DRAIN VALVE
FULL OPEN

1
SH-TC-8
CV(B) ABOVE SEAT SECONDARY DRAIN VALVE
FULL OPEN

1
SH-TC-12
CRV(A) BELOW SEAT SECONDARY DRAIN VALVE
FULL OPEN

1
SH-TC-13
CRV(B) BELOW SEAT SECONDARY DRAIN VALVE
FULL OPEN

A

ATS
(CRI:1.25)

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
OPEN FAULT

ANN 1.

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
CLOSE FAULT

ANN 2.

TCS

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
MOTOR ELECTRICAL TROUBLE

MCC

ANN 3.

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
TORQUE SWITCH OPEN

ANN 4.

MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE
TORQUE SWITCH CLOSE

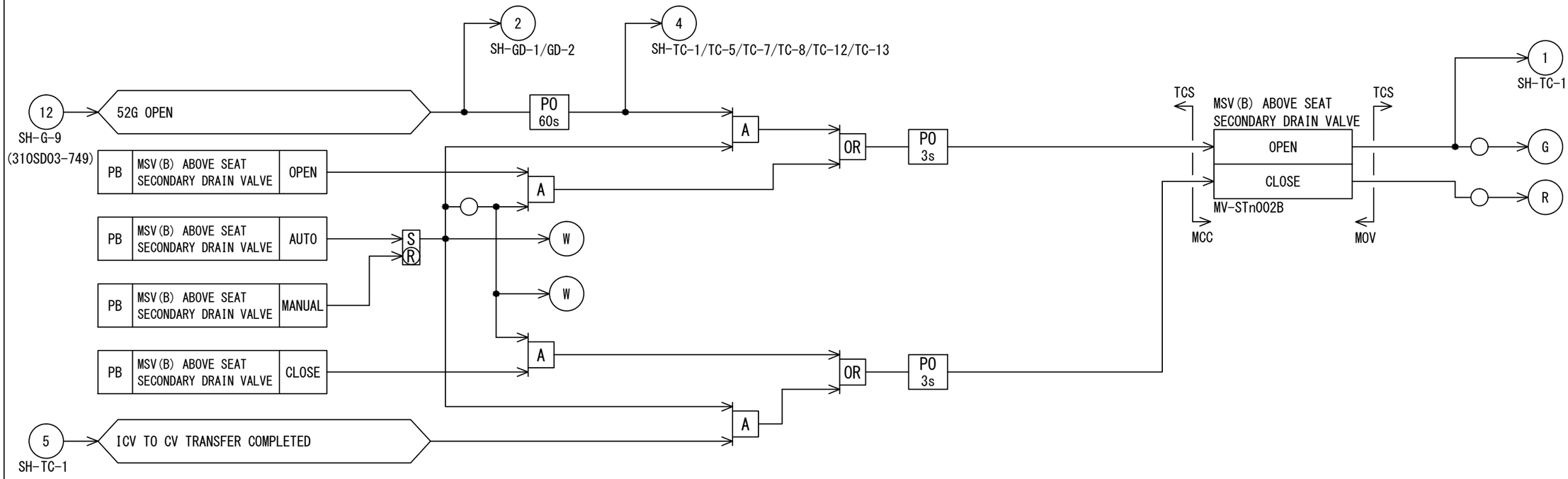
ANN 5.

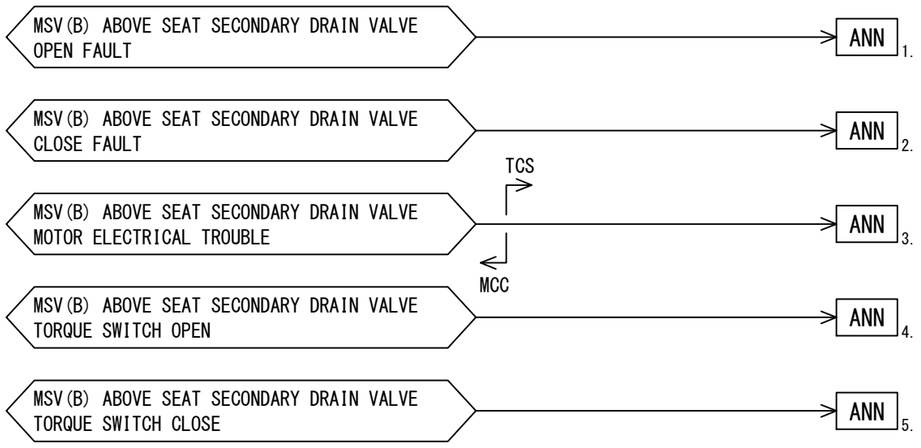
1. MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE	TORQUE SEAT (SEAL-IN)					PAGE	TC-1
SERVICE	MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	64	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



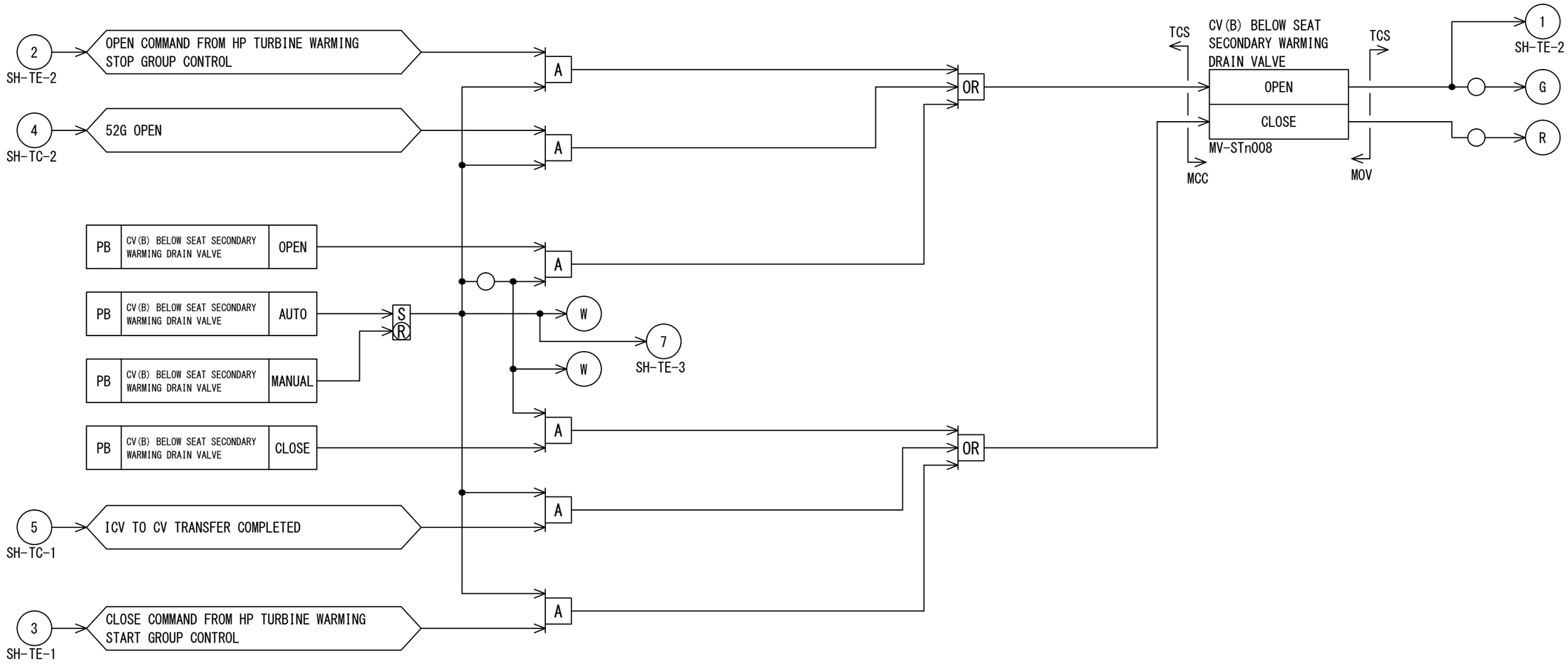


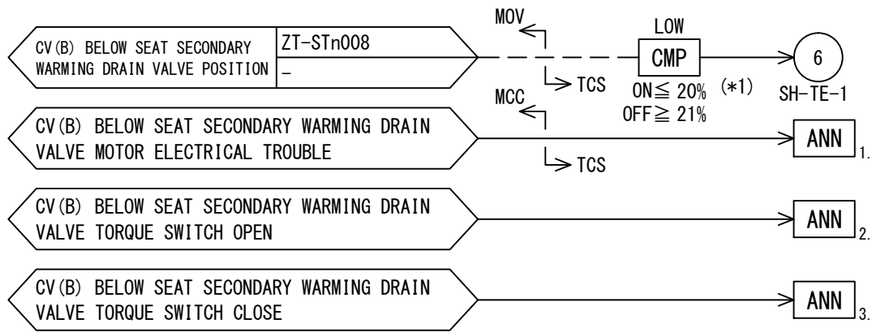
1. MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE	TORQUE SEAT (SEAL-IN)				PAGE	TC-2
SERVICE	MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	65	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





1. CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE MOTOR ELECTRICAL TROUBLE
2. CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE TORQUE SWITCH OPEN
3. CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE TORQUE SWITCH CLOSE

NOTE:
(*1) SETPOINT WILL BE ADJUSTED AT COMMISSIONING STAGE.

TYPE: TORQUE SEAT (INCHING)					PAGE: TC-5	
SERVICE: CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 66	Date of issue: 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

CV CHEST WARMING MODE SET

PO 30s

7 SH-TA-31 — MSV(A) FULL OPEN

5 SH-TA-32 — MSV(B) FULL OPEN

4 SH-TA-14 — TURBINE TRIP OIL PRESS. VERY LOW TRIP

4 SH-TC-2 — 52G OPEN

A — DI 5s — A — PO 60s — 4 SH-TC-8

OR — A — OR — PO 3s

TCS — CV(A) ABOVE SEAT SECONDARY DRAIN VALVE — TCS

OPEN — 1 SH-TC-1 — G

CLOSE — R

MV-STn004A

MCC — MOV

PB | CV(A) ABOVE SEAT SECONDARY DRAIN VALVE | OPEN → A

PB | CV(A) ABOVE SEAT SECONDARY DRAIN VALVE | AUTO → S/R → W

PB | CV(A) ABOVE SEAT SECONDARY DRAIN VALVE | MANUAL → W

PB | CV(A) ABOVE SEAT SECONDARY DRAIN VALVE | CLOSE → A → OR → PO 3s

5 SH-TC-1 — ICV TO CV TRANSFER COMPLETED → A

CV CHEST WARMING COMPLETE → A → OR → A → OR

5 SH-TC-8

CV(A) ABOVE SEAT SECONDARY DRAIN VALVE OPEN FAULT → ANN 1.

CV(A) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE FAULT → ANN 2.

CV(A) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE (TCS / MCC) → ANN 3.

CV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN → ANN 4.

CV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE → ANN 5.

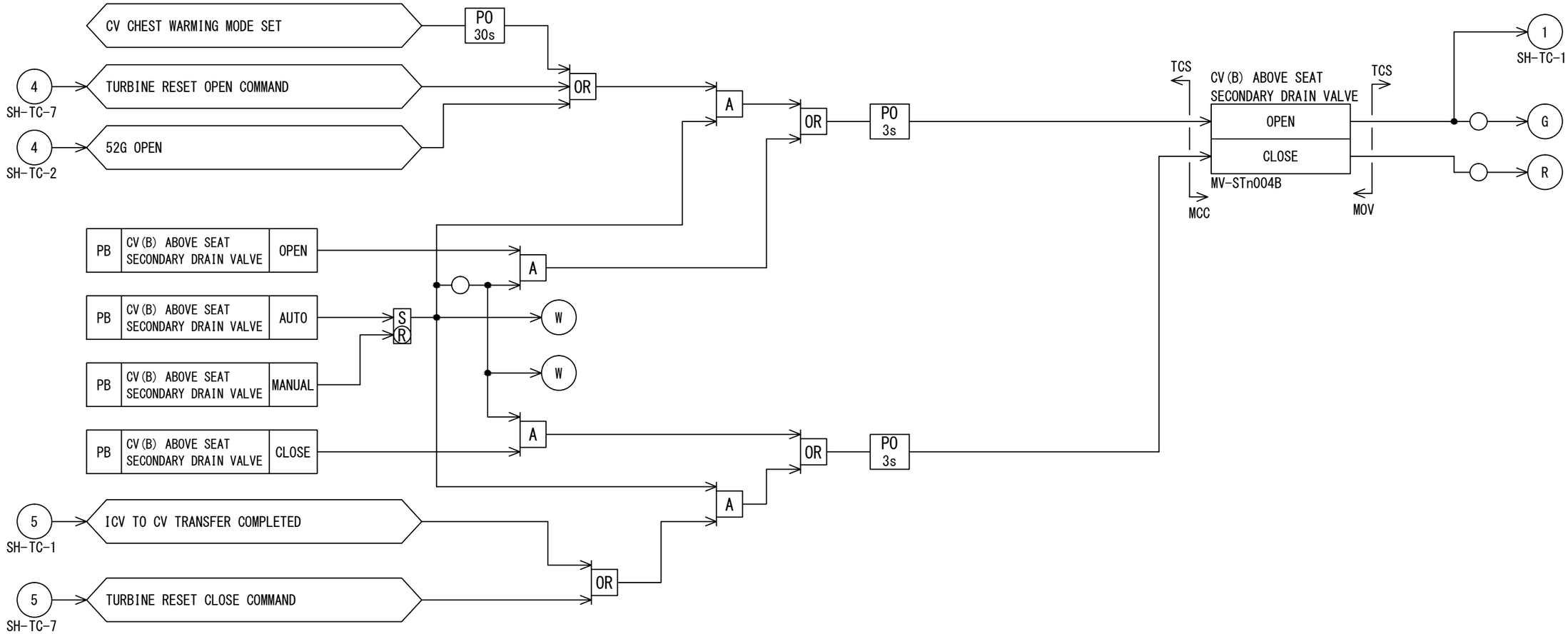
1. CV(A) ABOVE SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. CV(A) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. CV(A) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. CV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. CV(A) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE	TORQUE SEAT (SEAL-IN)	PAGE	TC-7
SERVICE	CV(A) ABOVE SEAT SECONDARY DRAIN VALVE		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	67	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

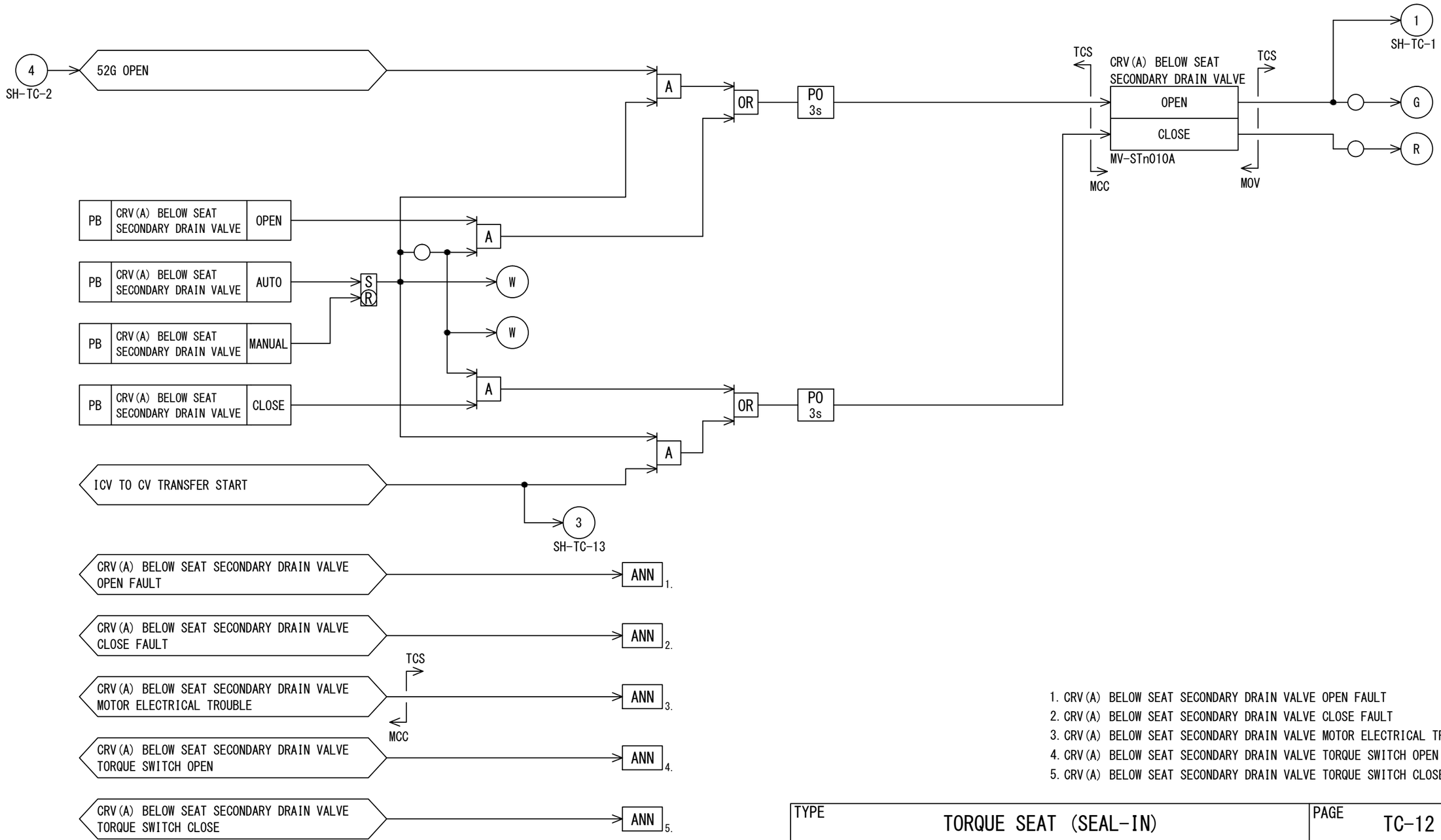


1. CV(B) ABOVE SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. CV(B) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. CV(B) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. CV(B) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. CV(B) ABOVE SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE	TORQUE SEAT (SEAL-IN)					PAGE	TC-8
SERVICE	CV(B) ABOVE SEAT SECONDARY DRAIN VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 68	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

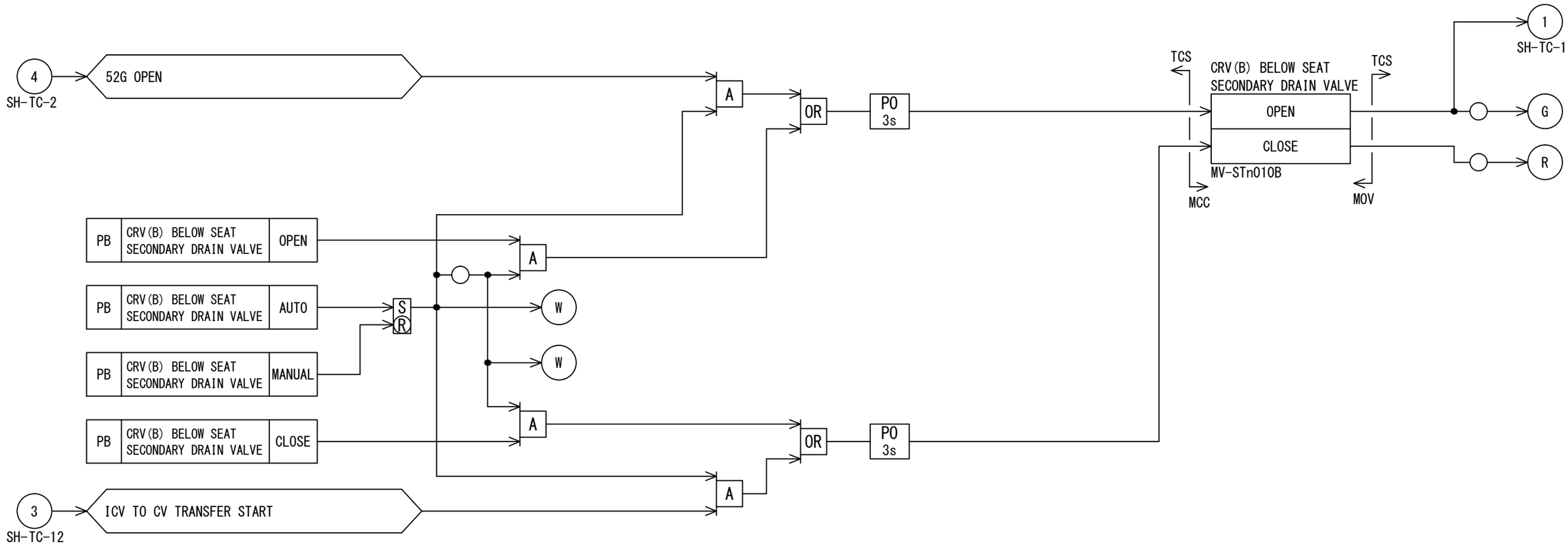


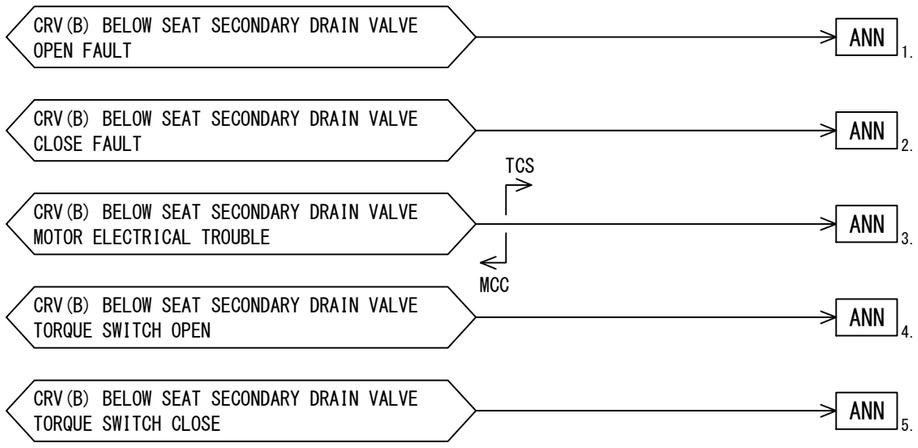
1. CRV(A) BELOW SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. CRV(A) BELOW SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. CRV(A) BELOW SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. CRV(A) BELOW SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. CRV(A) BELOW SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE	TORQUE SEAT (SEAL-IN)					PAGE	TC-12
SERVICE	CRV(A) BELOW SEAT SECONDARY DRAIN VALVE						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	69	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



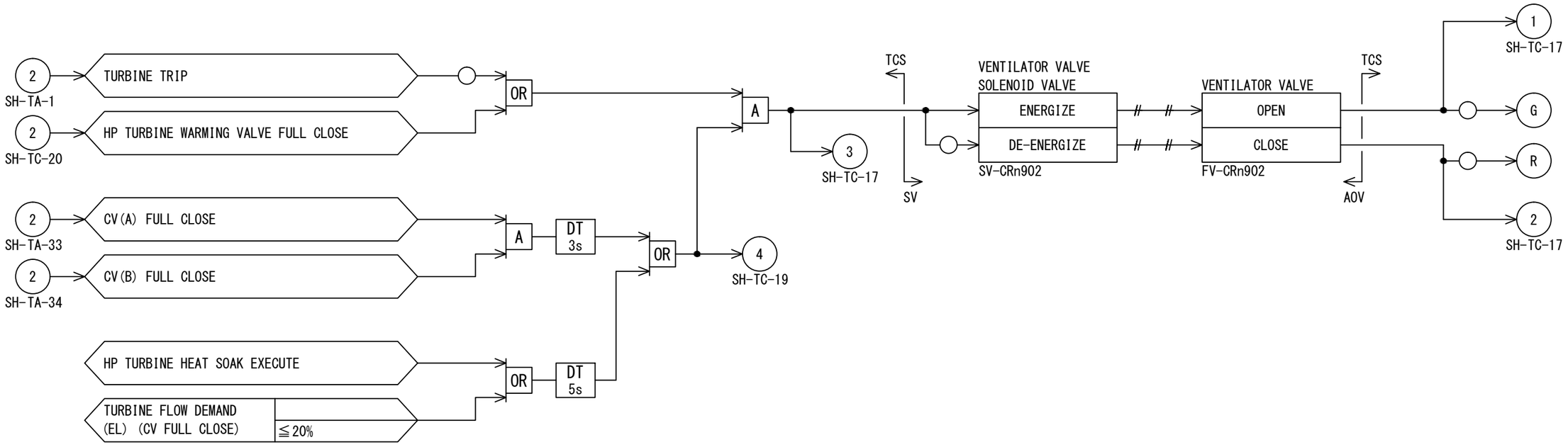


1. CRV(B) BELOW SEAT SECONDARY DRAIN VALVE OPEN FAULT
2. CRV(B) BELOW SEAT SECONDARY DRAIN VALVE CLOSE FAULT
3. CRV(B) BELOW SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE
4. CRV(B) BELOW SEAT SECONDARY DRAIN VALVE TORQUE SWITCH OPEN
5. CRV(B) BELOW SEAT SECONDARY DRAIN VALVE TORQUE SWITCH CLOSE

TYPE: TORQUE SEAT (SEAL-IN)					PAGE: TC-13	
SERVICE: CRV(B) BELOW SEAT SECONDARY DRAIN VALVE						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 70	Date of issue: 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

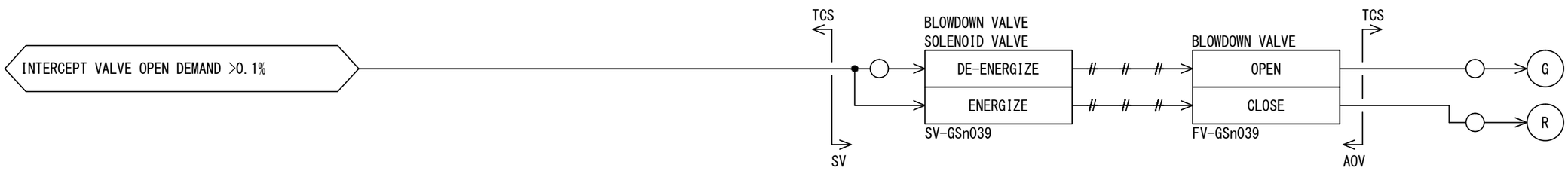


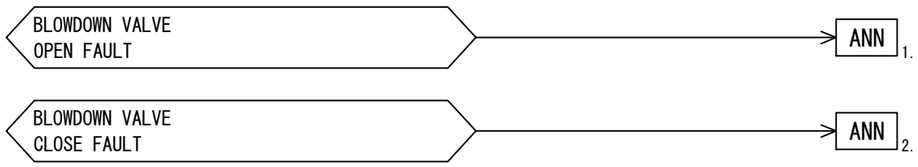
1. VENTILATOR VALVE OPEN FAULT
2. VENTILATOR VALVE CLOSE FAULT
3. VENTILATOR VALVE TROUBLE

TYPE –					PAGE TC-17	
SERVICE VENTILATOR VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 71	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



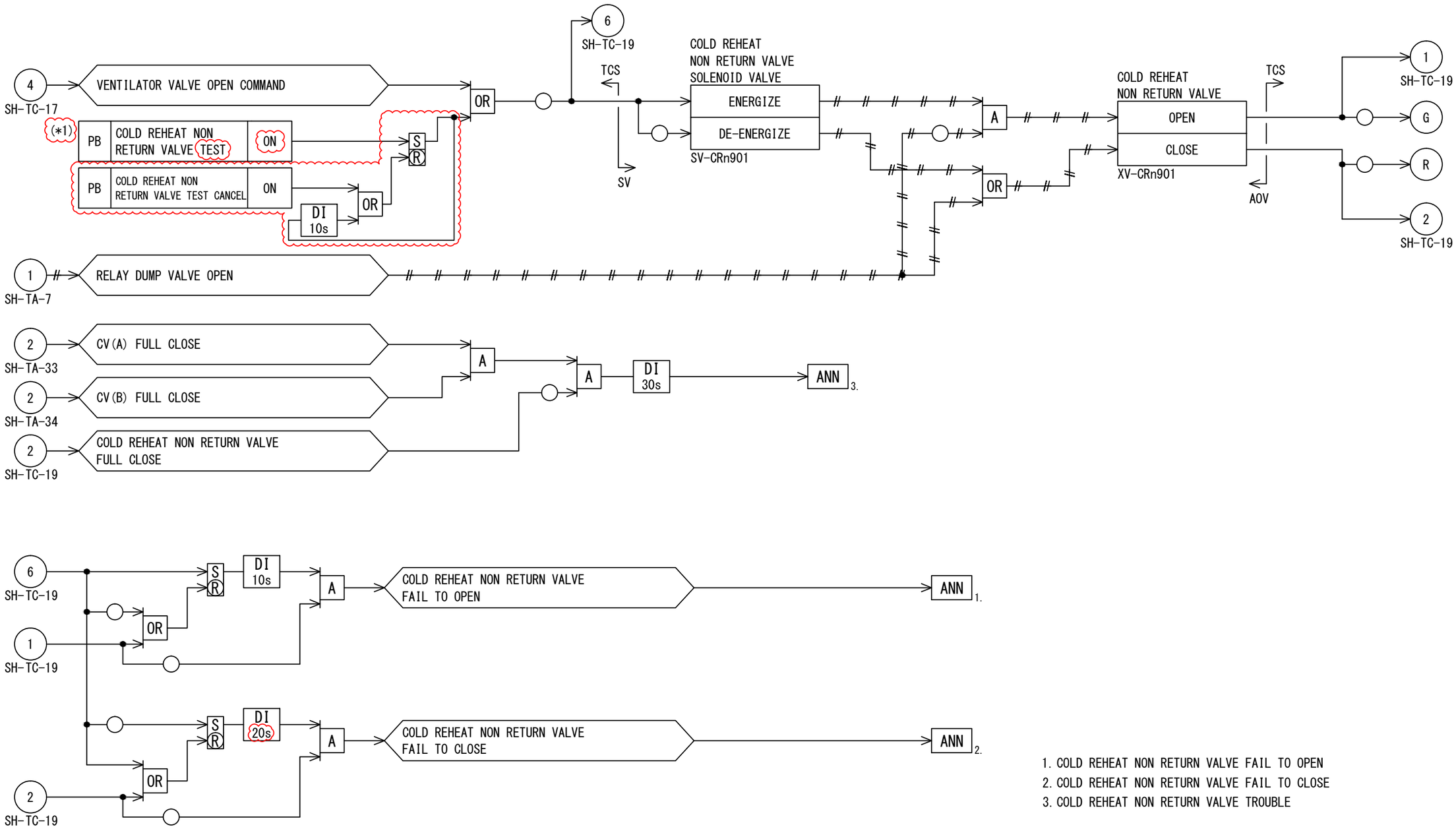


1. BLOWDOWN VALVE OPEN FAULT
2. BLOWDOWN VALVE CLOSE FAULT

TYPE –					PAGE TC-18	
SERVICE BLOWDOWN VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 72	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. COLD REHEAT NON RETURN VALVE FAIL TO OPEN
2. COLD REHEAT NON RETURN VALVE FAIL TO CLOSE
3. COLD REHEAT NON RETURN VALVE TROUBLE

NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

TYPE	–					PAGE	TC-19
SERVICE	COLD REHEAT NON RETURN VALVE						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-794	2	73	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

AUX. STEAM HEADER PRESS. (LOWER)
DCS
TCS
NORMAL
CMP
ON≧ 7.0 kg/cm²g
OFF≦ 6.0 kg/cm²g
HP TURBINE WARMING PERMISSIVE
7
SH-TC-20/TE-1

4
SH-TA-14
TURBINE TRIP OIL PRESS. VERY LOW TRIP

PB | HP TURBINE WARMING VALVE | OPEN

1
SH-TE-1
OPEN COMMAND FROM HP TURBINE WARMING START GROUP CONTROL

3
SH-TA-21
HP TURBINE EXHAUST STEAM PRESS. (MEDIAN)

LOW
CMP
ON≦ 4.0 kg/cm²g
OFF≧ 5.0 kg/cm²g

5
SH-TC-20
HP TURBINE WARMING VALVE POSITION

CMP
ON≦ 40%
OFF≧ 41%

PB | HP TURBINE WARMING VALVE | PROGRAM CONTROL

6
SH-TA-10
CONDENSER VACUUM LOW

4
SH-TE-1
PROGRAM CONTROL START COMMAND FROM HP TURBINE WARMING START GROUP CONTROL

PB | HP TURBINE WARMING VALVE | AUTO

PB | HP TURBINE WARMING VALVE | MANUAL

7
SH-TC-20
HP TURBINE WARMING PERMISSIVE

3
SH-TE-2
CLOSED COMMAND FROM HP TURBINE WARMING STOP GROUP CONTROL

PB | HP TURBINE WARMING VALVE | CLOSE

3
SH-TA-21
HP TURBINE EXHAUST STEAM PRESS. (MEDIAN)

HIGH
CMP
ON≧ 5.0 kg/cm²g
OFF≦ 4.0 kg/cm²g

HP TURBINE WARMING VALVE POSITION | ZT-ASn011 | –

TCS
MOV

CMP
ON≧ 5%
OFF≦ 4%

CMP
ON≧ 10%
OFF≦ 9%

5
SH-TC-20

6
SH-TE-1

A, A, A, OR

(*1)
FLK
T1 : 1s
T2 : 15s

4
SH-TE-1

W, W, W

OR, S/R

8
SH-TE-3

A, A, OR

A
FLK
T1 : 1s
T2 : 15s
(*1)

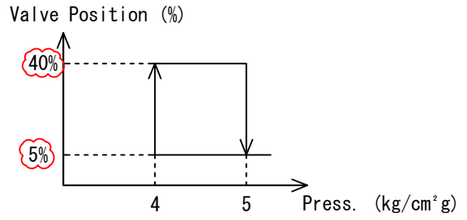
TCS
HP TURBINE WARMING VALVE
OPEN
CLOSE
MV-ASn011
MCC
MOV
TCS

G
R
OPI
2
SH-TC-17/TE-2

TCS
HP TURBINE WARMING VALVE MOTOR ELECTRICAL TROUBLE
MCC
ANN 1.

HP TURBINE WARMING VALVE TORQUE SWITCH OPEN
ANN 2.

HP TURBINE WARMING VALVE TORQUE SWITCH CLOSE
ANN 3.



1. HP TURBINE WARMING VALVE MOTOR ELECTRICAL TROUBLE
2. HP TURBINE WARMING VALVE TORQUE SWITCH OPEN
3. HP TURBINE WARMING VALVE TORQUE SWITCH CLOSE

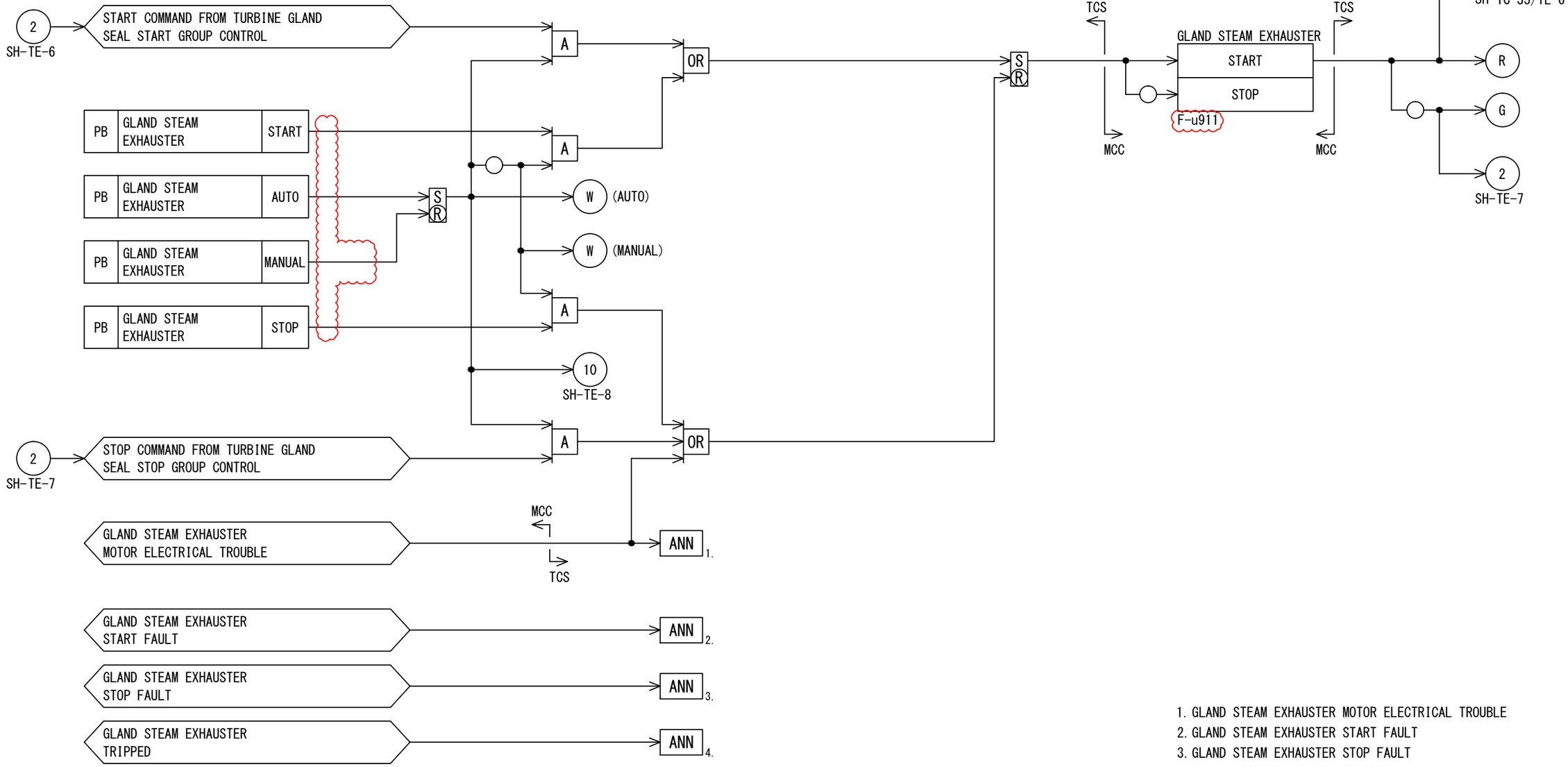
NOTE:
(*1) THIS TIMER SHOULD BE ADJUSTED AT COMMISSIONING STAGE ACCORDING TO THE WARMING STEAM PRESSURE.

TYPE	TORQUE SEAT (INCHING)	PAGE	TC-20
SERVICE	HP TURBINE WARMING VALVE		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	74	2016-07-18

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Engineering Headquarters / Plant Engineering Division

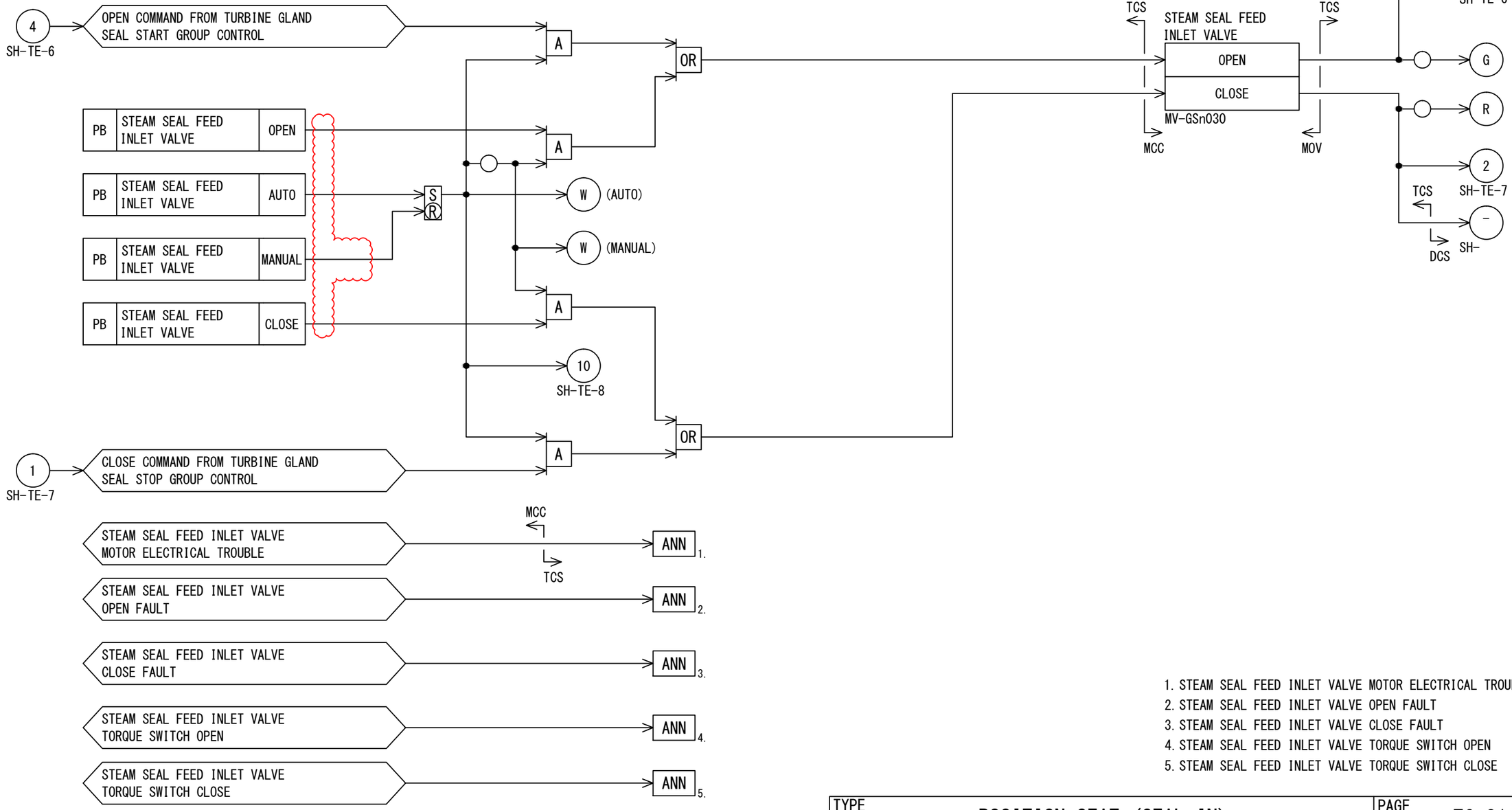
DWG. No. 310SC90-794



1. GLAND STEAM EXHAUSTER MOTOR ELECTRICAL TROUBLE
2. GLAND STEAM EXHAUSTER START FAULT
3. GLAND STEAM EXHAUSTER STOP FAULT
4. GLAND STEAM EXHAUSTER TRIPPED

TYPE –						PAGE TC-28	
SERVICE GLAND STEAM EXHAUSTER							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 75	Date of issue 2016-07-18	
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division							

DWG. No. 310SC90-794

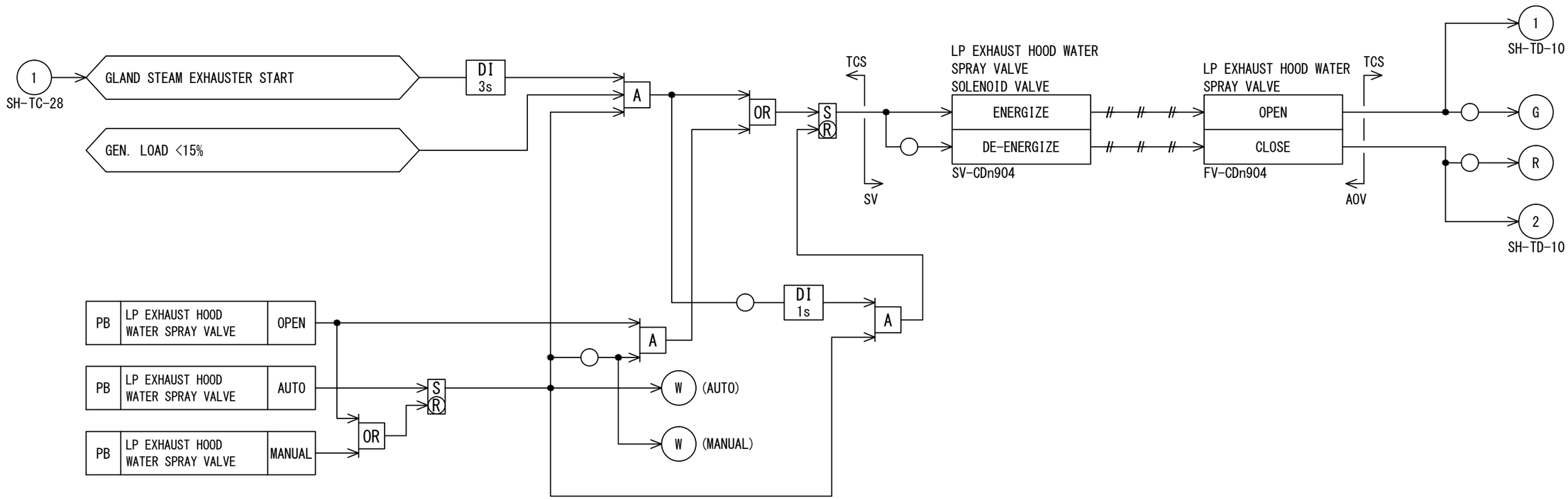


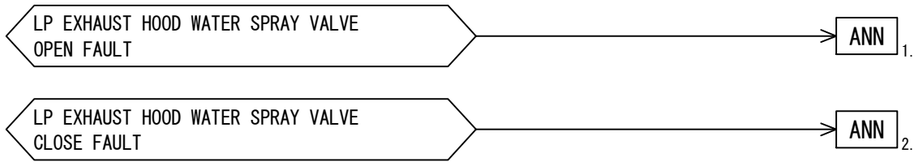
1. STEAM SEAL FEED INLET VALVE MOTOR ELECTRICAL TROUBLE
2. STEAM SEAL FEED INLET VALVE OPEN FAULT
3. STEAM SEAL FEED INLET VALVE CLOSE FAULT
4. STEAM SEAL FEED INLET VALVE TORQUE SWITCH OPEN
5. STEAM SEAL FEED INLET VALVE TORQUE SWITCH CLOSE

TYPE: POSITION SEAT (SEAL-IN)					PAGE: TC-31	
SERVICE: STEAM SEAL FEED INLET VALVE						
Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department	Scale: 1:1		DWG. No.: 310SC90-794	Rev.: 2	Sheet: 76	Date of issue: 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

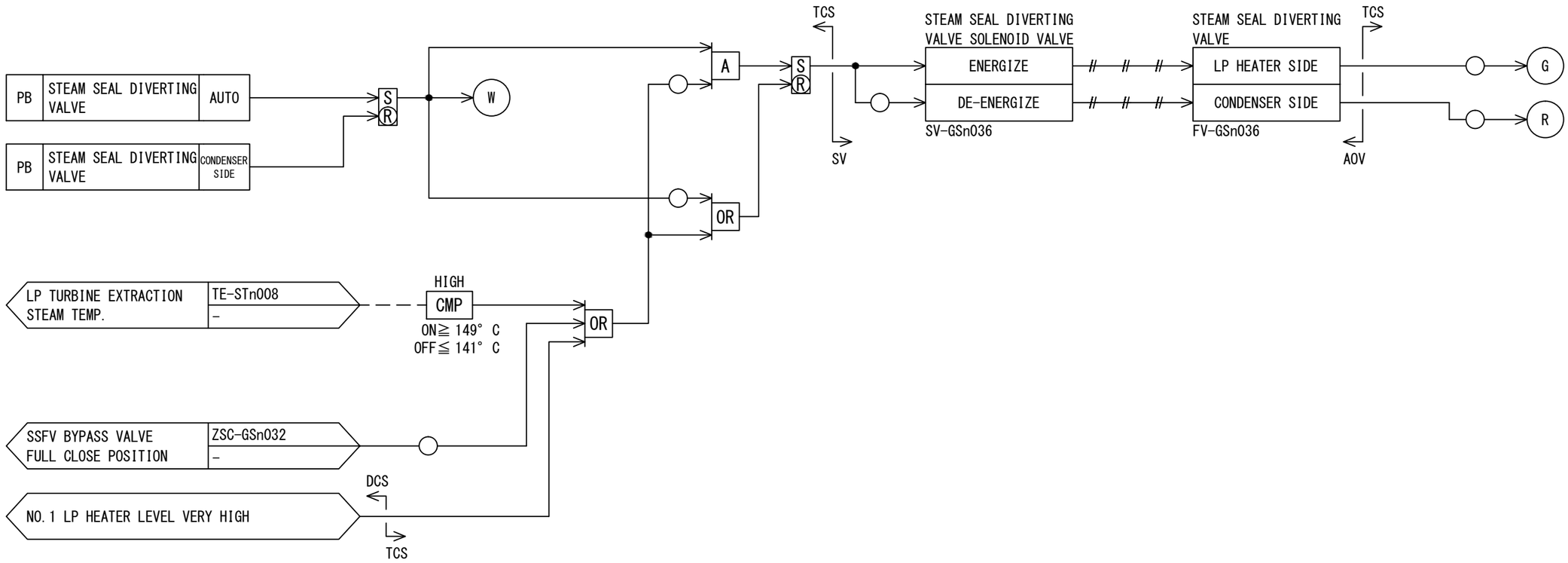




1. LP EXHAUST HOOD WATER SPRAY VALVE OPEN FAULT
2. LP EXHAUST HOOD WATER SPRAY VALVE CLOSE FAULT

TYPE –					PAGE TC-35	
SERVICE LP EXHAUST HOOD WATER SPRAY VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 77	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794

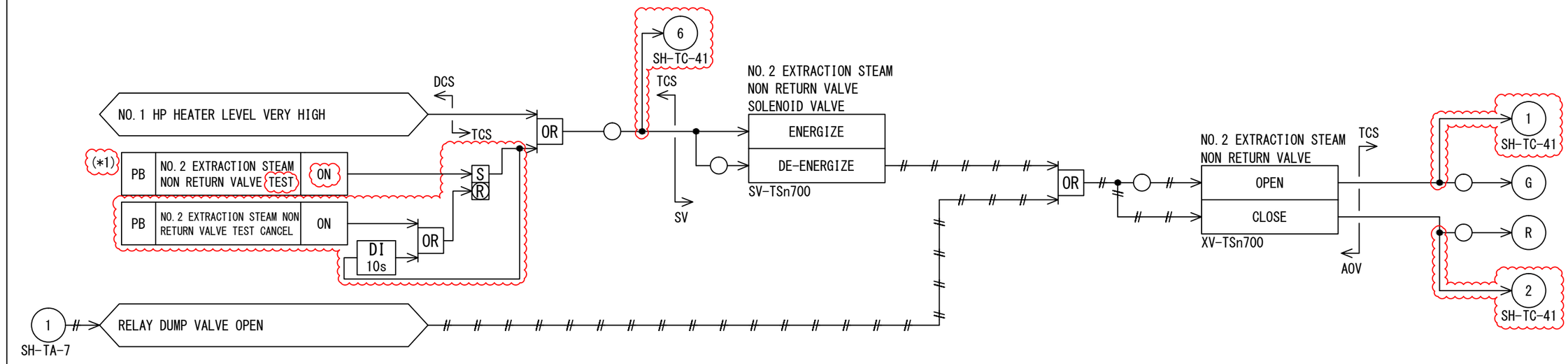


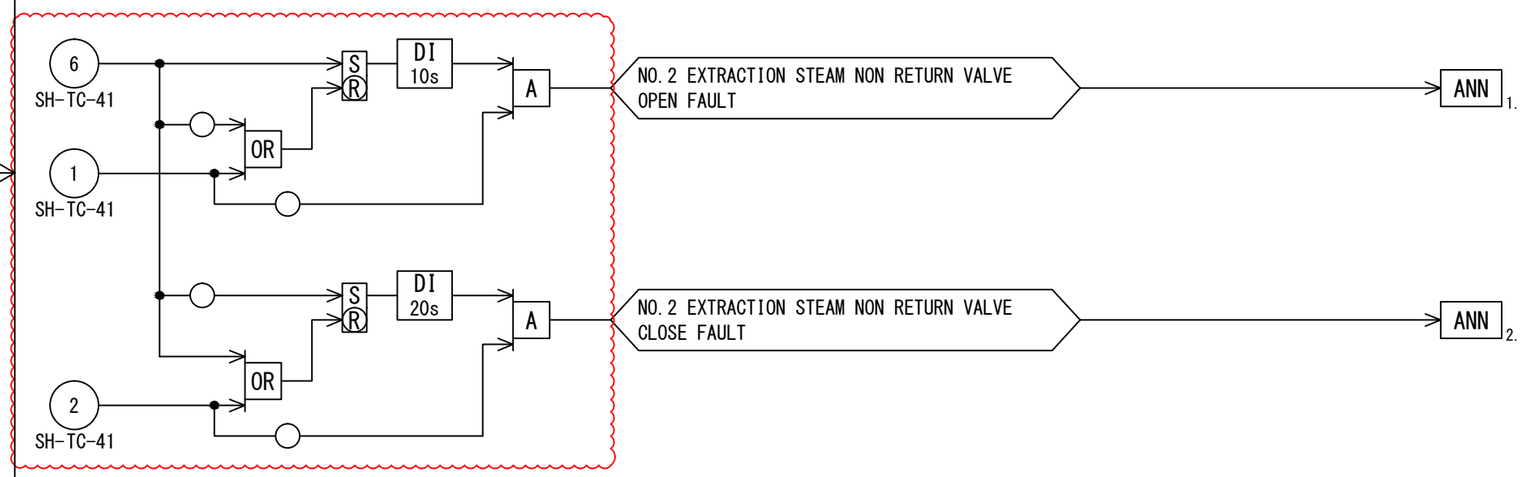
1. STEAM SEAL DIVERTING VALVE OPEN FAULT
2. STEAM SEAL DIVERTING VALVE CLOSE FAULT

TYPE –					PAGE TC-36	
SERVICE STEAM SEAL DIVERTING VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 78	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



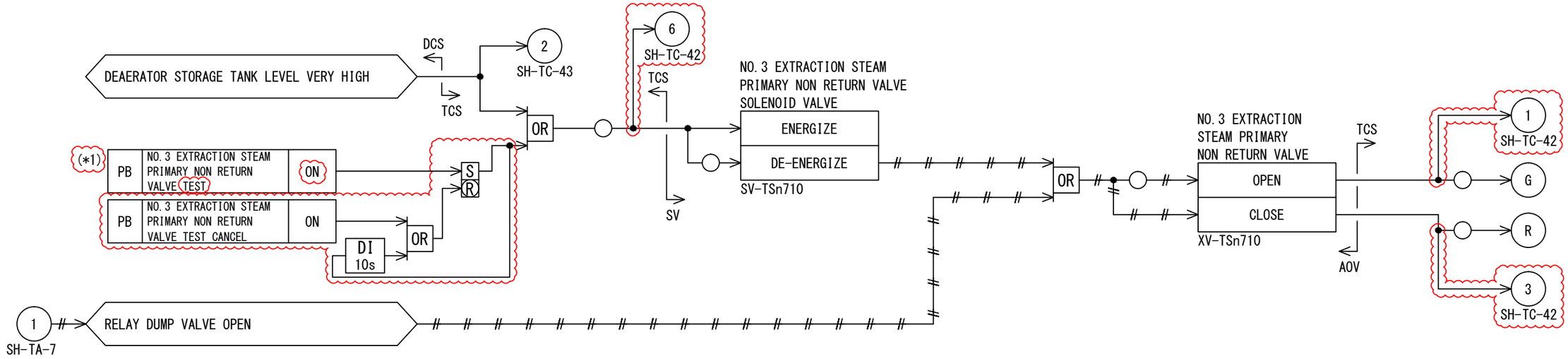


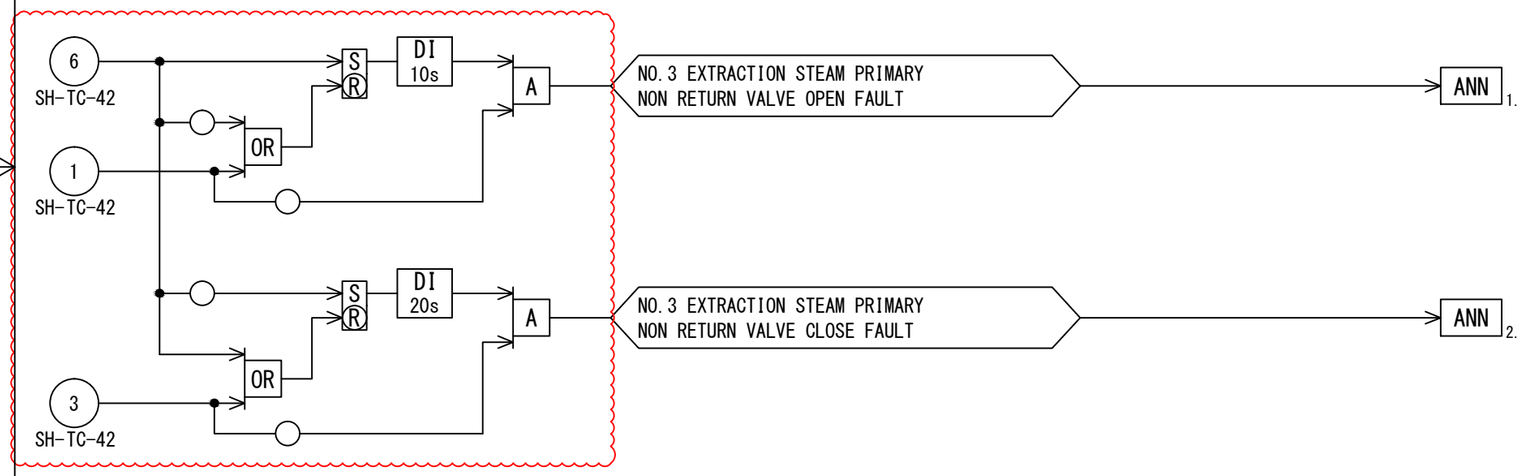
1. NO. 2 EXTRACTION STEAM NON RETURN VALVE OPEN FAULT
2. NO. 2 EXTRACTION STEAM NON RETURN VALVE CLOSE FAULT

NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

TYPE –					PAGE TC-41	
SERVICE NO. 2 EXTRACTION STEAM NON RETURN VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1 : 1		DWG. No. 310SC90-794	Rev. 2	Sheet 79	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794





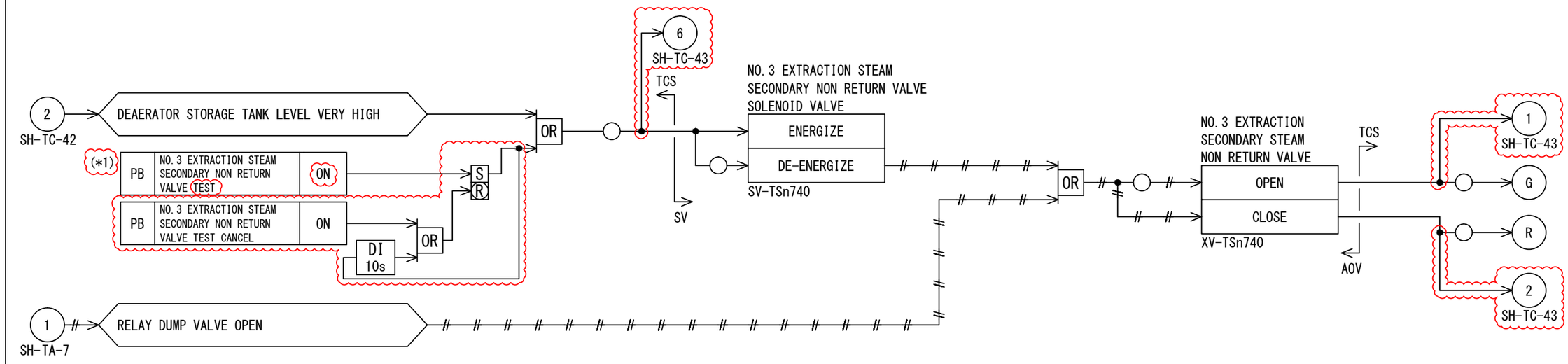
NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

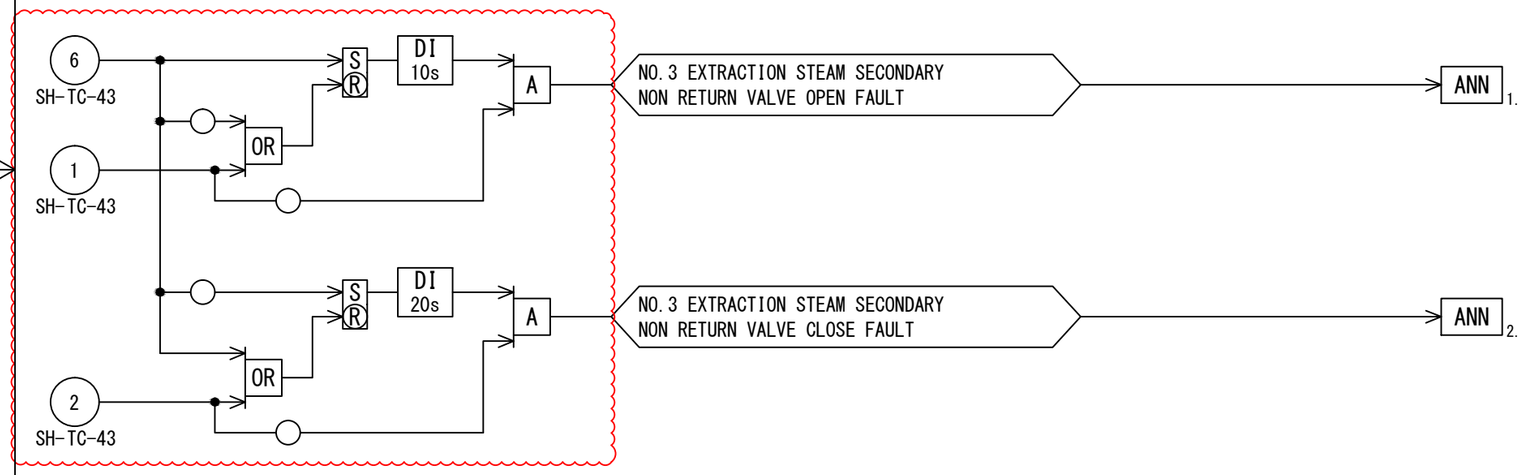
1. NO.3 EXTRACTION STEAM PRIMARY NON RETURN VALVE OPEN FAULT
2. NO.3 EXTRACTION STEAM PRIMARY NON RETURN VALVE CLOSE FAULT

TYPE	–				PAGE	TC-42
SERVICE	NO.3 EXTRACTION STEAM PRIMARY NON RETURN VALVE					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	80	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





1. NO. 3 EXTRACTION STEAM SECONDARY NON RETURN VALVE OPEN FAULT
2. NO. 3 EXTRACTION STEAM SECONDARY NON RETURN VALVE CLOSE FAULT

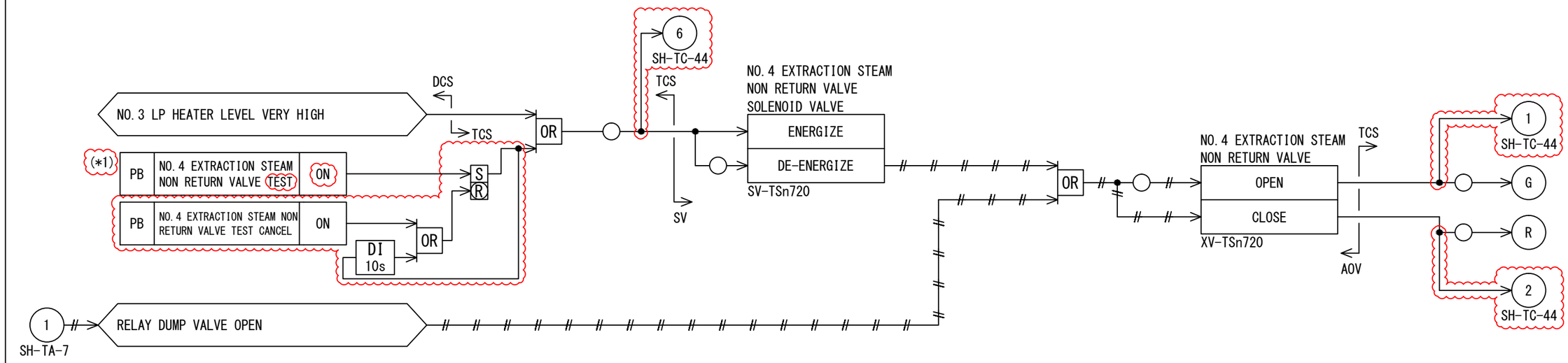
NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

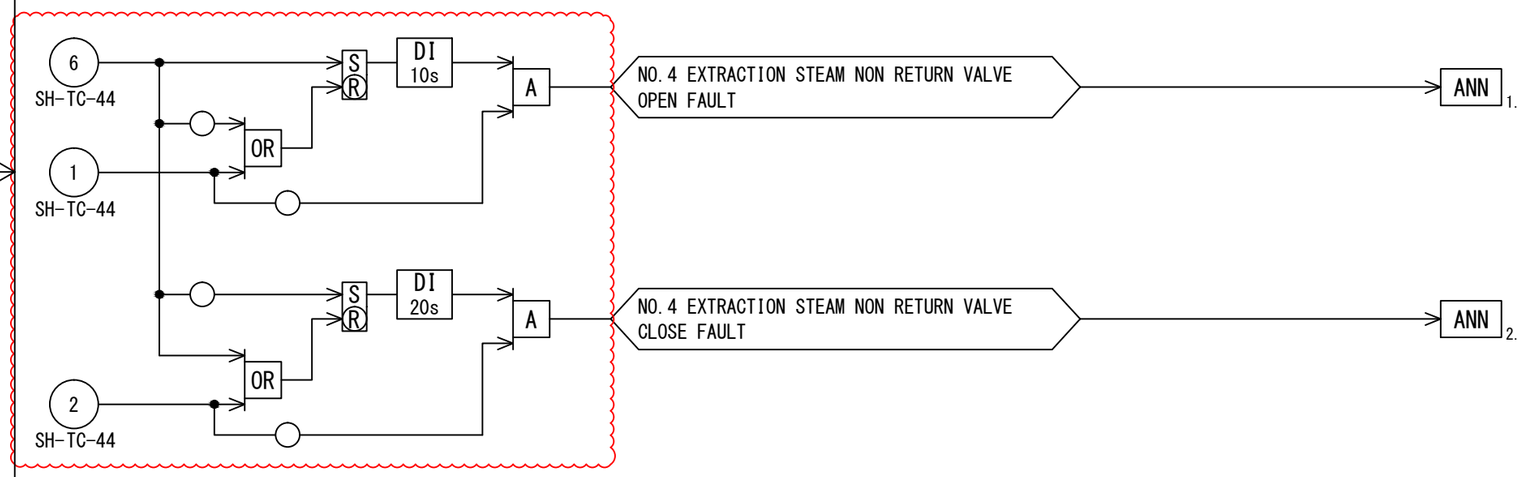
TYPE	–	PAGE	TC-43
SERVICE	NO. 3 EXTRACTION STEAM SECONDARY NON RETURN VALVE		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	81	2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





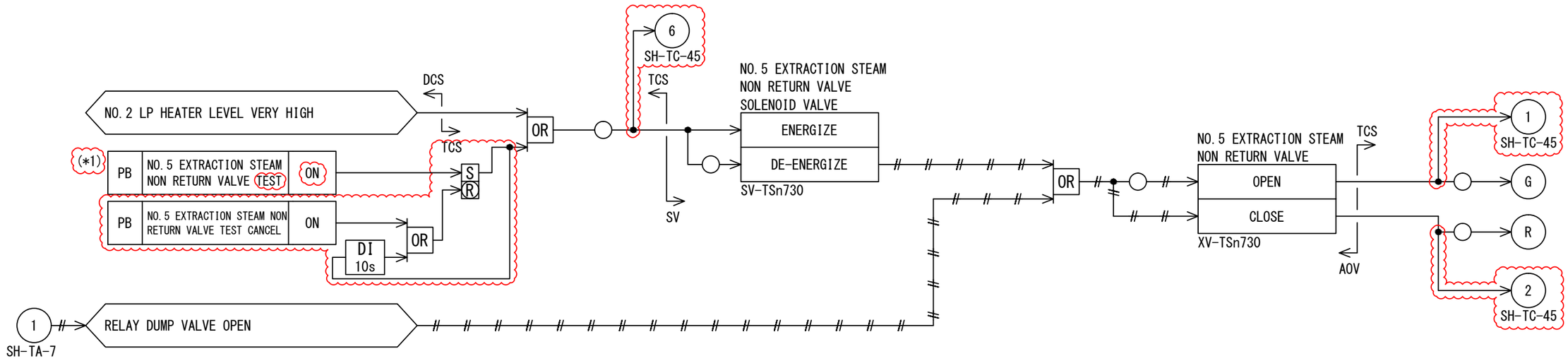
1. NO. 4 EXTRACTION STEAM NON RETURN VALVE OPEN FAULT
2. NO. 4 EXTRACTION STEAM NON RETURN VALVE CLOSE FAULT

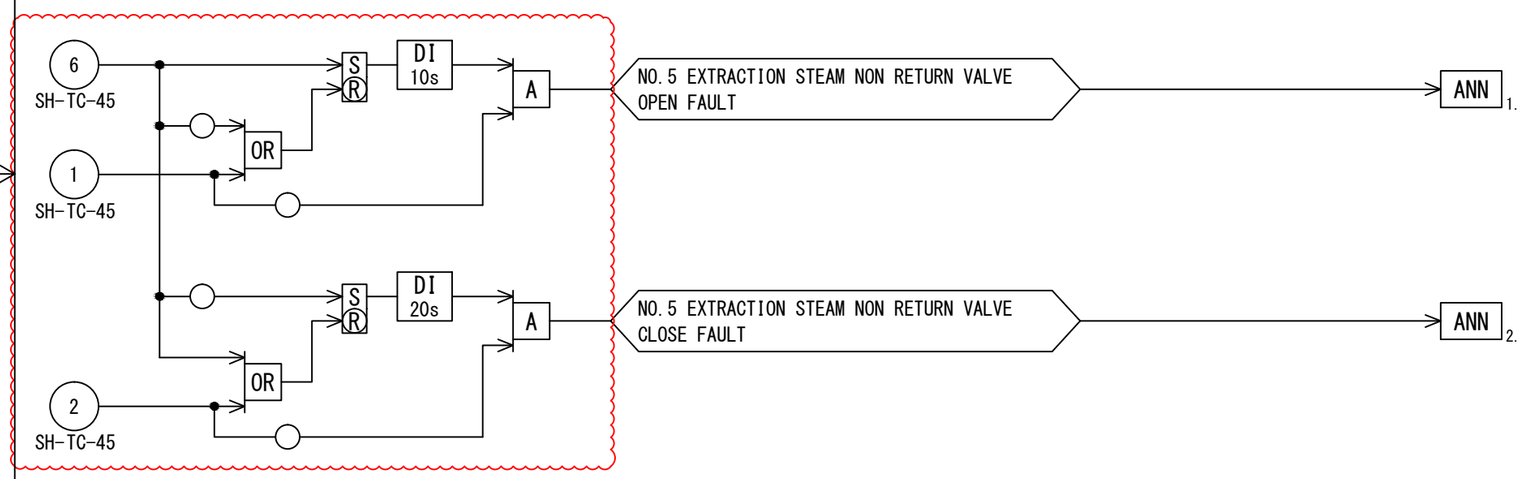
NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

TYPE –						PAGE TC-44	
SERVICE NO. 4 EXTRACTION STEAM NON RETURN VALVE							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 82	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794





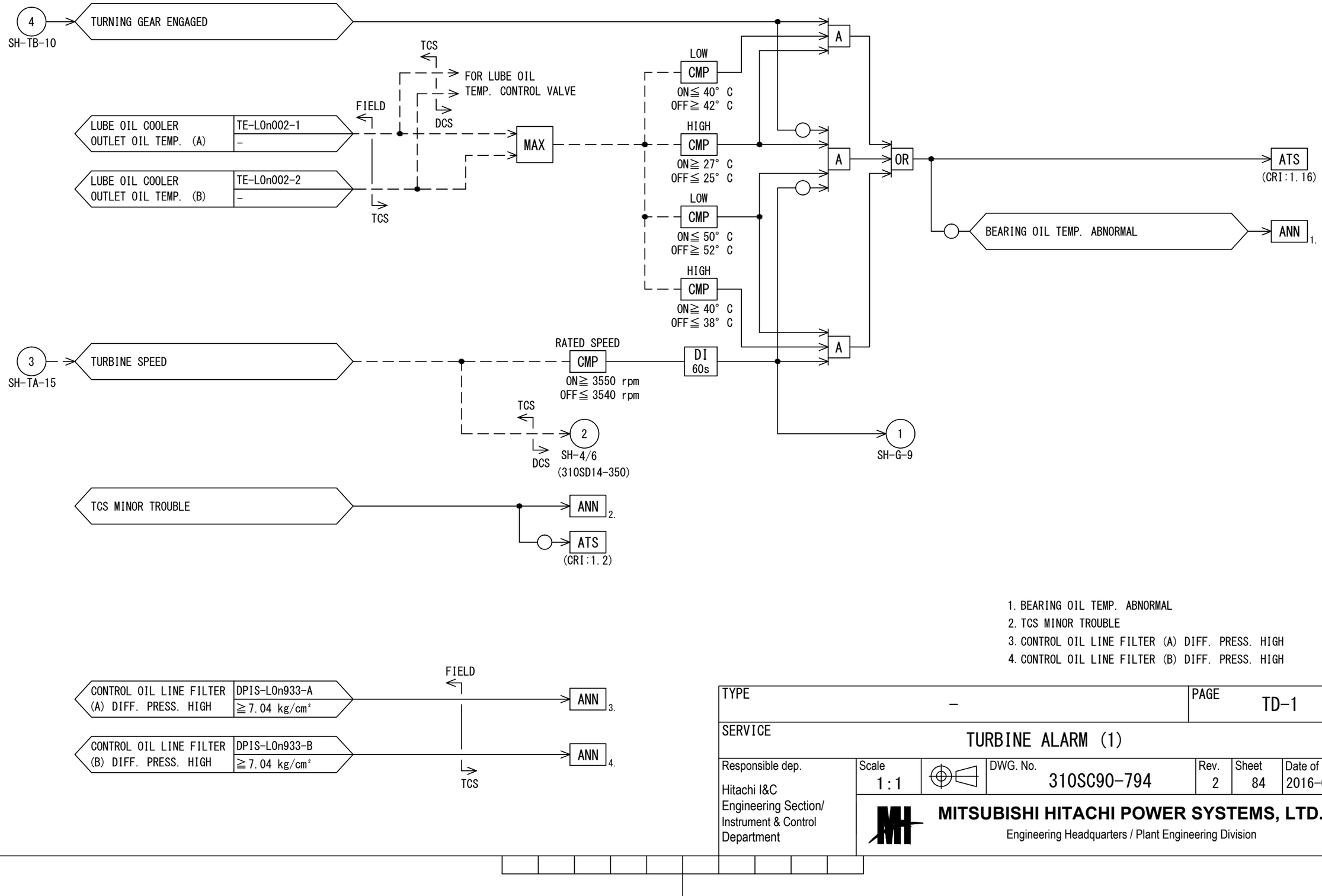
1. NO. 5 EXTRACTION STEAM NON RETURN VALVE OPEN FAULT
2. NO. 5 EXTRACTION STEAM NON RETURN VALVE CLOSE FAULT

NOTE:
(*1) DURING THE VALVE TEST, PLEASE CHECK THAT THE VALVE CLOSES PARTIALLY BY THE INDICATION OF BOTH GREEN AND RED LIGHT AND THEN CANCEL THE VALVE TEST.
FOR THE DETAIL OF VALVE TEST, PLEASE REFER TO "INSTRUCTION MANUAL FOR STEAM TURBINE AND AUXILIARIES (OPERATION MANUAL) (TEQ-15-228) SECTION 8"

TYPE –					PAGE TC-45	
SERVICE NO. 5 EXTRACTION STEAM NON RETURN VALVE						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 83	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

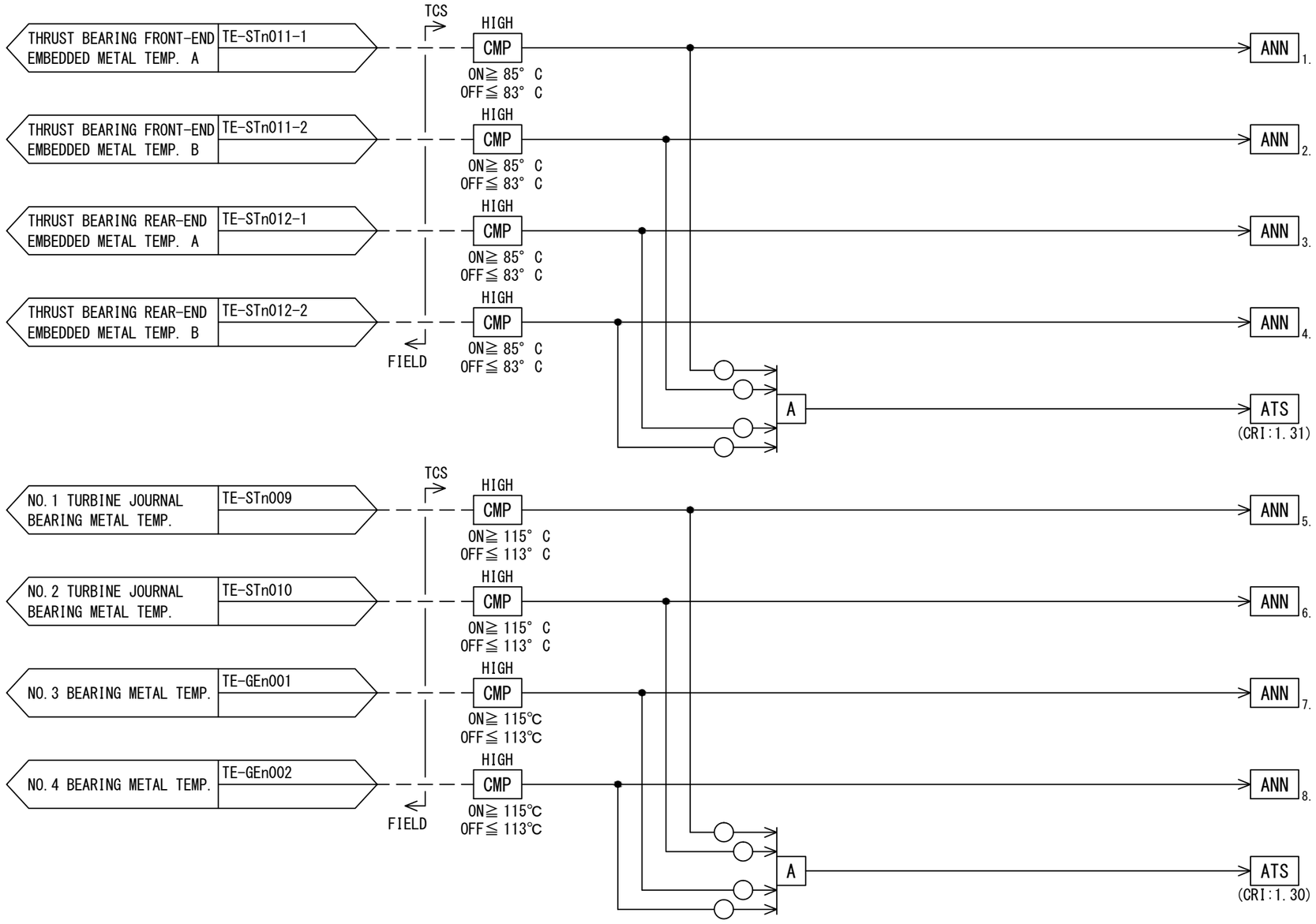


1. BEARING OIL TEMP. ABNORMAL
2. TCS MINOR TROUBLE
3. CONTROL OIL LINE FILTER (A) DIFF. PRESS. HIGH
4. CONTROL OIL LINE FILTER (B) DIFF. PRESS. HIGH

TYPE –			PAGE TD-1			
SERVICE TURBINE ALARM (1)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 84	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

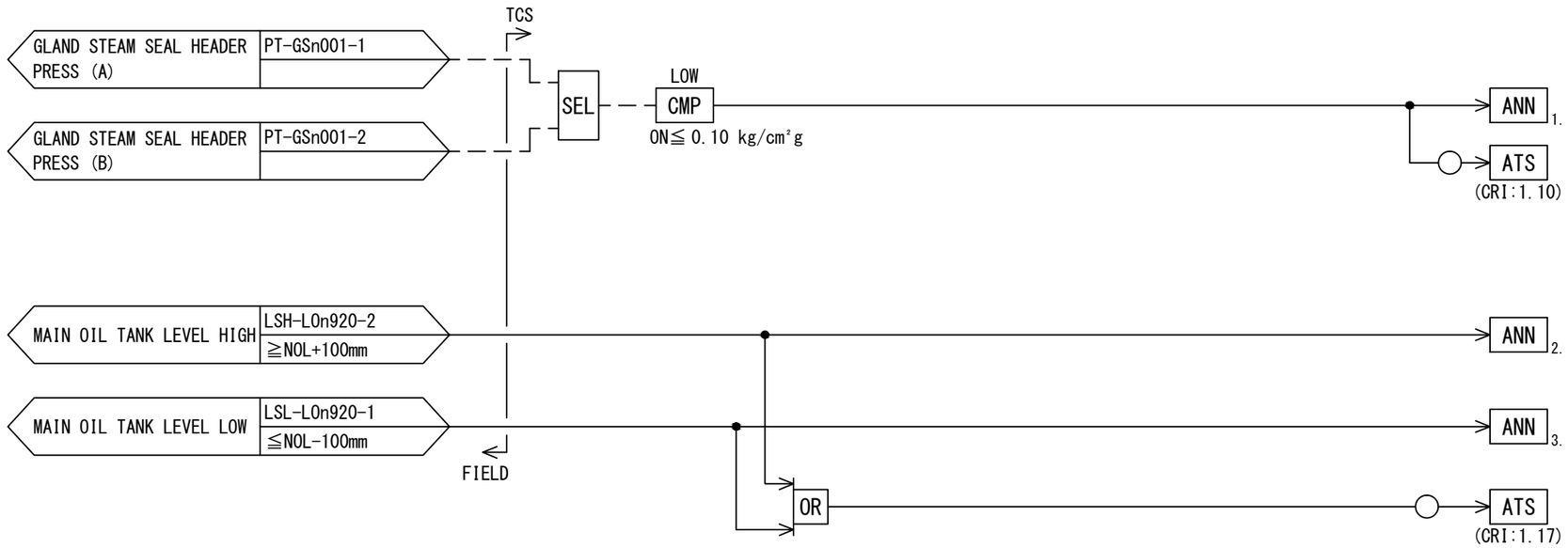
DWG. No. 310SC90-794

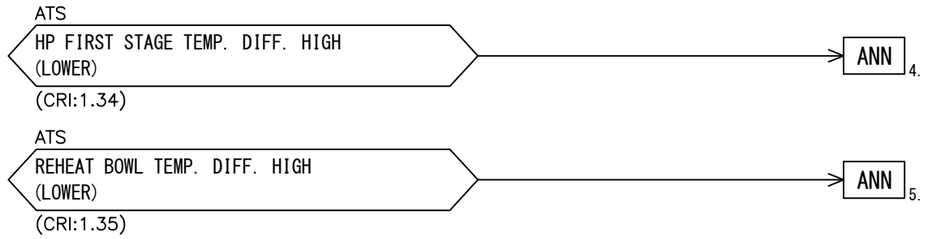


1. THRUST BEARING FRONT-END EMBEDDED METAL TEMP. 1 HIGH
2. THRUST BEARING FRONT-END EMBEDDED METAL TEMP. 2 HIGH
3. THRUST BEARING REAR-END EMBEDDED METAL TEMP. 1 HIGH
4. THRUST BEARING REAR-END EMBEDDED METAL TEMP. 2 HIGH
5. NO.1 TURBINE JOURNAL BEARING METAL TEMP. HIGH
6. NO.2 TURBINE JOURNAL BEARING METAL TEMP. HIGH
7. NO.3 BEARING METAL TEMP. HIGH
8. NO.4 BEARING METAL TEMP. HIGH

TYPE –					PAGE TD-2	
SERVICE TURBINE ALARM (2)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 85	Date of issue 2016-07-18
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-794

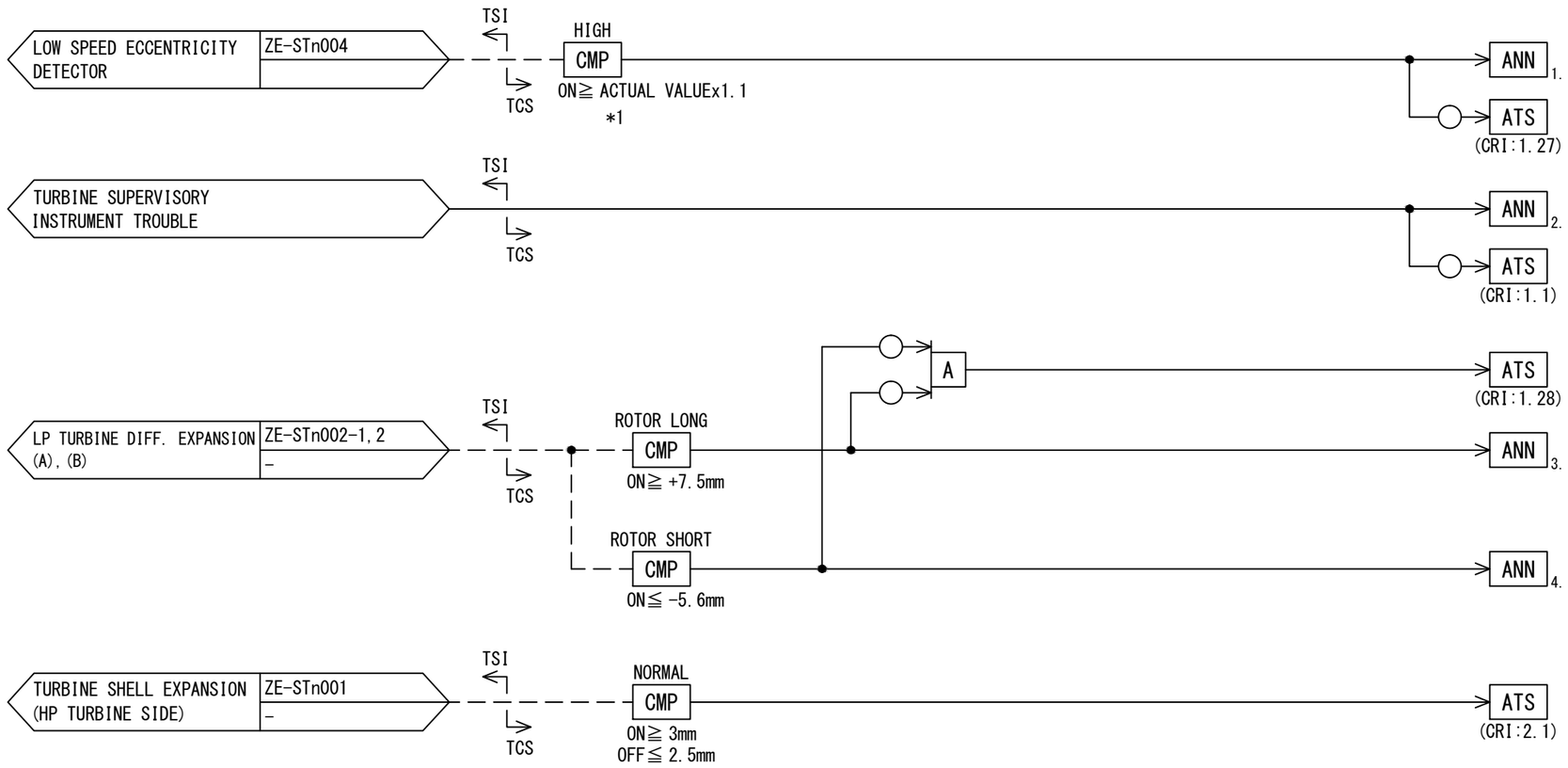




1. GLAND STEAM SEAL HEADER PRESS LOW
2. MAIN OIL TANK LEVEL HIGH
3. MAIN OIL TANK LEVEL LOW
4. HP FIRST STAGE TEMP. DIFF. HIGH (LOWER)
5. REHEAT BOWL TEMP. DIFF. HIGH (LOWER)

TYPE –					PAGE TD-3	
SERVICE TURBINE ALARM (3)						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 86	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division



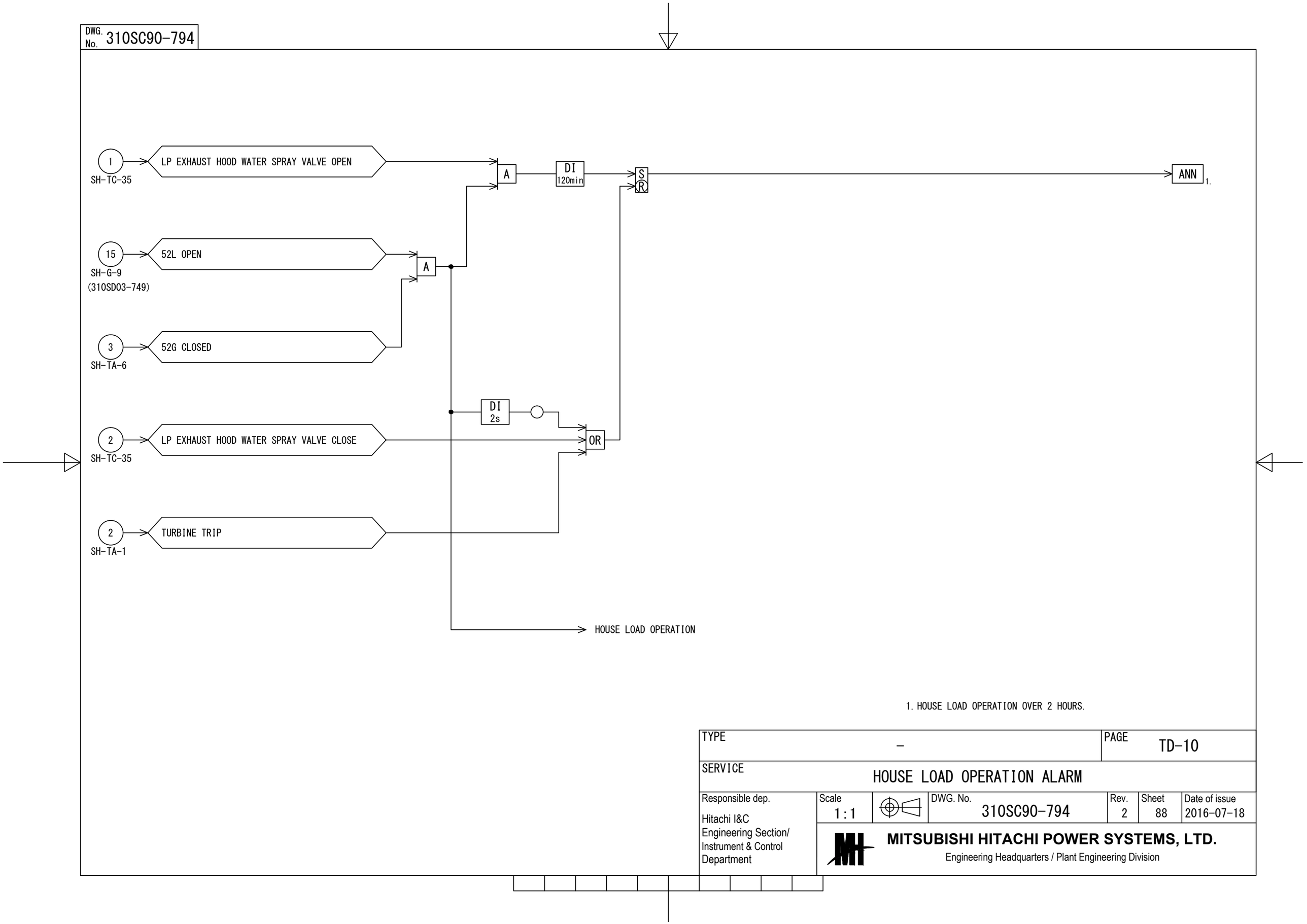
1. LOW SPEED ECCENTRICITY DETECTOR HIGH
2. TURBINE SUPERVISORY INSTRUMENT TROUBLE
3. LP TURBINE DIFF. EXPANSION HIGH (ROTOR LONG)
4. LP TURBINE DIFF. EXPANSION HIGH (ROTOR SHORT)

NOTE:
(*1) INITIAL SET POINT IS 0.05mm,
FINAL SET POINT WILL BE ADJUSTED DURING COMMISSIONING.

TYPE –						PAGE TD-4	
SERVICE TURBINE ALARM (4)							
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 87	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



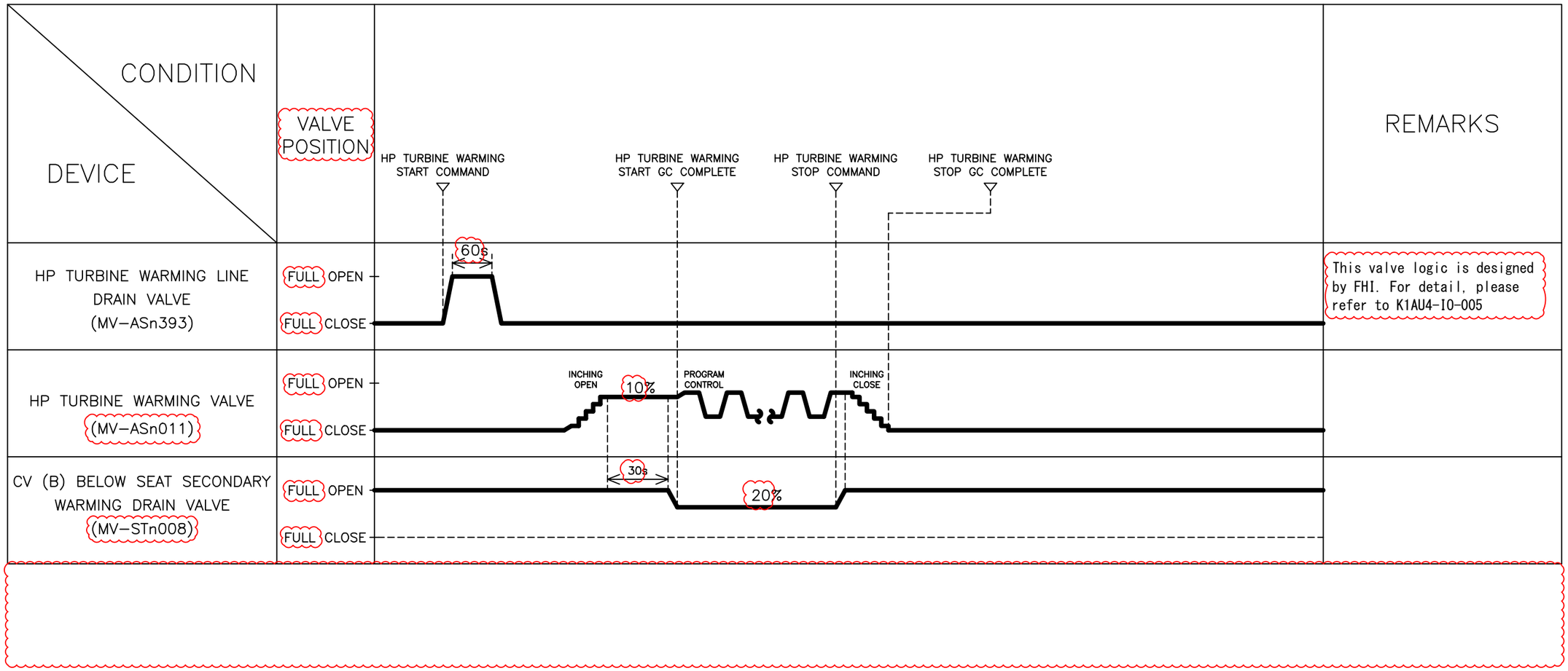
1. HOUSE LOAD OPERATION OVER 2 HOURS.

TYPE –					PAGE TD-10	
SERVICE HOUSE LOAD OPERATION ALARM						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 88	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

DEVICE \ CONDITION	VALVE POSITION	HP TURBINE WARMING START COMMAND / HP TURBINE WARMING START GC COMPLETE / HP TURBINE WARMING STOP COMMAND / HP TURBINE WARMING STOP GC COMPLETE	REMARKS
HP TURBINE WARMING LINE DRAIN VALVE (MV-ASn393)	FULL OPEN / FULL CLOSE	60s	This valve logic is designed by FHI. For detail, please refer to K1AU4-IO-005
HP TURBINE WARMING VALVE (MV-ASn011)	FULL OPEN / FULL CLOSE	INCHING OPEN 10% → PROGRAM CONTROL → INCHING CLOSE	
CV (B) BELOW SEAT SECONDARY WARMING DRAIN VALVE (MV-STn008)	FULL OPEN / FULL CLOSE	30s, 20%	



TYPE	–				PAGE	TE-0
SERVICE	HP TURBINE WARMING CHART					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	89	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

TCS
HP TURBINE WARMING START COMMAND FROM ATS
PB | HP TURBINE WARMING | AUTO
14 SH-TB-9 TURBINE TURNING
9 SH-TE-1 HP TURBINE WARMING AUTO CONTROL RESET CONDITION
7 SH-TE-2 HP TURBINE WARMING STOP PB ON
5 SH-TE-2 HP TURBINE WARMING STOP GROUP CONTROL COMPLETE
HP TURBINE FIRST STAGE INNER WALL METAL TEMP. | TE-STn001-1 | –
FIELD TCS CMP ON≧150℃ OFF≦145℃
DI VARIABLE (*1)
M ℃
CMP ON≧151℃ OFF≦156℃
OR A S R
DI 300s DT 10s A
A 10 SH-TE-2

(*1) : Set value of variable timer will be fixed at HP turbine warming valve not full close condition.

HP TURBINE FIRST STAGE INNER WALL METAL TEMP (LOWER) (℃)	0	95	110	125	135	140	145	150
HOLDING TIME (hour)	3.8	2.5	2.2	1.75	1.25	0.95	0.55	0

PB | HP TURBINE WARMING | START
14 SH-TB-9 TURBINE TURNING
6 SH-TE-1 HP TURBINE WARMING START GROUP CONTROL COMPLETE
4 SH-TE-2 HP TURBINE WARMING STOP GROUP CONTROL PROGRESS
7 SH-TC-20 HP TURBINE WARMING PERMISSIVE
7 SH-TA-10 CONDENSER VACUUM NORMAL
2 SH-TA-1 TURBINE TRIP
PB | HP TURBINE WARMING | RESET
7 SH-TE-2
3 SH-TE-3 HP TURBINE WARMING START GC RELATED AUX. IN AUTO
OR A
PO 2s
OR S R
OR 9 SH-TE-1
DI 500s A ANN 1.

(*2) The open position and timer to be adjusted during the site commissioning and is basically determined by the HP turbine exhaust pressure.

1. HP TURBINE WARMING START GROUP CONTROL TROUBLE

HP TURBINE WARMING AUTO CONTROL PROGRESS W
DI 230m
2 SH-TA-1 TURBINE TRIP
PO 2s
DI 120m
OR A S R OR ATS (CRI:2.92)

HP TURBINE WARMING START GROUP CONTROL PROGRESS W 5 SH-TE-2

(WO) TCS DCS 81 HP TURBINE WARMING LINE DRAIN VALVE | OPEN | MV-ASn393 DWG NO.SH- 1 TCS DCS OR A DI 60s

(WO) TCS DCS 82 HP TURBINE WARMING LINE DRAIN VALVE | CLOSE | MV-ASn393 DWG NO.SH- 2 TCS DCS A

A FLK T1 : 2s T2 : 60s DI 280s OR (WO) 1 SH-TC-20 HP TURBINE WARMING VALVE | 10% OPENING | MV-ASn011 6 SH-TC-20 (*2)

A (*2) DI 30s (WO) 3 SH-TC-5 CV(B) BELOW SEAT SECONDARY WARMING DRAIN VALVE | 20% OPENING | MV-STn008 6 SH-TC-5 (*2)

A 4 SH-TC-20 HP TURBINE WARMING VALVE | PROGRAM CONTROL | MV-ASn011 4 SH-TC-20

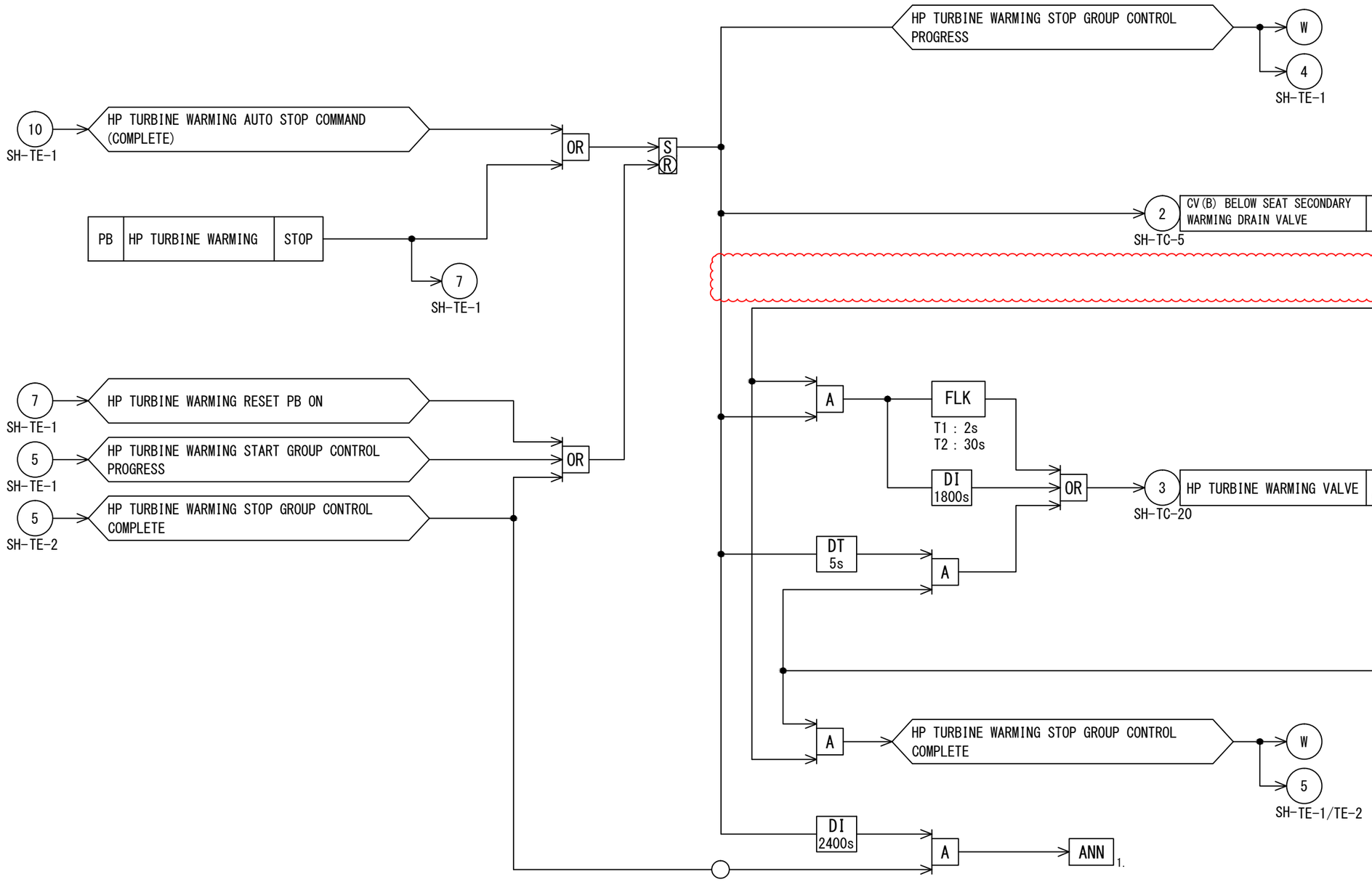
A HP TURBINE WARMING START GROUP CONTROL COMPLETE W 6 SH-TE-1

TYPE	–	PAGE	TE-1
SERVICE	HP TURBINE WARMING START GROUP CONTROL		

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-794	2	90	2016-07-18

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Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

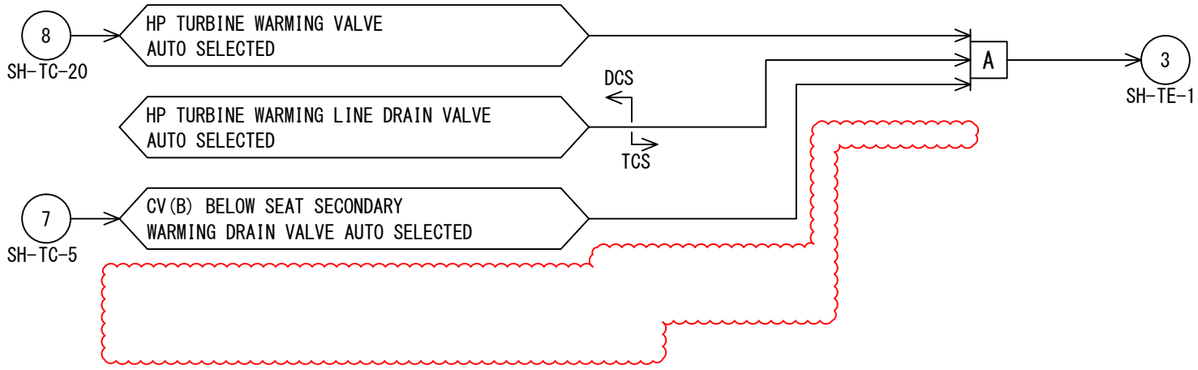


1. HP TURBINE WARMING STOP GROUP CONTROL TROUBLE

TYPE	–					PAGE	TE-2
SERVICE	HP TURBINE WARMING STOP GROUP CONTROL						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-794	2	91	2016-07-18	

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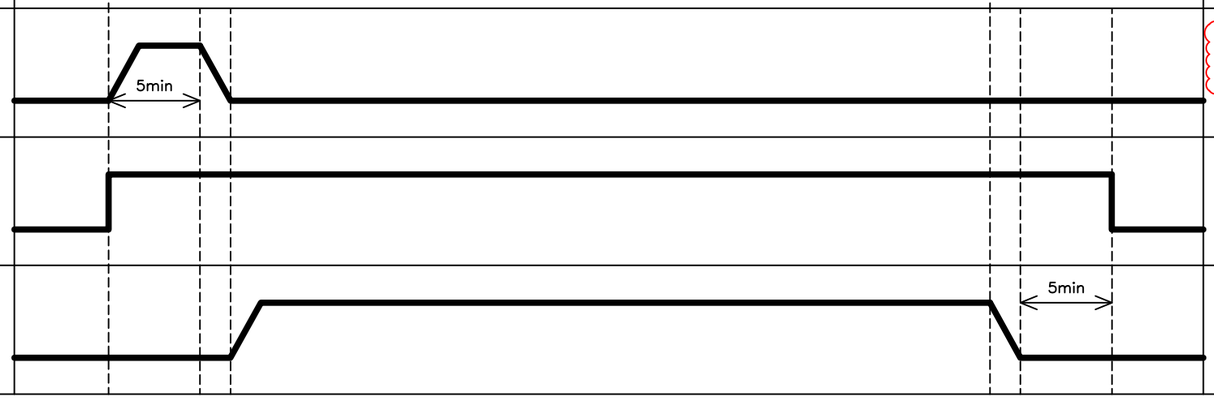
DWG. No. 310SC90-794



TYPE –					PAGE TE-3	
SERVICE HP TURBINE WARMING GC RELATED AUX. IN AUTO						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 92	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

CONDITION / DEVICE	STATUS	KICK SIGNAL FROM TURBINE GLAND SEAL START GROUP CONTROL / KICK SIGNAL FROM TURBINE GLAND SEAL STOP GROUP CONTROL	REMARKS
TURBINE GLAND STEAM LINE DRAIN VALVE (MV-ASn397)	OPEN / CLOSE		This valve logic is designed by FHI. For detail, please refer to K1AU4-IO-005
GLAND STEAM EXHAUSTER (F-u911)	START / STOP		
STEAM SEAL FEED INLET VALVE (MV-GSn030)	OPEN / CLOSE		

TYPE –					PAGE TE-5	
SERVICE TURBINE GLAND SEAL TIME CHART						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 93	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

TURBINE GLAND SEAL GROUP START COMMAND
DCS
TCS
OR
PB | TURBINE GLAND SEAL | GROUP START
31
SH-TE-7
STEAM SEAL AUXILIARY FEED VALVE IN AUTO
310SD14-314 (SH-5)
33
SH-TE-7
STEAM PACKING UNLOADING VALVE IN AUTO
310SD14-314 (SH-5)
CONDENSER VACUUM BREAK COMPLETE
30
SH-TE-7
A
S
R
(WO)
TURBINE GLAND SEAL START GROUP CONTROL IN PROGRESS
W
6
SH-TE-7
8
TURBINE GLAND STEAM LINE DRAIN VALVE
FULL OPEN
62
SH-AS01
MV-ASn397
SH-AS01
OR
2
GLAND STEAM EXHAUSTER
START
1
SH-TC-28
F-u911
SH-TC-28
Ambient Cold Mode Temp. (For Gland Steam Desuperheater Temp. Control (TCV-CDn126))
HP TURBINE FIRST STAGE INNER WALL METAL TEMP. | TE-STn001-1 | –
REHEAT BOWL INNER WALL METAL TEMP. (A)/(B) | TE-STn004-1, 2 | –
LS
AMB. COLD
CMP
ON < 150° C
OFF > 155° C
OR
W
(PERMISSIVE)
GLAND STEAM DRAIN SEPARATOR TURBINE SIDE TEMP. | ASn017 | 0-300° C
(*1)
CMP
ON ≦ 270° C
OFF ≧ 275° C
(*1)
CMP
ON ≧ 200° C
OFF ≦ 195° C
DI 300s
11
TURBINE GLAND STEAM LINE DRAIN VALVE
FULL CLOSE
61
SH-AS01
MV-ASn397
SH-AS01
W
(COMPLETE)
DI 10s
4
STEAM SEAL FEED INLET VALVE
FULL OPEN
1
SH-TC-31
MV-GSn030
SH-TC-31
DI 480s
ANN 1
PB | TURBINE GLAND SEAL | GROUP START RESET
14
SH-TB-9
TURBINE TURNING
1
SH-TE-8
TURBINE GLAND SEAL START GC RELATED AUX IN AUTO
6
SH-TE-7
TURBINE GLAND SEAL STOP GROUP CONTROL IN PROGRESS
7
SH-TE-6
TURBINE GLAND SEAL START GC COMPLETE

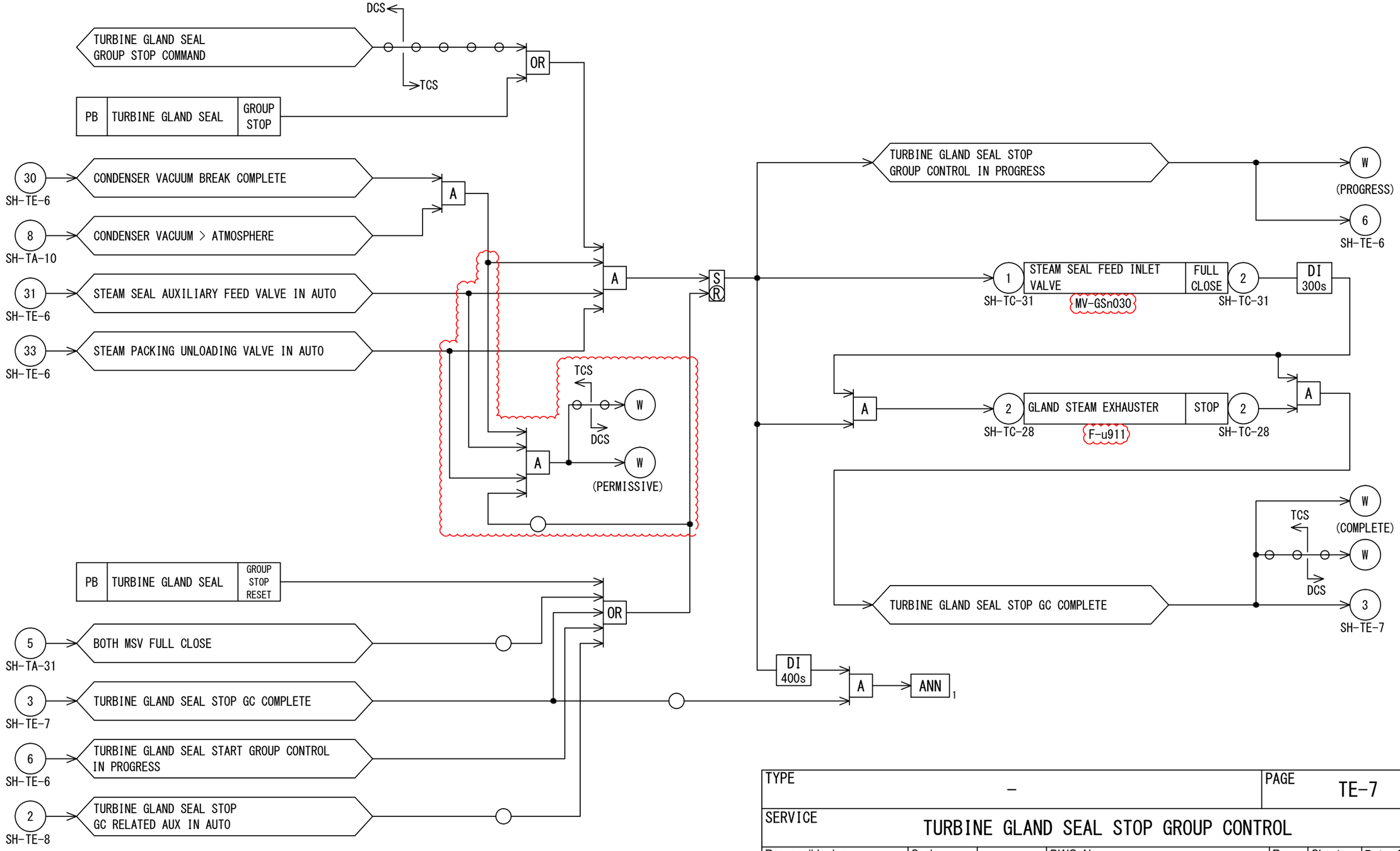
1. TURBINE GLAND SEAL START GROUP CONTROL TROUBLE

NOTE:
(*1) THE COMPLETE AUXILIARY STEAM REQUIREMENT IS SHOWN THE FOLLOWING DOCUMENTS.
AUXILIARY STEAM CONSUMPTION PER 1 UNIT (PPE-13-0251)

TYPE –					PAGE TE-6	
SERVICE TURBINE GLAND SEAL START GROUP CONTROL						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 94	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. TURBINE GLAND SEAL STOP GROUP CONTROL TROUBLE

TYPE	–					PAGE	TE-7
SERVICE	TURBINE GLAND SEAL STOP GROUP CONTROL						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-794	2	95	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

51
SH-AS01
TURBINE GLAND STEAM LINE
DRAIN VALVE AUTO (MV-ASn937)

DCS
TCS

10
SH-TC-31
STEAM SEAL FEED INLET VALVE AUTO

10
SH-TC-28
GLAND STEAM EXHAUSTER AUTO

A
TURBINE GLAND SEAL START
GC RELATED AUX IN AUTO
1
SH-TE-6

A
TURBINE GLAND SEAL STOP
GC RELATED AUX IN AUTO
2
SH-TE-7

HP TURBINE FIRST STAGE INNER WALL METAL
TEMP. (TE-STn001-1) ABNORMAL

REHEAT BOWL INNER WALL METAL TEMP. (A)
(TE-STn004-1) ABNORMAL

REHEAT BOWL INNER WALL METAL TEMP. (B)
(TE-STn004-2) ABNORMAL

OR

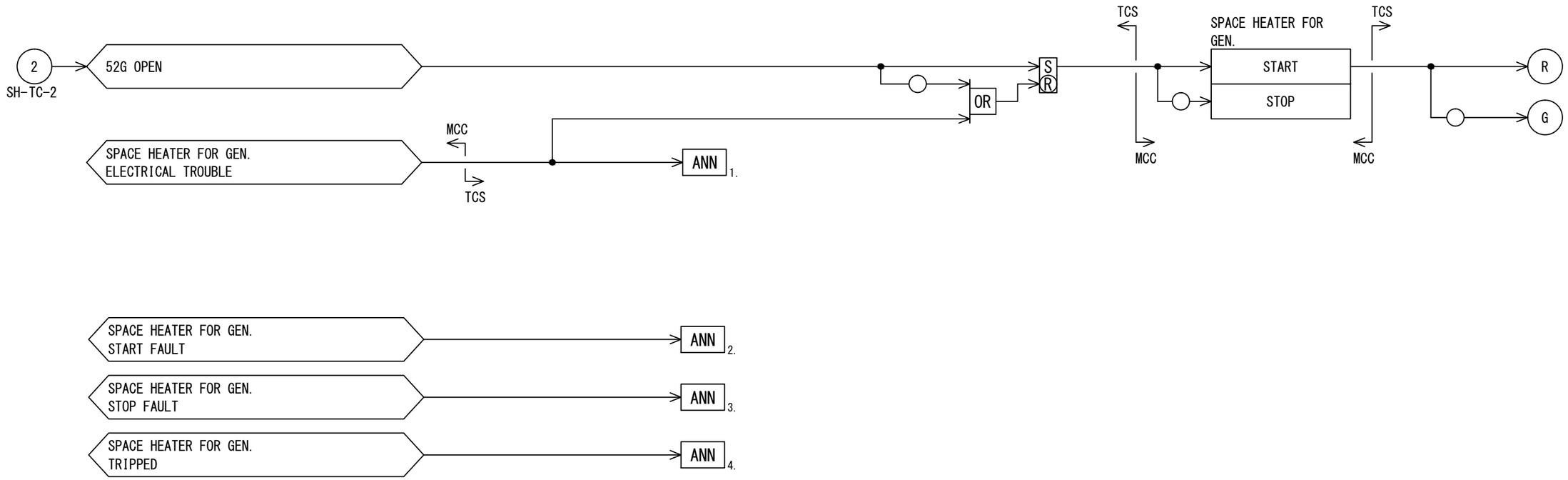
A
TCS
DCS

Mode Judgment Instrument Normal
(For Gland Steam Desuperheater Temp. Control
(TCV-CDn126))

TYPE –					PAGE TE-8	
SERVICE TURBINE GLAND SEAL GC RELATED AUX. IN AUTO						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-794	Rev. 2	Sheet 96	Date of issue 2016-07-18

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

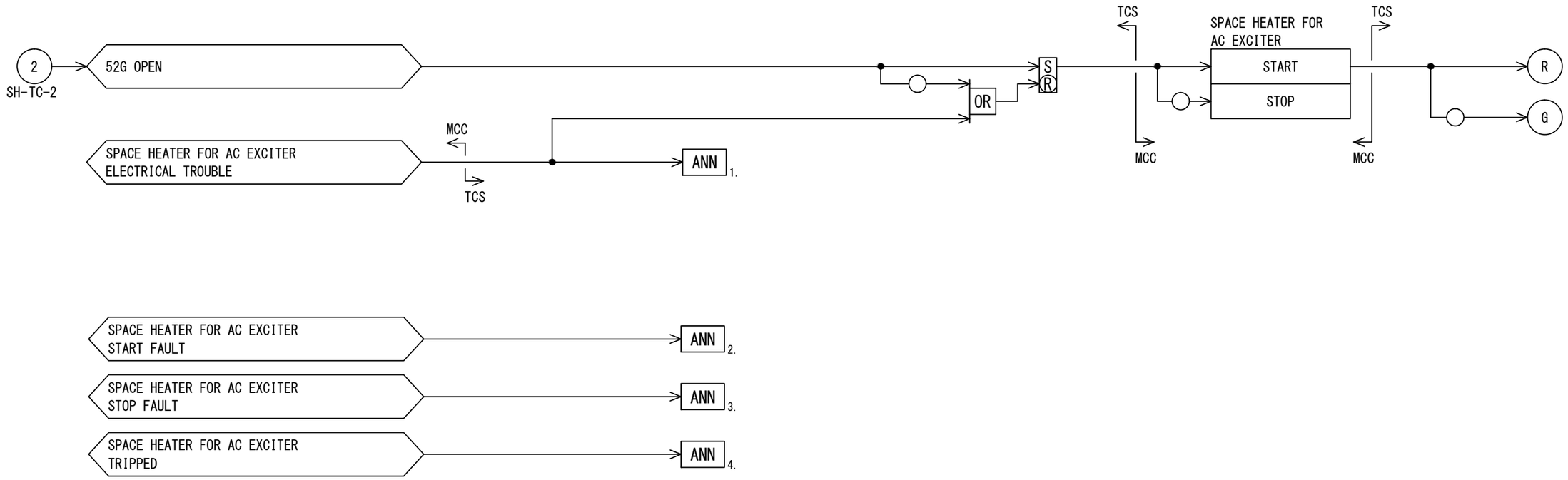


1. SPACE HEATER FOR GEN. ELECTRICAL TROUBLE
2. SPACE HEATER FOR GEN. START FAULT
3. SPACE HEATER FOR GEN. STOP FAULT
4. SPACE HEATER FOR GEN. TRIPPED

TYPE	–					PAGE	GD-1
SERVICE	SPACE HEATER FOR GEN.						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-794	2	97	2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794

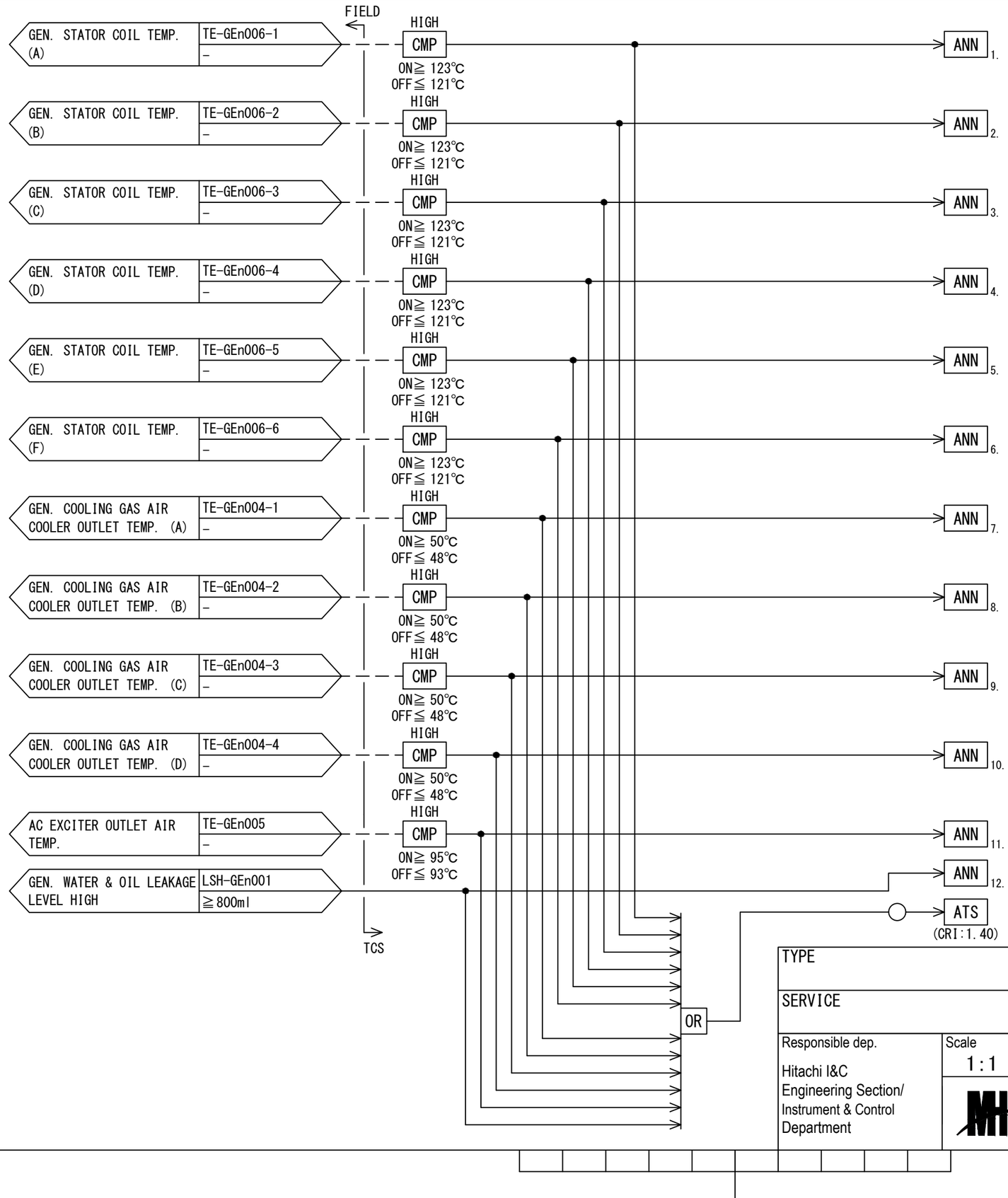


1. SPACE HEATER FOR AC EXCITER ELECTRICAL TROUBLE
2. SPACE HEATER FOR AC EXCITER START FAULT
3. SPACE HEATER FOR AC EXCITER STOP FAULT
4. SPACE HEATER FOR AC EXCITER TRIPPED

TYPE –					PAGE GD-2	
SERVICE SPACE HEATER FOR AC EXCITER						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1	DWG. No. 310SC90-794	Rev. 2	Sheet 98	Date of issue 2016-07-18	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-794



1. GEN. STATOR COIL TEMP. (A) HIGH
2. GEN. STATOR COIL TEMP. (B) HIGH
3. GEN. STATOR COIL TEMP. (C) HIGH
4. GEN. STATOR COIL TEMP. (D) HIGH
5. GEN. STATOR COIL TEMP. (E) HIGH
6. GEN. STATOR COIL TEMP. (F) HIGH
7. GEN. COOLING GAS AIR COOLER OUTLET TEMP. (A) HIGH
8. GEN. COOLING GAS AIR COOLER OUTLET TEMP. (B) HIGH
9. GEN. COOLING GAS AIR COOLER OUTLET TEMP. (C) HIGH
10. GEN. COOLING GAS AIR COOLER OUTLET TEMP. (D) HIGH
11. AC EXCITER OUTLET AIR TEMP. HIGH
12. GEN. WATER & OIL LEAKAGE LEVEL HIGH

TYPE	–
PAGE	GD-5
SERVICE	GENERATOR ALARM (1)

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-794	2	99	2016-07-18

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